MATHEMATICS

CIRCLE

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EQUATION OF CIRCLE

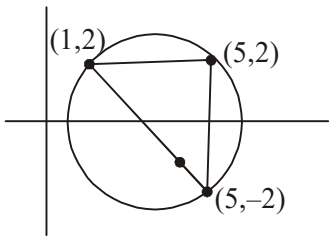
1. The radius of the circle passing through the points (1, 2), (5, 2) and (5, –2) is :

   बिन्दुओं (1, 2), (5, 2) एवं (5, –2) से गुजरने वाले वृत्त की त्रिज्या है :

   (A) $5\sqrt{2}$ (B) $2\sqrt{5}$ (C) $3\sqrt{2}$ (D) $2\sqrt{2}$

Ans. D

Sol.



diameter = $4\sqrt{2}$ $\Rightarrow$ r = $2\sqrt{2}$

2. The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle having area as 154 sq. units. Then the equation of the circle is : (Take $\pi = 22/7$)

   रेखा $2x - 3y = 5$ तथा $3x - 4y = 7$ क्षेत्रफल 154 वर्ग इकाई वाले वृत्त के व्यास है। तब वृत्त का समीकरण है : ($\pi = 22/7$)

   (A) $x^2 + y^2 - 2x + 2y = 62$ (B) $x^2 + y^2 + 2x - 2y = 62$

   (C) $x^2 + y^2 + 2x - 2y = 47$ (D) $x^2 + y^2 - 2x + 2y = 47$

Ans. D

Sol. Centre of circle = Point of intersection of diameters = (1, –1)

Now area = 154 $\Rightarrow \pi r^2 = 154 \Rightarrow r = 7$

Hence the equation of required circle is

$(x - 1)^2 + (y + 1)^2 = 7^2 \Rightarrow x^2 + y^2 - 2x + 2y = 47$

3. The equation of the circle passing through the points (1, 0) and (0, 1) and having the smallest radius is :

[AIEEE-2011]

उस वृत्त का समीकरण जो बिन्दुओं (1, 0) तथा (0, 1) से होकर जाता है तथा जिसकी त्रिज्या न्यूनतम है, हैं : **[AIEEE 2011]**

(A) $x^2 + y^2 + x + y - 2 = 0$ (B) $x^2 + y^2 - 2x - 2y + 1 = 0$

(C) $x^2 + y^2 - x - y = 0$ (D) $x^2 + y^2 + 2x + 2y - 7 = 0$

Ans. C

Sol. (1, 0) and (0, 1) will be ends of diameter

So equation of circle

$(x-1)(x-0)+(y-0)(y-1)$

$x^2+y^2-x-y=0$

INTERCEPT OF CIRCLE

4. The two lines through (2, 3) from which the circle $x^2+y^2=25$ intercepts chords of length 8 units have equations:

(A) $2x+3y=13,\ x+5y=17$ (B) $y=3,\ 12x+5y=39$

(C) $x=2,\ 9x-11y=51$ (D) none of these

बिन्दु (2, 3) से गुजरने वाली उन दो सरल रेखाओं के समीकरण जिन पर वृत्त $x^2+y^2=25$, 8 इकाई लम्बाई के अन्तःखण्ड की जीवा बनाता है, होंगे :

(A) $2x+3y=13,\ x+5y=17$ (B) $y=3,\ 12x+5y=39$

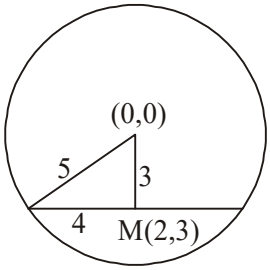
(C) $x=2,\ 9x-11y=51$ (D) इनमें से कोई नहीं

Ans. B

Sol. Let slope of required line is m

$y-3=m(x-2) \quad \Rightarrow mx-y+(3-2m)=0$

length of $\perp$ from origin $=3$



$\Rightarrow 9+4m^2-12m=9+9m^2 \Rightarrow 5m^2+12m=0 \Rightarrow m=0,-\frac{12}{5}$

Hence lines are $y-3=0 \Rightarrow y=3$

$y-3=-\frac{12}{5}(x-2) \Rightarrow 5y-15=-12x+24 \Rightarrow 12x+5y=39.$

5. A variable circle passes through the fixed point A (p, q) and touches x –axis. The locus of the other end of the diameter through A is :

एक चर वृत्त स्थिर बिन्दु A (p, q) से गुजरता है। एवं x –अक्ष को स्पर्श करता है। A से गुजरने वाले व्यास के दूसरे सिरे का बिन्दुपथ है:

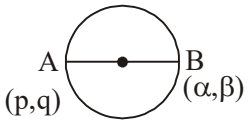
(A) $(y-q)^2=4px$ (B) $(x-q)^2=4py$ (C) $(y-p)^2=4qx$ (D) $(x-p)^2=4qy$

Ans. D

Sol. Equation of circle having AB as diameter is

$(x-p)(x-\alpha)+(y-q)(y-\beta)=0$

or $x^2+y^2-(p+\alpha)x-(q+\beta)y+p\alpha+q\beta=0$ ..... (i)



as it touches x-axis putting $y=0$,

we get $x^2-(p+\alpha)x+p\alpha+q\beta=0$ ..... (ii)

Since, circle (i) touches x-axis

Discriminant of equation (ii) $=0$

$\Rightarrow (p+\alpha)^2=4(p\alpha+q\beta) \Rightarrow (p-\alpha)^2=4q\beta$

$\therefore$ Locus of $B(\alpha,\beta)$ is $(p-x)^2=4qy$

or $(x-p)^2=4qy$

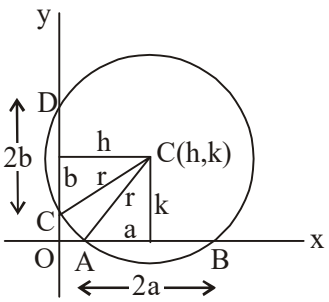
6. Two thin rods AB & CD of lengths 2a & 2b move along OX & OY respectively, when 'O' is the origin. The equation of the locus of the centre of the circle passing through the extremities of the two rods is :

दो पतली छड़े AB और CD जिनकी लम्बाई 2a और 2b है क्रमशः OX और OY अक्षों के अनुदिश गतिशील है। जहाँ 'O' मूलबिन्दु हैं, तो दोनों छड़ों के सिरों से गुजरने वाले वृत्त के केन्द्र का बिन्दुपथ होगा :

(A) $x^2 + y^2 = a^2 + b^2$ (B) $x^2 - y^2 = a^2 - b^2$ (C) $x^2 + y^2 = a^2 - b^2$ (D) $x^2 - y^2 = a^2 + b^2$

Ans. B

Sol. $h^2 + b^2 = r^2$ and $k^2 + a^2 = r^2$ $\Rightarrow h^2 - k^2 = a^2 - b^2$



$\therefore$ locus is $x^2 - y^2 = a^2 - b^2$

PARAMETRIC FORM OF CIRCLE

7. The centre of a set of circles, each of radius 3, lie on the circle $x^2+y^2= 25$. The locus of any point in the set is :

प्रत्येक 3 त्रिज्या के वृत्त के समुच्चय का केन्द्र वृत्त $x^2+y^2= 25$ पर स्थित है। समुच्चय किसी बिन्दु का बिन्दुपथ होगा :

(A) $4 \le x^2+y^2 \le 64$ (B) $x^2+y^2 \le 25$ (C) $x^2+y^2 \ge 25$ (D) $3 \le x^2+y^2 \le 9$

Ans. A

Sol. Let C lie on $x^2 + y^2 = 25$

Let S be the set of circles with radius 3 and center (h, k).

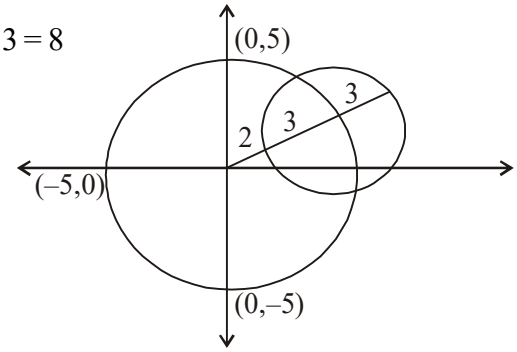
Any point P(h, k) on S will be 3 units from center C.

Distance from origin will br at least $5 - 3 = 2$. and at most $5 + 3 = 8$

$\Rightarrow 2 \le \sqrt{h^2 + k^2} \le 8$

$4 \le h^2 + k^2 \le 64$

Locus is $4 \le x^2 + y^2 \le 64$



8. If $\frac{x^2 + y^2}{x + y} = 4$, then all possible values of $(x - y)$ is given by :

यदि $\frac{x^2+y^2}{x+y}=4$ तो (x – y) के सम्भावित मान है :

(A) $\left[-2\sqrt{2},2\sqrt{2}\right]$ (B) {– 4, 4} (C) [– 4, 4] (D) [– 2, 2]

Ans. C

Sol. We have $x^2+y^2=4x+4y$

$\Rightarrow \quad (x-2)^2+(y-2)^2=8=\left(2\sqrt{2}\right)^2$

Hence x, y lie on a circle with parametric coordinates

$$\left.\begin{aligned}x&=2+2\sqrt{2}\cos\theta\\ y&=2+2\sqrt{2}\sin\theta\end{aligned}\right] \quad \Rightarrow \quad x-y=2\sqrt{2}(\cos\theta-\sin\theta)$$

$\therefore \quad x-y\in[-4,4]$

POSITION OF POINT W.R.T. A CIRCLE

9. The circle $x^2+y^2-6x-10y+c=0$ does not intersect or touch either axis & the point (1, 4) is inside the circle. Then the range of possible values of c is given by :

वृत्त $x^2+y^2-6x-10y+c=0$ न तो अक्षों को काटता है और न ही स्पर्श करता है तथा बिन्दु (1, 4) वृत्त के अन्दर स्थित हो, तो c के संभावित मानों का परिसर है :

(A) c > 9 (B) c > 25 (C) c > 29 (D) 25 < c < 29

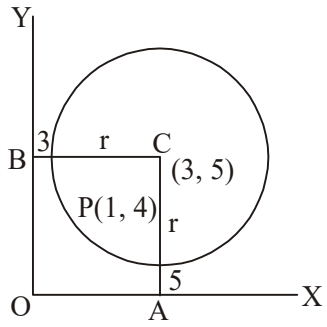
Ans. D

Sol. $S=x^2+y^2-6x-10y+k=0$

Centre C(3,5), $r^2=9+25k=34-k$

Point P(1,4) lies inside the circle therefore

//S = –ve// $-29+k<0$ or $k<29$ ...(1)



Since the circle neither touches nor intersects the axes of co-ordinates,

therefore from thee figure it is clear that r < 3 (x coordinate of centre) and r < 5 (y co-ordinate of centre).

Hence, $r<3 \Rightarrow r^2<9$

or $34-k<9$ or $25<k$ ...(2)

from (1) and (2), we have $25<k<29$.

TANGENT AND NORMAL

10. The tangent lines to the circle $x^2 + y^2 - 6x + 4y = 12$ which are parallel to the line $4x + 3y + 5 = 0$ are given by :

(A) $4x + 3y - 7 = 0, 4x + 3y + 15 = 0$ (B) $4x + 3y - 31 = 0, 4x + 3y + 19 = 0$

(C) $4x + 3y - 17 = 0, 4x + 3y + 13 = 0$ (D) none of these

सरल रेखा $4x + 3y + 5 = 0$ के समान्तर वृत्त $x^2 + y^2 - 6x + 4y = 12$ की स्पर्श रेखाओं के समीकरण है :

(A) $4x + 3y - 7 = 0, 4x + 3y + 15 = 0$ (B) $4x + 3y - 31 = 0, 4x + 3y + 19 = 0$

(C) $4x + 3y - 17 = 0, 4x + 3y + 13 = 0$ (D) इनमें से कोई नहीं

Ans. B

Sol. Line parallel to given line $4x + 3y + 5 = 0$ is $4x + 3y + k = 0$

This is tangent to $x^2 + y^2 - 6x + 4y - 12 = 0$

$$\left|\frac{12 - 6 + k}{5}\right| = 5 \Rightarrow 6 + k = \pm 25 \Rightarrow k = 19, -31$$

Hence required line $4x + 3y - 31 = 0,\ 4x + 3y + 19 = 0$

11. The condition so that the line $(x + g)\cos\theta + (y + f)\sin\theta = k$ is a tangent to $x^2 + y^2 + 2gx + 2fy + c = 0$ is :

सरल रेखा $(x + g)\cos\theta + (y + f)\sin\theta = k$ के वृत्त $x^2 + y^2 + 2gx + 2fy + c = 0$ की स्पर्श रेखा होने के लिए प्रतिबन्ध है :

(A) $g^2 + f^2 = c + k^2$ (B) $g^2 + f^2 = c^2 + k$ (C) $g^2 + f^2 = c^2 + k^2$ (D) $g^2 + f^2 = c + k$

Ans. A

Sol. $$p = \left|\frac{(-g + g)\cos\theta + (-f + f)\sin\theta - k}{\sqrt{\cos^2\theta + \sin^2\theta}}\right| = \sqrt{g^2 + f^2 - c} \Rightarrow g^2 + f^2 = c + k^2$$

12. The tangent to the circle $x^2 + y^2 = 5$ at the point $(1, -2)$ also touches the circle $x^2 + y^2 - 8x + 6y + 20 = 0$ at :

बिन्दु $(1,-2)$ पर वृत्त $x^2 + y^2 = 5$ की स्पर्श रेखा वृत्त $x^2 + y^2 - 8x + 6y + 20 = 0$ को किस बिन्दु पर स्पर्श करती हैं :

(A) $(-2, 1)$ (B) $(-3, 0)$ (C) $(-1, -1)$ (D) $(3, -1)$

Ans. D

Sol. Equation of tangent $x - 2y = 5$

Let required point be (α,β)

$\alpha x + \beta y - 4(x + \alpha) + 3(y + \beta) + 20 = 0 \Rightarrow x(\alpha - 4) + y(\beta + 3) - 4\alpha + 3\beta + 20 = 0$

Comparing

$$\frac{\alpha - 4}{1} = \frac{\beta + 3}{-2} = \frac{4\alpha - 3\beta - 20}{5}$$

Similarly $(\alpha,\beta) \equiv (3, -1)$

POWER OF A POINT

13. The lengths of the tangents from any point of the circle $15x^2 + 15y^2 - 48x + 64y = 0$ to the two circles $5x^2 + 5y^2 - 24x + 32y + 75 = 0$, $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ are in the ratio :

वृत $15x^2 + 15y^2 - 48x + 64y = 0$ पर स्थित किसी एक बिन्दु से वृत $5x^2 + 5y^2 - 24x + 32y + 75 = 0$ तथा $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ पर डाली गई स्पर्श रेखाओं की लम्बाई का अनुपात है :

(A) 1 : 2 (B) 2 : 3 (C) 3 : 4 (D) 1 : 4

Ans. A

Sol. Let any point $P(x_1, y_1)$ to the circle $x^2 + y^2 - \frac{16x}{5} + \frac{64y}{15} = 0 \Rightarrow x_1^2 + y_1^2 - \frac{16}{5}x_1 + \frac{64}{15}y_1 = 0$

Length of tangent from $P(x_1, y_1)$ to the circle are in ration

$$\frac{\sqrt{S_1}}{\sqrt{S_2}} = \frac{\sqrt{x_1^2 + y_1^2 - \frac{24}{5}x_1 + \frac{32}{5}y_1 + 15}}{\sqrt{x_1^2 + y_1^2 - \frac{48}{5}x_1 + \frac{64}{5}y_1 + 60}} = \sqrt{\frac{\frac{16}{5}x_1 - \frac{64}{15}y_1 - \frac{24}{5}x_1 + \frac{32}{5}y_1 + 15}{\frac{16}{5}x_1 - \frac{64}{15}y_1 - \frac{48}{5}x_1 + \frac{64}{5}y_1 + 60}}$$

$$= \sqrt{\frac{-24x_1 + 32y_1 + 225}{-96x_1 + 128y_1 + 900}} = \sqrt{\frac{-24x_1 + 32y_1 + 225}{4(-24x_1 + 32y_1 + 225)}} = \frac{1}{2}$$

Alter :

Circle passes through origin (0, 0) so

$$\sqrt{\frac{S_1}{S_2}} = \sqrt{\frac{15}{60}} = \frac{1}{2}$$

14. If L_1 and L_2 are the length of the tangent from (0, 5) to the circles $x^2 + y^2 + 2x - 4 = 0$ and $x^2 + y^2 - y + 1 = 0$ then :

यदि L_1 तथा L_2 वृतो $x^2 + y^2 + 2x - 4 = 0$ तथा $x^2 + y^2 - y + 1 = 0$ पर बिन्दु (0, 5) से डाली गयी स्पर्श रेखाओं की लम्बाई है तो :

(A) $L_1 = 2L_2$ (B) $L_2 = 2L_1$ (C) $L_1 = L_2$ (D) $L_1^2 = L_2$

Ans. C

Sol. $L_1 = L_2 = \sqrt{21}$

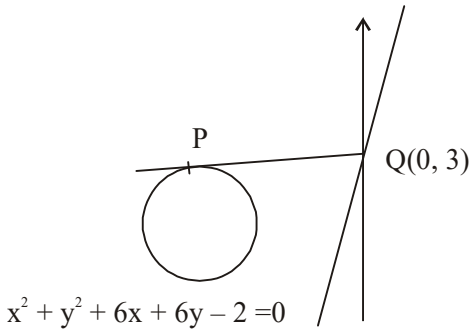
15. If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y–axis, then the length of PQ is :

यदि वृत्त $x^2 + y^2 + 6x + 6y = 2$ पर स्थित बिन्दु P पर स्पर्श रेखा, रेखा $5x - 2y + 6 = 0$ को y अक्ष पर बिन्दु Q पर मिलती है, तो PQ की लम्बाई है :

(A) 4 (B) $2\sqrt{5}$ (C) 5 (D) $3\sqrt{5}$

Ans. C

Sol.



$PQ = \sqrt{S_1} = 9 + 18 - 2$

16. Tangents drawn from the point P(l, 8) to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. The equation of the circumcircle of the triangle PAB is : **[JEE-2009]**

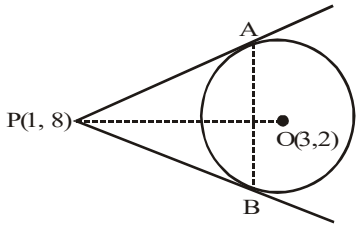
बिन्दु P(l, 8) से वृत $x^2 + y^2 - 6x - 4y - 11 = 0$ पर खीचीं गई स्पर्श रेखायें वृत्त को A व B पर स्पर्श करती है। त्रिभुज PAB के परिवृत का समीकरण होगा : **[JEE-2009]**

(A) $x^2 + y^2 + 4x - 6y + 19 = 0$ (B) $x^2 + y^2 - 4x - 10y + 19 = 0$

(C) $x^2 + y^2 - 2x + 6y - 29 = 0$ (D) $x^2 + y^2 - 6x - 4y + 19 = 0$

Ans. B

Sol.



The required circle is a circle described on OP as diameter.

DIRECTOR CIRCLE

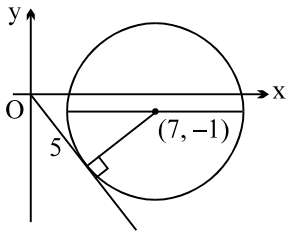
17. The angle between the two tangents from the origin to the circle $(x-7)^2 + (y+1)^2 = 25$ equals :

मूल बिन्दु से वृत $(x-7)^2 + (y+1)^2 = 25$ पर डाली गई स्पर्श रेखाओं के मध्य कोण होगा :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Ans. C

Sol. radius = 5
length of tangent = 5
$\therefore$ the figure is square
angle = 90°
] origin lies on the director circle



CHORD WHOSE MID POINT IS GIVEN

18. The equation of the diameter of the circle $(x-2)^2+(y+1)^2=16$ which bisects the chord cut off by the circle on the line $x-2y-3=0$ is :

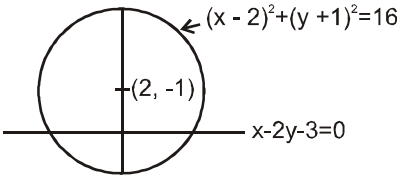
(A) $x+2y=0$ (B) $2x+y-3=0$ (C) $3x+2y-4=0$ (D) none

वृत्त $(x-2)^2+(y+1)^2=16$ के व्यास का वह समीकरण जो वृत्त द्वारा सरल रेखा $x-2y-3=0$ पर काटी गई जीवा को समद्विभाजित करता है :

(A) $x+2y=0$ (B) $2x+y-3=0$ (C) $3x+2y-4=0$ (D) इनमें से कोई नहीं

Ans. B

Sol. Required diameter is $\perp$ to given line.



Hence $y+1=-2(x-2)$ $\Rightarrow 2x+y-3=0$.

19. The co–ordinates of the middle point of the chord cut off on $2x-5y+18=0$ by the circle $x^2+y^2-6x+2y-54=0$ are :

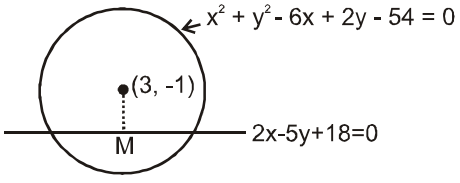
वृत्त $x^2+y^2-6x+2y-54=0$ द्वारा सरल रेखा $2x-5y+18=0$ पर काटी गई जीवा के मध्य बिन्दु के निर्देशांक है :

(A) (1, 4) (B) (2, 4) (C) (4, 1) (D) (1, 1)

Ans. A

Sol. Required point is foot of $\perp$

$$\frac{x-3}{2}=\frac{y+1}{-5}=-\left(\frac{6+5+18}{4+25}\right)=-1 \Rightarrow x=1,\ y=4$$



20. If the chord $y=mx+1$ of the circle $x^2+y^2=1$ subtends an angle of measure 45º at the major segment of the circle then value of m is :

(A) $2\pm\sqrt{2}$ (B) $-2\pm\sqrt{2}$ (C) ±1 (D) none of these

यदि वृत्त $x^2 + y^2 = 1$ की जीवा $y = mx + 1$ वृत्त के दीर्घ चाप पर 45° का कोण अन्तरित करता है, तब m का मान है :

(A) $2 \pm \sqrt{2}$ (B) $-2 \pm \sqrt{2}$ (C) ± 1 (D) कोई नहीं

Ans. C

Sol. Correct option is C.

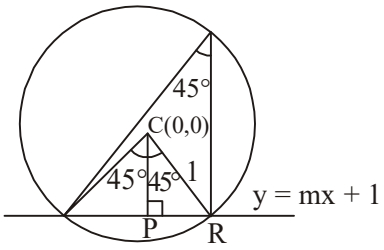
Given circle is $x^2 + y^2 = 1$

C(0,0) and radius = 1 and chord is $y = mx + 1$

$$\cos 45^\circ = \frac{CP}{CR}$$

CP = perpendicular distance from (0,0) to chord $y = mx + 1$

$$CP = \frac{1}{\sqrt{m^2+1}} \left(CR = \text{radius} = 1\right)$$



$$\Rightarrow \cos 45^\circ = \frac{\frac{1}{\sqrt{m^2+1}}}{1}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{m^2+1}}$$

$\Rightarrow m^2 + 1 = 2$

$\Rightarrow m = \pm 1$

21. Let C be the circle with centre (0, 0) and radius 3 units. The equation of the locus of the mid points of the chords of the circle C that subtend an angle of $\frac{2\pi}{3}$ at its centre is :

माना C केन्द्र (0, 0) एवं त्रिज्या 3 इकाई वाला वृत्त है। तब वृत्त C के केन्द्र पर $\frac{2\pi}{3}$ कोण अन्तरित करने वाली जीवा के मध्य बिन्दुओं का बिन्दुपथ का समीकरण है :

(A) $x^2 + y^2 = \frac{3}{2}$ (B) $x^2 + y^2 = 1$ (C) $x^2 + y^2 = \frac{27}{4}$ (D) $x^2 + y^2 = \frac{9}{4}$

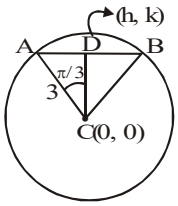
Ans. D

Sol. Let AB be the chord subtending angle $2\pi/3$ at the centre C of circle

Now, $\angle ACD = \pi/3$

Let the coordinates of midpoint D be (h, k)

In ΔACD, $\cos\frac{\pi}{3} = \frac{CD}{CA}$



$\Rightarrow \frac{1}{2} = \frac{\sqrt{h^2+k^2}}{3}$

$\Rightarrow x^2 + y^2 = \frac{9}{4}$, which is the required locus.

CHORD OF CONTACT

22. The chord of contact of the tangents drawn from a point on the circle, $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$ then a, b, c are in :

वृत $x^2 + y^2 = a^2$ पर स्थित बिन्दु से वृत $x^2 + y^2 = b^2$ पर स्पर्श जीवा वृत $x^2 + y^2 = c^2$ को स्पर्श करती है जो a, b, c है :

(A) A.P. (B) G.P. (C) H.P. (D) A.G.P.

Ans. B

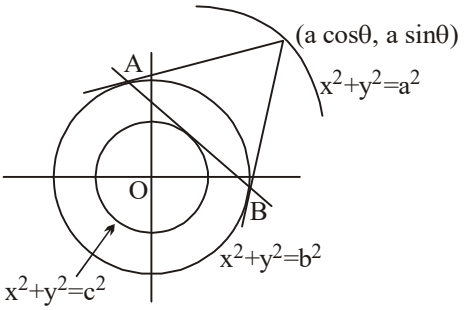
Sol. equation of chord of contact AB

$x\,a\cos\theta + y\,a\sin\theta = b^2$ .....(1)

this is tangent to the circle $x^2 + y^2 = c^2$

$\therefore$ perpendicular from (0, 0) on (1) is equal to c

$\Rightarrow \frac{b^2}{\sqrt{a^2\cos^2\theta + b^2\sin^2\theta}} = c \quad \Rightarrow \quad b^2 = ac$



23. If tangent at (1, 2) to the circle $c_1 : x^2 + y^2 = 5$ intersects the circle $c_2 : x^2 + y^2 = 9$ at A and B and tangents at A and B to the second circle meet at point C, then the co–ordinates of C are :

यदि वृत $c_1 : x^2 + y^2 = 5$ पर (1, 2) पर स्पर्श रेखा वृत $c_2 : x^2 + y^2 = 9$ को A तथा B पर प्रदर्शित करती है एवं A तथा B पर खिचीं गई स्पर्श रेखायें द्वितीय वृत को बिन्दु C पर मिलती है तो C के निर्देशांक है :

(A) (4, 5) (B) (9/15, 18/5) (C) (4, –5) (D) (9/5, 18/5)

Ans. D

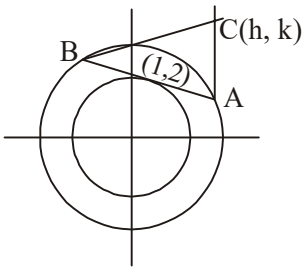
Sol. Tangent at (1, 2) to the circle $x^2 + y^2 = 5$ is :

$x + 2y - 5 = 0$

chord of contact from C(h, k) to $x^2 + y^2 = 9$ is :

$hx + ky - 9 = 0$

Comparing both equations, we get $\frac{h}{1} = \frac{k}{2} = \frac{9}{5}$



$\Rightarrow (h,k) \equiv \left(\frac{9}{5}, \frac{18}{5}\right)$

PAIR OF TANGENTS

24. Combined equation to the pair of tangents drawn from the origin to the circle $x^2+y^2+4x+6y+9=0$ is :

वृत $x^2+y^2+4x+6y+9=0$ पर मूल बिन्दु से डाली गई स्पर्श रेखा युग्म का संयुक्त समीकरण है :

(A) $3(x^2+y^2)=(x+2y)^2$ (B) $2(x^2+y^2)=(3x+y)^2$

(C) $9(x^2+y^2)=(2x+3y)^2$ (D) $x^2+y^2=(2x+3y)^2$

Ans. C

Sol. one of the tangent is y–axis i.e. $x=0$ and only (C) satisfies $x=0$

25. If $x=3$ is the chord of contact of the circle $x^2+y^2=81$, then the equation of the corresponding pair of tangents, is :

यदि $x=3$ वृत $x^2+y^2=81$ की स्पर्श जीवा है, तो इसके सापेक्ष स्पर्श रेखा युग्म का समीकरण है :

(A) $x^2-8y^2+54x+729=0$ (B) $x^2-8y^2-54x+729=0$

(C) $x^2-8y^2-54x-729=0$ (D) $x^2-8y^2=729$

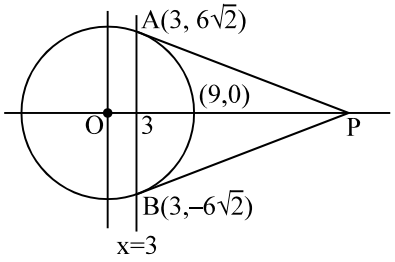
Ans. B

Sol. Equation of tangent to the circle at $A\left(3, 6\sqrt{2}\right)$

$3x+6\sqrt{2}\,y=81$ ....(1)

and at $B\left(3,-6\sqrt{2}\right)$

$3x-6\sqrt{2}\,y=81$ ....(2)



hence equation of the line pair

$(x+2\sqrt{2}\,y-27)(x-2\sqrt{2}\,y-27)=0$

$x^2-8y^2-27(x+2\sqrt{2}\,y)-27(x-2\sqrt{2}\,y)+729=0$

$x^2-8y^2-54x+729=0$

FAMILY OF CIRCLES

26. Intercept on the line $y=x$ by the circle $x^2+y^2-2x=0$, is AB. Equation of the circle on AB as a diameter is :

रेखा $y=x$ पर वृत्त $x^2+y^2-2x=0$ द्वारा बनाया गया अन्तःखण्ड AB है। AB को व्यास लेकर बनाये गये वृत्त का समीकरण है :

(A) $x^2+y^2+x-y=0$ (B) $x^2+y^2-x+y=0$ (C) $x^2+y^2+x+y=0$ (D) $x^2+y^2-x-y=0$

Ans. D

Sol. Given, circle is $x^2 + y^2 - 2x = 0$ ..... (i)

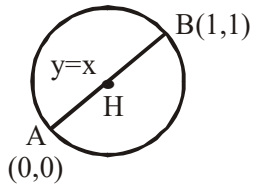
and line is $y = x$ ..... (ii)

Puting $y = x$ in (i),

We get $2x^2 - 2x = 0 \Rightarrow x = 0, 1$

From (i), $y = 0, 1$

Let $A = (0, 0)$, $B = (1, 1)$



Equation of required circle is

$$(x - 0)(x - 1) + (y - 0)(y - 1) = 0$$

or $x^2 + y^2 - x - y = 0$

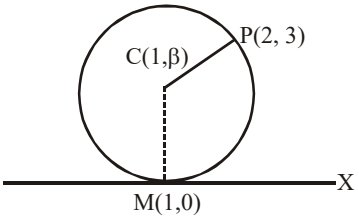
27. The length of the diameter of the circle which touches the x–axis at the point (1, 0) and passes through the point (2, 3) is : **[AIEEE-2012]**

एक वृत्त जो x–अक्ष को बिन्दु (1, 0) पर स्पर्श करता है तथा बिन्दु (2, 3) से होकर जाता है तो व्यास की लम्बाई होगी : **[AIEEE-2012]**

(A) $\frac{5}{3}$ (B) $\frac{10}{3}$ (C) $\frac{3}{5}$ (D) $\frac{6}{5}$

Ans. B

Sol.



Let center of the circle be $C(1, \beta)$

$\beta^2 = (2 - 1)^2 + (3 - \beta)^2$

$\Rightarrow \beta^2 = -6\beta + 10 + \beta^2$

$\Rightarrow \beta = \frac{5}{3}$

$\therefore r = \frac{5}{3}$

$\text{diameter} = \frac{10}{3}$

28. The circle passing through the point (–1, 0) and touching the y–axis at (0, 2) also passes through the point : **[JEE-2011]**

बिन्दु (–1, 0) से गुजरने वाले तथा y–अक्ष को (0, 2) पर स्पर्श करने वाला वृत्त किस बिन्दु से गुजरता है : **[JEE-2011]**

(A) $\left(-\frac{3}{2},0\right)$ (B) $\left(-\frac{5}{2},0\right)$ (C) $\left(-\frac{3}{2},\frac{5}{2}\right)$ (D) (–4, 0)

Ans. D

Sol. Family of circle which touches y-axis at (0,2) is

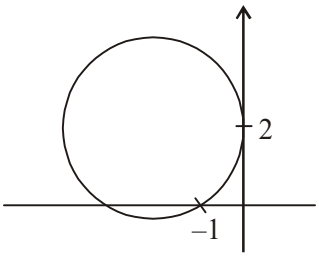
$x^2 + (y-2)^2 + \lambda x = 0$

Passing through (–1,0)

$\Rightarrow \quad 1 + 4 - \lambda = 0 \quad \Rightarrow \quad \lambda = 5$

$\therefore \quad x^2 + y^2 + 5x - 4y + 4 = 0$

which satisfy the point (–4,0).



29. A circle touches the x – axis and also touches the circle with centre at (0, 3) and radius 2. The locus of the centre of the circle is :

(A) an ellipse (B) a circle (C) a hyperbola (D) a parabola

एक वृत्त x –अक्ष को स्पर्श करता है तथा केन्द्र (0, 3) व त्रिज्या 2 वाले वृत्त को भी स्पर्श करता है, तब वृत्त के केन्द्र का बिन्दुपथ है :

(A) दीर्घ वृत्त (B) वृत्त (C) अतिपरवलय (D) परवलय

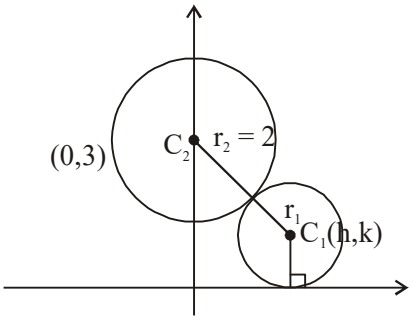
Ans. D

Sol. Let centre ≡ (h, k); As $C_1C_2 = r_1 + r_2$, (Given)

$\Rightarrow \quad \sqrt{(h-0)^2 + (k-3)^2} = |k + 2|$

$\Rightarrow \quad h^2 = 5(2k - 1)$

Hence locus, $x^2 = 5(2y - 1)$, which is parabola



POSITION OF TWO CIRCLES AND COMMON TANGENTS BETWEEN THEM

30. The centre of the circle passing through (0,0) and (1, 0) and touching the circle $x^2+y^2=9$ is :

(0,0) तथा (1, 0) से गुजरने वाली एवं वृत्त $x^2+y^2 = 9$ को स्पर्श करने वाले वृत्त का केन्द्र है :

(A) $\left(\frac{1}{2},\frac{1}{2}\right)$ (B) $\left(\frac{1}{2},-\sqrt{2}\right)$ (C) $\left(\frac{3}{2},\frac{1}{2}\right)$ (D) $\left(\frac{1}{2},\frac{3}{2}\right)$

Ans. B

Sol. Correct option is B.

Let the equation of the circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$

which passes through (0,0) and (1,0).

$\therefore\ c = 0$ and $1 + 2g + c = 0$

$$\Rightarrow c = 0,\ g = -\frac{1}{2}$$

The circle $x^2 + y^2 + 2gx + 2fy + c = 0$ touches the circle $x^2 + y^2 = 9$

$$\therefore \sqrt{g^2 + f^2} = 3 \pm \sqrt{g^2 + f^2 - c} \qquad \left[\because C_1C_2 = I_1 \pm I_2\right]$$

$$\Rightarrow \sqrt{\frac{1}{4} + f^2} = 3 \pm \sqrt{\frac{1}{4} + f^2}$$

$$\Rightarrow 3 = 2\sqrt{\frac{1}{4} + f^2}$$

$$\Rightarrow f^2 = \frac{9}{4} - \frac{1}{4} = 2$$

$$\Rightarrow f = \pm\sqrt{2}$$

Hence, the coordinates of the centre are $\left(\frac{1}{2}, \pm\sqrt{2}\right)$

31. Let C be the circle with centre at (1, 1) and radius = 1. If T is the circle centred at (0, y), passing through origin and touching the circle C externally, then the radius of T is equal to : **[JEE(Main)2014]**

माना C एक वृत्त है जिसका केन्द्र (1, 1) पर है तथा त्रिज्या = 1 है। यदि T केन्द्र (0, y) वाला वृत्त है जो मूल बिन्दु से हो कर जाता है तथा वृत्त C को बाह्य रूप से स्पर्श करता है, तो T की त्रिज्या बराबर है : **[JEE(Main)2014]**

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{\sqrt{3}}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

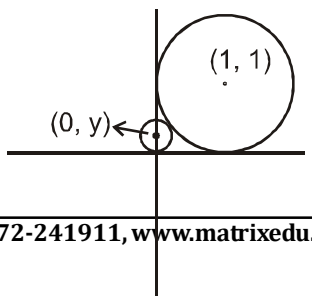
Ans. B

Sol. $c_1\ (1, 1)$ $r_1 = 1$

$c_2\ (0, y)$ $r_2 = |y|$

$c_1c_2 = r_1 + r_2$

$$\sqrt{(1-0)^2 + (1-y)^2} = 1 + |y|$$



$2 - 2y + y^2 = y^2 + 2|y| + 1$

$4|y| = 1$

$|y| = \frac{1}{4}$

$y = \frac{1}{4}$

RADICAL AXIS AND ITS PROPERTIES

32. The equation of the common tangent to the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$ at their point of contact is :

वृत्त $x^2 + y^2 - 4x - 6y - 12 = 0$ और $x^2 + y^2 + 6x + 18y + 26 = 0$ के स्पर्श बिन्दु पर उभयनिष्ठ स्पर्श रेखा का समीकरण है :

(A) $12x + 5y + 19 = 0$ (B) $5x + 12y + 19 = 0$ (C) $5x - 12y + 19 = 0$ (D) $12x - 5y + 19 = 0$

Ans. B

Sol. Equation of common tangent at point of contact is $S_1 - S_2 = 0$

$\Rightarrow 10x + 24y + 38 = 0 \Rightarrow 5x + 12y + 19 = 0$

33. The locus of the centre of the circle which bisects the circumferences of the circles $x^2 + y^2 = 4$ & $x^2 + y^2 - 2x + 6y + 1 = 0$ is :

(A) a straight line (B) a circle (C) a parabola (D) none of these

वृत्तों $x^2 + y^2 = 4$ और $x^2 + y^2 - 2x + 6y + 1 = 0$ की परिधियों को समद्विभाजित करने वाले वृत्त के केन्द्र का बिन्दु पथ है :

(A) सरल रेखा (B) वृत्त (C) परवलय (D) इनमें से कोई नहीं

Ans. A

Sol. Let required circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

Hence common chord with $x^2 + y^2 - 4 = 0$

is $2gx + 2fy + c + y = 0$

This is diameter of circle $x^2 + y^2 = 4$ hence $c = -4$.

Now again common chord with other circle

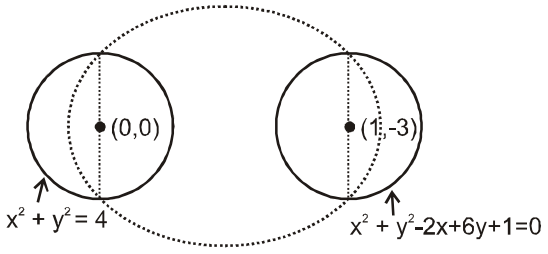
$2x(g + 1) + 2y(f - 3) + (c - 1) = 0$

This is diameter of $x^2 + y^2 - 2x + 6y + 1 = 0$

$2(g + 1) - 6(f - 3) + 5 = 0$

$2g - 6f + 15 = 0$

locus $2x - 3y - 15 = 0$ which is st. line.



34. The circumference of the circle $x^2 + y^2 - 2x + 8y - q = 0$ is bisected by the circle $x^2 + y^2 + 4x + 12y + p = 0$, then $p + q$ is equal to :

वृत्त $x^2 + y^2 - 2x + 8y - q = 0$ की परिधि को वृत्त $x^2 + y^2 + 4x + 12y + p = 0$ समद्विभाजित करता हो, तो $p + q$ का मान होगा:

(A) 25 (B) 100 (C) 10 (D) 48

Ans. C

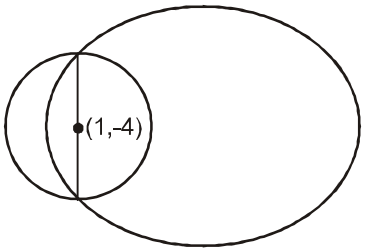
Sol. Common chord of given circle

$6x + 4y + (p + q) = 0$

This is diameter of $x^2 + y^2 - 2x + 8y - q = 0$

centre $(1, -4)$

$6 - 16 + (p + q) = 0 \Rightarrow p + q = 10$



35. If P and Q are the points of intersection of the circles $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ and $x^2 + y^2 + 2x + 2y - p^2 = 0$, then there is a circle passing through P, Q and (1, 1) for : **[AIEEE-2009]**

(A) all except one value of p (B) all except two values of p

(C) exactly one value of p (D) all values of p

यदि वृत्तों $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ तथा $x^2 + y^2 + 2x + 2y - p^2 = 0$ के प्रतिच्छेद बिन्दु P एवं Q हो, तो P, Q और (1, 1) से गुजरने वाला एक वृत्त होगा यदि : **[AIEEE-2009]**

(A) p के एक मान के अलावा सभी मानों के लिए (B) p के दो मानों के अलावा सभी मानों के लिए

(C) p के ठीक एक मान के लिए (D) p के सभी मानों के लिए

Ans. A

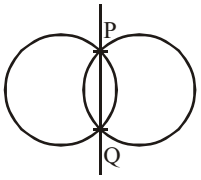
Sol. Eq$^{n.}$ of line PQ

$x + 5y + 2p - 5 + p^2 = 0$

P, Q and (1, 1)

will not lie on a circle of (1, 1)

Lies on the line



$x + 5y + p^2 + 2p - 5 = 0$

$\Rightarrow \quad 1 + 5 + p^2 + 2p - 5 = 0$

$p^2 + 2p + 1 = 0$

$\Rightarrow \quad p = -1$

Therefore their is a circle passing through P, Q and (1, 1) for all values of p.

Except $p = -1$.

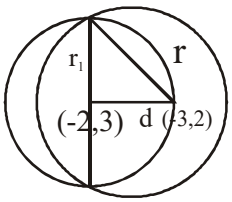
36. If one of the diameters of the circle, given by the equation, $x^2 + y^2 - 4x + 6y - 12 = 0$, is a chord of a circle S, whose centre is at (–3, 2), then the radius of S is : **[JEE(Main)2016]**

यदि समीकरण $x^2 + y^2 - 4x + 6y - 12 = 0$ द्वारा प्रदत्त एक वृत्त का एक व्यास एक अन्य वृत्त S, जिसका केन्द्र (–3, 2) है, की जीवा है, तो वृत्त S की त्रिज्या है : **[JEE(Main)2016]**

(A) 5 (B) 10 (C) $5\sqrt{2}$ (D) $5\sqrt{3}$

Ans. D

Sol.



$r_1 = 5$

$d = \sqrt{50}$

$r = \sqrt{r_1^2 + d^2}$

$= \sqrt{75} = 5\sqrt{3}$

37. If the two circles, $x^2 + y^2 + 2g_1x + 2f_1y = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y = 0$ touch each other then :

(A) $f_1g_1 = f_2g_2$ (B) $\frac{f_1}{g_1} = \frac{f_2}{g_2}$ (C) $f_1f_2 = g_1g_2$ (D) none

यदि दो वृत्त $x^2 + y^2 + 2g_1x + 2f_1y = 0$ एवं $x^2 + y^2 + 2g_2x + 2f_2y = 0$ एक दूसरे को स्पर्श करते हैं, तो :

(A) $f_1g_1 = f_2g_2$ (B) $\frac{f_1}{g_1} = \frac{f_2}{g_2}$ (C) $f_1f_2 = g_1g_2$ (D) इनमें से कोई नहीं

Ans. B

Sol. If two circles touch each other, then

$C_1C_2 = r_1 + r_2$

$\sqrt{(-g_1 + g_2)^2 + (-f_1 + f_2)^2} = \sqrt{g_1^2 + f_1^2} + \sqrt{g_2^2 + f_2^2}$ squaring both sides

$-2g_1g_2 - 2f_1f_2 = 2\sqrt{\left(g_1^2 + f_1^2\right)\left(g_2^2 + f_2^2\right)} \Rightarrow (g_1f_2)^2 + (g_2f_1)^2 - 2g_1g_2f_1f_2 = 0 \Rightarrow \frac{g_1}{g_2} = \frac{f_1}{f_2}$

ANGLE BETWEEN TWO CIRCLES & ORTHOGONALITY OF CIRCLES

38. Equation of the circle cutting orthogonally the three circles $x^2 + y^2 - 2x + 3y - 7 = 0$, $x^2 + y^2 + 5x - 5y + 9 = 0$ and $x^2 + y^2 + 7x - 9y + 29 = 0$ is :

(A) $x^2 + y^2 - 16x - 18y - 4 = 0$ (B) $x^2 + y^2 - 7x + 11y + 6 = 0$

(C) $x^2 + y^2 + 2x - 8y + 9 = 0$ (D) none of these

तीन वृत्तों $x^2 + y^2 - 2x + 3y - 7 = 0$, $x^2 + y^2 + 5x - 5y + 9 = 0$ एवं $x^2 + y^2 + 7x - 9y + 29 = 0$ को लम्बकोणीय काटने वाले वृत्त का समीकरण है :

(A) $x^2 + y^2 - 16x - 18y - 4 = 0$ (B) $x^2 + y^2 - 7x + 11y + 6 = 0$

(C) $x^2 + y^2 + 2x - 8y + 9 = 0$ (D) इनमें से कोई नहीं

Ans. A

Sol. $S_1 - S_2 = 0 \quad \Rightarrow \quad 7x - 8y + 16 = 0$

$S_2 - S_3 = 0 \quad \Rightarrow \quad 2x - 4y + 20 = 0$

$S_3 - S_1 = 0 \quad \Rightarrow \quad 9x - 12y + 36 = 0$

On solving centre (8, 9)

Length of tangent $= \sqrt{S_1} = \sqrt{64 + 81 - 16 + 27 - 7} = \sqrt{149} = (x-8)^2 + (y-9)^2 = 149$

$= x^2 + y^2 - 16x - 18y - 4 = 0$

39. If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the locus of its centre is:

यदि एक वृत्त बिन्दु (a, b) से गुजरता है तथा वृत्त $x^2 + y^2 = 4$ को लम्बकोणीय प्रतिच्छेद करता है, तब इसके केन्द्र का बिन्दुपथ है :

(A) $2ax - 2by - (a^2 + b^2 + 4) = 0$ (B) $2ax + 2by - (a^2 + b^2 + 4) = 0$

(C) $2ax - 2by + (a^2 + b^2 + 4) = 0$ (D) $2ax + 2by + (a^2 + b^2 + 4) = 0$

Ans. B

Sol. Let the variable circle be

$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \ldots\ldots (i)$

Circle (i) cuts circle $x^2 + y^2 - 4 = 0$ orthogonally

$\Rightarrow 2g.0 + 2f.0 = c - 4 \Rightarrow c = 4$

Since circle (i) passes through (a, b)

$\therefore\ a^2 + b^2 + 2ga + 2fb + 4 = 0$

$\therefore$ Locus of centre $(-g, -f)$ is

$2ax + 2by - (a^2 + b^2 + 4) = 0$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

EQUATION OF CIRCLE

1. Determine the nature of the quadrilateral formed by four lines $3x + 4y – 5 = 0$; $4x – 3y – 5 = 0$; $3x + 4y + 5 = 0$ and $4x – 3y + 5 = 0$. Find the equation of the circle inscribed and circumscribing this quadrilateral.

   रेखाओं $3x + 4y – 5 = 0$; $4x – 3y – 5 = 0$; $3x + 4y + 5 = 0$ तथा $4x – 3y + 5 = 0$ से बनने वाले चतुर्भुज की प्रकृति ज्ञात करें। इस चतुर्भुज के अन्तःवृत्त तथा परिवृत्त का समीकरण ज्ञात कीजिये।

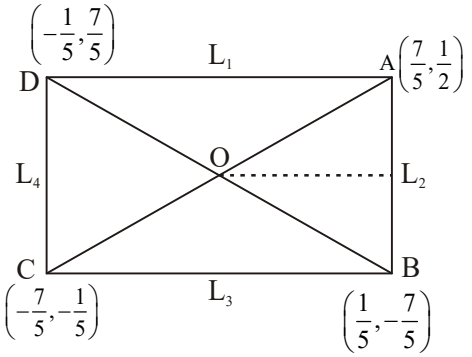
Ans. square of side 2; $x^2 + y^2 = 1$; $x^2 + y^2 = 2$

Sol. $L_1 = 3x + 4y = 5 = m = -\frac{3}{4}$

$L_2 = 4x – 3y = 5 = m = \frac{4}{3}$

$L_3 = 3x + 4y = –5 = m = -\frac{3}{4}$

$L_4 = 4x – 3y = – 5 = m = \frac{4}{3}$



$$\text{Diagonal AC} = \frac{\frac{1}{5}+\frac{1}{5}}{\frac{7}{5}+\frac{7}{5}} = \frac{\frac{2}{5}}{\frac{14}{5}} = \frac{1}{7}$$

$$\text{Diagonal BD} = \frac{\frac{7}{5}+\frac{7}{5}}{-\frac{1}{5}-\frac{1}{5}} = \frac{\frac{14}{5}}{-\frac{2}{5}} = -7$$

$$M_{AC} \times M_{BD} = \frac{1}{7} \times -7$$

Diagonals are at right angle.

ABCD is square.

2. A circle $S = 0$ is drawn with its centre at $(–1, 1)$ so as to touch the circle $x^2 + y^2 – 4x + 6y – 3 = 0$ externally. Find the intercept made by the circle $S = 0$ on the coordinate axes.

वृत्त $x^2 + y^2 - 4x + 6y - 3 = 0$ के बाह्य स्पर्श करने वाले वृत $S = 0$ का समीकरण ज्ञात करें यदि वृत्त $S = 0$ का केन्द्र $(-1, 1)$ है। वृत $S = 0$ द्वारा अक्षों पर काटे गये अन्तः खण्ड ज्ञात कीजियें।

Ans. zero, zero

Sol. Let the radius of the circle be r.

Since, they touch each other externally

Then, $C_1C_2 = r_1 + r_2$

$$\Rightarrow \sqrt{(-1-2)^2 + (1+3)^2} = r + 4$$

$$\Rightarrow r = 1$$

Therefore, equation of circle is $(x + 1)^2 + (y - 1)^2 = 1$

which touches both the co-ordinate axis.

Therefore, both the intercept are zero.

3 The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q. The circle on PQ as diameter passes through the origin. Prove that $n^2(a + b) = l^2 + m^2$.

वक्र $ax^2 + 2hxy + by^2 = 1$ को रेखा $lx + my + n = 0$ बिन्दु P तथा Q पर काटती है। PQ को व्यास लेकर खीचां गया वृत मूल बिन्दु से गुजरता है। सिद्ध करों कि $n^2(a + b) = l^2 + m^2$.

Ans.

Sol. curve : $ax^2 + 2hxy + by^2 = 1$

$$\text{line : } ln + my + n = 0 \Rightarrow \frac{ln + my}{-n} = 1$$

homogenizing given curve with the given line

$$ax^2 + 2hxy + by^2 = \left(\frac{ln + my}{-n}\right)^2$$

$$\left(a - \frac{l^2}{n^2}\right)x^2 + \left(2h - 2\frac{l}{n}\frac{m}{n}\right)xy + \left(b - \frac{m^2}{n^2}\right)y^2 = 0 \qquad \text{_____(1)}$$

equation (1) give combine equation of OA = OB

as AB is diameter

$\angle AOB = 90^\circ$

Thus $m_1m_2 = -1$

$m_1 \rightarrow$ slope of OB $\qquad$ $m_2 \rightarrow$ slope of OA

$$\left(\frac{y}{x}\right)^2\left(b-\frac{m^2}{n^2}\right)+\frac{y}{x}\left(2h-\frac{2l}{n}\frac{m}{n}\right)+\left(a-\frac{l^2}{n^2}\right)=0$$

Let $\frac{y}{x}=m$

This quad has 2 roots m_1 & m_2.

we have $m_1m_2=-1$

$$\frac{a-\frac{l^2}{n^2}}{b-\frac{m^2}{n^2}}=-1$$

$\Rightarrow (a+b)\,n^2=l^2+m^2$

4 One of the diameters of the circle circumscribing the rectangle ABCD in order is $4y=x+7$. If A & B are the points $(-3, 4)$ & $(5,4)$ respectively, then find the area of the rectangle.

आयत ABCD के परिवृत का एक व्यास $4y = x + 7$ है। यदि A $(-3, 4)$ तथा B $(5,4)$ हो तो आयत का क्षेत्रफल ज्ञात कीजिए।

Ans. 32 sq. unit

Sol. Clearly, the centre of the circle lies on the line $4y=x+7$.

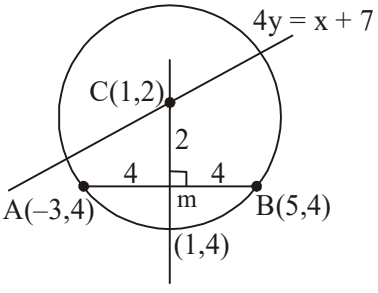
The circle passes through A(–3,4) and B(5,4)

The slope of the segment joining A and B is zero.

$\therefore$ the slope of the perpendicular bisector of AB is not defined.

Hence the perpendicular bisector of AB will be parallel to the y–axis and will pass through

$$\left(\frac{-3+5}{2},\frac{4+4}{2}\right)=(1,4)$$



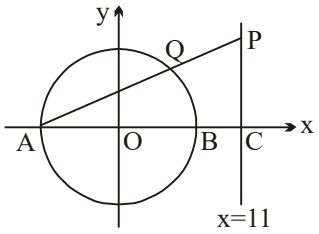
The equation of the perpendicular bisector is $x=1$.

The intersection point of the perpendicular bisector and $4y=x+7$ is (1,2).

$\therefore$ Centre = (1,2)

Area of rectangle = AB × 2CM = 8 × 2 × 2 = 32 square units.

5. In the given figure, the circle $x^2 + y^2 = 25$ intersects the x-axis at the point A and B. The line $x = 11$ intersects the x-axis at the point C. Point P moves along the line $x = 11$ above the x-axis and AP intersects the circle at Q. Find

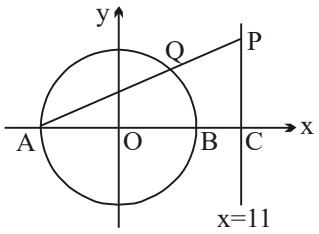


(i) The coordinates of the point P if the triangle AQB has the maximum area.

(ii) The coordinates of the point P if Q is the middle point of AP.

(iii) The coordinates of P if the area of the triangle AQB is $(1/4)^{th}$ of the area of the triangle APC.

दिये गये चित्र में वृत $x^2 + y^2 = 25$, x–अक्ष को बिन्दुओं A व B पर प्रतिच्छेद करता है। रेखा $x = 11$, x–अक्ष को बिन्दु C पर काटता है। बिन्दु P रेखा $x = 11$ पर x अक्ष के ऊपर विचरण करता है तथा AP वृत को Q बिन्दु पर काटता है। ज्ञात कीजिए।



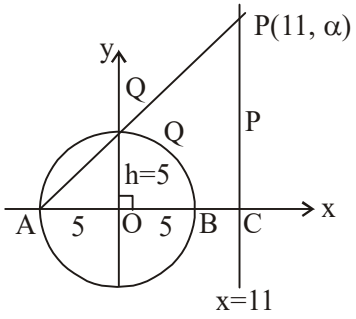
(i) यदि त्रिभुज AQB का क्षेत्रफल अधिकतम हो तो बिन्दु P के निर्देशांक

(ii) यदि AP का मध्य बिन्दु Q हो तो बिन्दु P के निर्देशांक

(iii) यदि त्रिभुज AQB का क्षेत्रफल त्रिभुज APC के क्षेत्रफल का एक चौथाई हो तो P के निर्देशांक

Ans. (i) (11, 16), (ii) (11, 8), (iii) (11, 12)

Sol. (i) Area of $\Delta AQB|_{max} = \frac{1}{2} \times AB \times QO = \frac{1}{2} \times 10 \times 5 = 25$



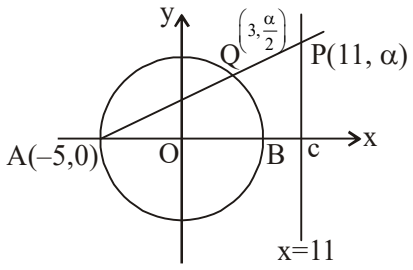
Equation of $AQ \Rightarrow y = x + 5$

$\Rightarrow P(11,16)$

(ii) Put $Q\left(3, \frac{\alpha}{2}\right)$ in $x^2 + y^2 = 25$

$\Rightarrow 9 + \frac{\alpha^2}{4} = 25 \Rightarrow \alpha^2 = 64 \Rightarrow \alpha = 8$

$\Rightarrow P(11,8)$



(iii) $\frac{\Delta APC}{\Delta AQB} = 4$

$\frac{AP^2}{AB^2} = 4$

$\frac{16^2 + \alpha^2}{10^2} = 4$

$\alpha^2 + 256 = 400$

$\alpha^2 = 144$

$\alpha = 12$

INTERCEPT OF CIRCLE

6 Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then find the equation(s) which represent L_1.

माना कि L_1 मूल बिन्दु से गुजरने वाली रेखा है तथा L_2 रेखा $x + y = 1$ है। यदि रेखाओं L_1 तथा $L_2 = 0$ पर वृत $x^2 + y^2 - x + 3y=0$ द्वारा काटे गये अन्तःखण्ड बराबर हो तो L_1 को प्रदर्शित करने वाली रेखा के समीकरण को ज्ञात कीजिए।

Ans. $x - y = 0$; $x + 7y = 0$

Sol. $\because$ L_1 passes through origin.

Let equation of $L_1 \cdot y = mx$

Intercepts made by L_1 and L_2 are equal

$\Rightarrow L_1$ and L_2 are at the same distance from centre of circle $x^2 + y^2 - x + 3y = 0$

centre of circle $x^2 + y^2 - x + 3y = 0$

$x^2 + y^2 - x + 3y = 0$

$$x^2 - x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + y^2 + 3y + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 = 0$$

$$\left(x-\frac{1}{2}\right)^2+\left(y+\frac{3}{2}\right)^2=\left(\frac{\sqrt{10}}{2}\right)^2$$

$\therefore$ Centre of circle $\left(\frac{1}{2},\frac{-3}{2}\right)$

Distance of a point (x_1, y_1) from line $Ax + By + C = 0$ is given by

$$d=\frac{|Ax_1+By_1+1|}{\sqrt{A^2+B^2}}$$

$\therefore$ Distance of $\left(\frac{1}{2},\frac{-3}{2}\right)$ from line L_1 and L_2 are

$$d_1=\frac{\left|m\left(\frac{1}{2}\right)+\frac{3}{2}\right|}{\sqrt{m^2+1}}$$

$$d_2=\frac{\left|\frac{1}{2}+\left(\frac{3}{2}\right)-1\right|}{\sqrt{1+1}}$$

$d_1 = d_2$

$$\Rightarrow \frac{m-3}{2\sqrt{m^2+1}}=\frac{3}{\sqrt{2}}$$

Squaring both sides we get

$$\frac{(m+3)^2}{4\left(m^2+1\right)}=\frac{4}{2}$$

$(m-3)^2 = 8(m^2 + 1)$

$m^2 + 9 + 6m = 8m^2 + 3$

$7m^2 - 6m - 1 = 0$

$$m=\frac{6\pm\sqrt{36+28}}{14}=\frac{+6\pm8}{14}=+1,-\frac{1}{7}$$

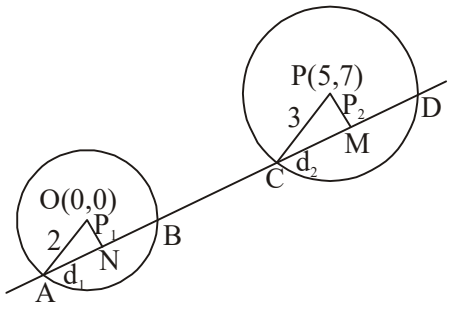
$\therefore$ $L_1 : -y + x = 0, x + 7y = 0.$

7 Find the equation of a line with gradient 1 such that the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 10x - 14y + 65 = 0$ intercept equal length on it.

उस रेखा का समीकरण ज्ञात करें जिसकी प्रवणता 1 हो तथा दो वृत्त $x^2 + y^2 = 4$ तथा $x^2 + y^2 - 10x - 14y + 65 = 0$ उस रेखा पर समान लम्बाई के अन्तः खण्ड काटते हों।

Ans. $2x - 2y - 3 = 0$

Sol. Equation of the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 10x - 14y + 65 = 0$ have centres, radius as (0,0), 2 and (5,7),3 respectively.



Let equation of line be $y - x = c$ ...(i)

Perpendicular from (0,0) on line (i) is $\left|\frac{-c}{\sqrt{2}}\right| = \frac{c}{\sqrt{2}}$

In ΔAON, $\sqrt{2^2 - \left(\frac{c}{\sqrt{2}}\right)^2} = AN$

Similarly in ΔCPM, $\sqrt{3^2 - \left(\frac{2-c}{\sqrt{2}}\right)^2} = CM$

Given, AB = CD

$\Rightarrow AN = CM$

$$\Rightarrow 4 - \frac{c^2}{2} = 9 - \frac{(2-c)^2}{2}$$

$$\Rightarrow c = -\frac{3}{2}$$

Therefore, equation of line :

$$y = x - \frac{3}{2}$$

$$\Rightarrow 2x - 2y - 3 = 0$$

8 Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres (–3, 0), (–1, 0), (1, 0) and (3, 0) respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C. If the length of this chord can be expressed as $\sqrt{x}$, find x.

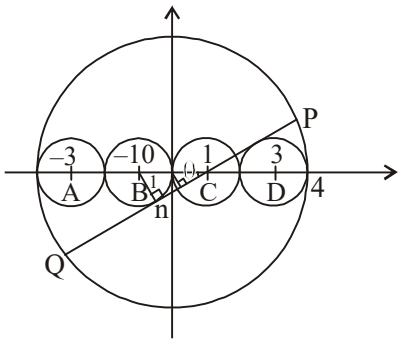
एक वृत्त S का केन्द्र मूल बिन्दु तथा त्रिज्या 4 है। चार वृत्त A, B, C तथा D जिनकी त्रिज्या 1 व केन्द्र क्रमशः (–3, 0), (–1, 0), (1, 0) तथा (3, 0) है खींचे गये है। वृत्त S की एक जीवा PQ वृत्त B को स्पर्श करती है तथा वृत्त C के केन्द्र से गुजरती है। यदि इस जीवा की लम्बाई $\sqrt{x}$ है तो x ज्ञात कीजिए।

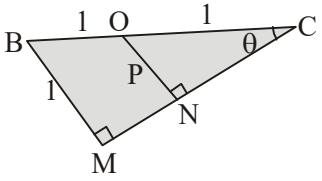
Ans. 63

Sol. $\sin\theta = \frac{1}{2} = \frac{ON}{1} \Rightarrow ON = \frac{1}{2} = P$

Length of chord = $2\sqrt{r^2 - P^2}$

$$PQ = 2\sqrt{16 - \frac{1}{4}} = \sqrt{63}$$





PARAMETRIC FORM OF CIRCLE

9. Real number x, y satisfies $x^2 + y^2 = 1$. If the maximum and minimum value of the expression $z = \frac{4-y}{7-x}$ are M and m respectively, then find the value $(2M + 6m)$.

वास्तविक संख्यायें x व y, $x^2 + y^2 = 1$ को संतुष्ट करती है। यदि $z = \frac{4-y}{7-x}$ का अधिकतम व न्यूनतम मान क्रमशः M व m है तो (2M + 6m) का मान ज्ञात कीजिए।

Ans. 4

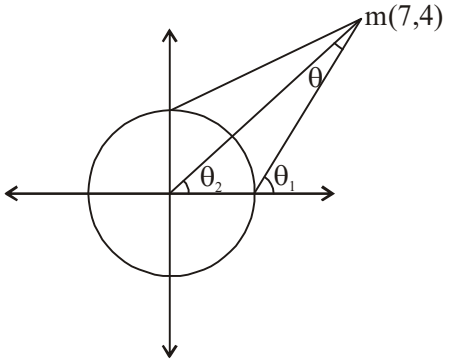
Sol. $x^2 + y^2 = 1$

$$Z = \frac{y-4}{x-7}$$

In this slope from the point (7,4) :

$$\tan\theta_2 = \frac{4}{7}$$

$$\tan\theta = \tan(\theta_1 - \theta_2) = \frac{m - \tan\theta_2}{1 + m\tan\theta_2}$$



$$\left|\frac{r}{\ell_{tangent}}\right| = \frac{m - \frac{4}{7}}{1 + m \times \frac{4}{7}}$$

$$\Rightarrow \pm\frac{1}{8} = \frac{m - \frac{4}{7}}{1 + \frac{4m}{7}} = \frac{7m-4}{7+4m}$$

$$\therefore M = \frac{3}{4} \text{ and } m = \frac{5}{12}$$

$$\therefore 2M + 6m = \frac{2\times3}{4} + \frac{6\times5}{12} = 4$$

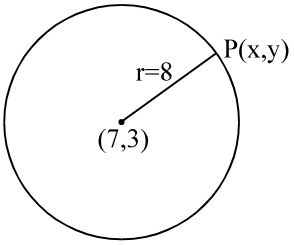
10 Given that $x^2 + y^2 = 14x + 6y + 6$, find the largest possible value of the expression $E = 3x + 4y$.

दिया गया है कि $x^2 + y^2 = 14x + 6y + 6$, तो व्यंजक $E = 3x + 4y$ का अधिकतम सम्भावित मान ज्ञात कीजिए।

Ans. 73

Sol. $(x-7)^2 + (y-3)^2 = 64 = 8^2$, a circle with centre (7, 3) and radius 8

now $\dfrac{x-7}{\cos\theta}=\dfrac{y-3}{\sin\theta}=8$



$\left.\begin{aligned}x&=8\cos\theta+7\\ y&=8\sin\theta+3\end{aligned}\right]$ lies on the circle for some θ

$\therefore$ $E = 3x + 4y = 3(8\cos\theta + 7) + 4(8\sin\theta + 3)$

$E = (21 + 12) + 24\cos\theta + 32\sin\theta$

$= 33 + 8(3\cos\theta + 4\sin\theta)$

$\therefore$ $E_{max} = 33 + 40 = 73$ Ans.

POSITION OF POINT WRT A CIRCLE

11. Find the shortest distance from the point M $(-7, 2)$ to the circle $x^2+y^2-10x-14y-151=0$.

बिन्दु M $(-7, 2)$ से वृत $x^2+y^2-10x-14y-151=0$ की न्यूनतम दूरी ज्ञात कीजिए।

Ans. 2

Sol. $S = x^2 + y^2 - 10x - 14y - 151 = 0$

Now putting the values of points (–7,2) in the equation to check whether the given point is on the circle, inside the circle or outside the circle.

$S = -7^2 + 2^2 - 10 \times (-7) - 14 \times (2) - 151 = 0$

$S = 49 + 4 + 70 - 28 - 151 = -56$

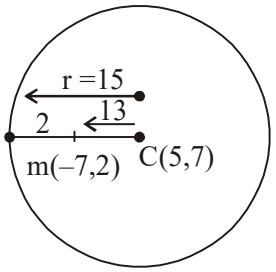
$S = -56 < 0$

Therefore, M(–7,2) does not lie on the circle.

The general equation of the circle is,

$ax^2 + by^2 + 2gx + 2gy + c = 0$

Now, $g = 5$, $f = 7$, $c = -151$



The radius of the circle is,

$r = \sqrt{g^2 + f^2 - c}$

$= \sqrt{(5^2 + (7)^2 - (-151)}$

$\sqrt{25 + 49 + 151} = \sqrt{225} = 15$

The centre of the circle is, $C = (-g, -f) = (5,7)$

Now, $CM = \sqrt{(5+7)^2 + (7-2)^2}$

$\Rightarrow CM = \sqrt{12^2 + 5^2}$

$\Rightarrow CM = \sqrt{144 + 25}$

$\Rightarrow CM = \sqrt{169} = 13$

The shortest distance is, r – CM = 15 – 13 = 2

Hence the shortest distance is 2 units.

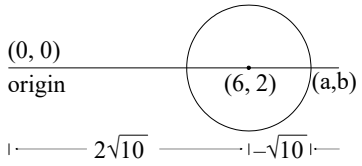
12. Find the co-ordinate of the point on the circle $x^2 + y^2 - 12x - 4y + 30 = 0$, which is farthest from the origin.

वृत $x^2 + y^2 - 12x - 4y + 30 = 0$ पर मूल से अधिकतम दूरी का बिन्दु ज्ञात कीजिए।

Ans. (9, 3)

Sol. radius = 36 + 4 – 30 = $\sqrt{10}$

Ue section formula



13. The straight line 2x – 3y = 1 divides the circular region $x^2 + y^2 \le 6$ into two parts. If $S = \left[\left(2, \frac{3}{4}\right), \left(\frac{5}{2}, \frac{3}{4}\right), \left(\frac{1}{4}, -\frac{1}{4}\right), \left(\frac{1}{8}, \frac{1}{4}\right)\right]$, then the number of point(s) in S lying inside the smaller part is :

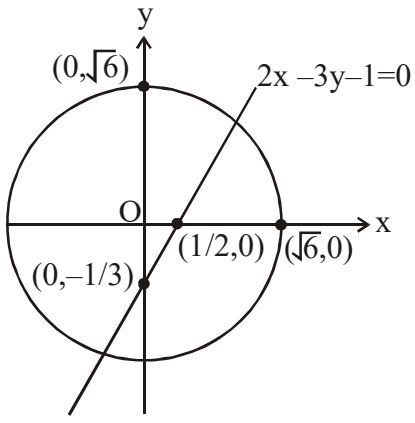
[JEE-2011]

रेखा 2x – 3y = 1 वृत्तीय क्षेत्र $x^2 + y^2 \le 6$ को दो भागों में विभाजित करती है यदि $S = \left[\left(2, \frac{3}{4}\right), \left(\frac{5}{2}, \frac{3}{4}\right), \left(\frac{1}{4}, -\frac{1}{4}\right), \left(\frac{1}{8}, \frac{1}{4}\right)\right]$,

तो समुच्चय S में स्थित बिन्दुओं की संख्या जो छोटे भाग में स्थित है। **[JEE-2011]**

Ans. 2

Sol.



If the point lies inside the smaller part, then origin and point should give opposite signs w.r.t. line & point should lie inside the circle.

for origin : $2 \times 0 - 3 \times 0 - 1 = -1$ (–ve)

for $(2, \frac{3}{4})$: $2 \times 2 - 3 \times \frac{3}{4} - 1$

$= \frac{3}{4}$ (+ve); point lies inside the circle

for $(\frac{5}{2}, \frac{3}{4})$: $2 \times \frac{5}{2} - 3 \times \frac{3}{4} - 1 = \frac{7}{4}$ (+ve) ; point lies outside the circle

For $\left(\frac{1}{4}, -\frac{1}{4}\right)$: $2 \times \frac{1}{4} - 3\left(-\frac{1}{4}\right) - 1 = \frac{1}{4}$ (+ve) ; point lies inside the circle

For $\left(\frac{1}{8}, \frac{1}{4}\right)$: $2 \times \frac{1}{8} - 3\left(\frac{1}{4}\right) - 1 = \frac{-3}{2}$ (–ve) ; point lies inside the circle.

$\therefore$ 2 points lie inside smaller part.

TANGENT AND NORMAL

14 A circle passes through the points (–1, 1), (0, 6) and (5, 5). Find the points on the circle the tangents at which are parallel to the straight line joining origin to the centre.

एक वृत्त बिन्दुओं (–1, 1), (0, 6) तथा (5, 5) से गुजरता है। वृत पर स्थित उन बिन्दुओं को ज्ञात करें जिन पर खीचीं गई स्पर्श रेखायें मूल बिन्दु को केन्द्र से मिलाने वाली रेखा के समान्तर होगें।

Ans. (5, 1) & (–1, 5)

Sol. Let the equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

It passes through (–1,1)

$\Rightarrow 1 + 1 - 2g + 2f + c = 0$

$\Rightarrow -2g + 2f + c = -2$ _____(i)

It passes through (0,6)

$\Rightarrow 0 + 36 + 0 + 12f + c = 0$

$\Rightarrow 12f + c = -36$ _____(ii)

It passes through (5,5)

$\Rightarrow 25 + 25 + 10g + 10f + c = 0$

$\Rightarrow 10g + 10f + c = -50$ _____(iii)

Subtracting (iii) from (i) we get

$-2g - 10f = 34$ _____(iv)

subtracting (iii) from (ii) we get

$-10g + 2f = 14$ _____(v)

Solving (iv) and (v)

$\Rightarrow g = -2,\ f = -3$

Substituting g and f in (i) we get $c = 0$

So the equation of circle is

$x^2 + y^2 - 4x - 6y = 0$ _____(vi)

Centre of the circle is (2,3)

Slope of line joining centre to origin $= \dfrac{3-0}{2-0} = \dfrac{3}{2}$

Equation of line passing through the centre and perpendicular to line joining centre to origin is

$$y - 3 = \frac{-2}{3}(x - 2)$$

$3y - 9 = -2x + 4$

$2x + 3y = 13$ _____(vii)

Tangents will be parallel to the line joining the origin to the centre at the points where (vii) cuts the circle.

Substituting y from (vii) in (vi) we get

$13x^2 - 52x - 65 = 0$

$x^2 - 4x - 5 = 0$

$x^2 - 5x + x - 5 = 0$

$x(x - 5) + (x - 5) = 0$

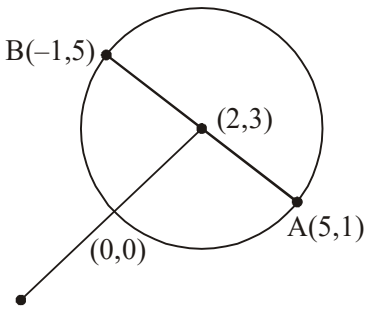
$(x + 1)(x - 5) = 0$

$x = -1,5$

Substituting x in (vii) we get

$y = 5,1$

So the point of contact of tangents are (–1,5) and (5,1)



15 A circle is drawn with its centre on the line $x + y = 2$ to touch the line $4x - 3y + 4 = 0$ and pass through the point (0, 1). Find its equation.

रेखा $x + y = 2$ पर केन्द्र लेते हुये वृत इस प्रकार से खींचा गया है कि वह $4x - 3y + 4 = 0$ को स्पर्श करता है तथा बिन्दु $(0, 1)$ से गुजरता है। वृत्त का समीकरण ज्ञात करें।

Ans. $x^2 + y^2 - 2x - 2y + 1 = 0$ OR $x^2 + y^2 - 42x + 38y - 39 = 0$

Sol. Let the circle be

$(x - h)^2 + (y - k)^2 = r^2$ ...(1)

We have $h + k = 2$ ...(2)

$(0 - h)^2 + (1 - k)^2 = r^2$ ...(3)

and $(4h - 3k + 4)^2 = 25r^2$ ...(4)

from (3) and (4)

$(4h - 3k + 4)^2 = 25[h^2 + (1 - k)^2]$

$\Rightarrow 9h^2 + 16k^2 + 24hk - 32h - 26k + 9 = 0$ ...(5)

Using (2) and (5) reduce to

$9h^2 + 16(2 - h)^2 + 24h(2 - h) - 32h - 25(2 - h) + 9 = 0$

$\Rightarrow h^2 - 22h + 21 = 0 \Rightarrow h = 1, 21 \Rightarrow k = 1, -19$

$\Rightarrow r^2 = h^2 + (1 - k)^2 = 1 \Rightarrow 21^2 + 20^2 = 1, 841$

Two circle are

$(x - 1)^2 + (y - 1)^2 = 1$ and $(x - 21)^2 + (y + 19)^2 = 841$

16 Find the locus of the middle points of portions of the tangents to the circle $x^2 + y^2 = a^2$ terminated by the coordinate axes.

वृत्त $x^2 + y^2 = a^2$ की उन स्पर्श रेखाओं के मध्य बिन्दुओं का बिन्दुपथ ज्ञात करें जिनके सिरे अक्षों पर स्थित हैं।

Ans. $a^2(x^2 + y^2) = 4x^2y^2$

Sol. equation of tangent of $x^2 + y^2 = a^2$ at (x_1, y_1)

$xx_1 + yy_1 = a^2$

$x_1 = a\cos\theta$

$y_1 = a\sin\theta$

$x\,a\cos\theta + y\,a\sin\theta = a^2$

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{a} = 1$$

intercept on x-axis $\dfrac{a}{\cos\theta} = 1$

intercept on y-axis $\frac{a}{\sin\theta} = m$

Let mid-point of AB be (h, k)

$h = \frac{1+0}{2}, 1 = 2h$

$k = \frac{m+0}{2}, m = 2k$

$\frac{a}{\cos\theta} = 2h$

$\frac{a}{\sin\theta} = 2k$

$\sin^2\theta + \cos^2\theta = 1$

$\left(\frac{a}{2h}\right)^2 + \left(\frac{a}{2k}\right)^2 = 1$

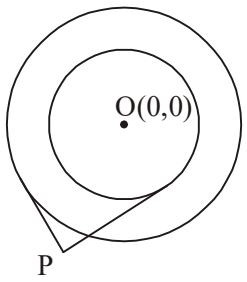
So locus $\frac{a}{x^2} + \frac{a^2}{y^2} = 4$

17 Tangents are drawn to the concentric circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ at right angle to one another. Show that the locus of their point of intersection is a 3rd concentric circle. Find its radius.

वृत्त $x^2 + y^2 = a^2$ तथा $x^2 + y^2 = b^2$ पर परस्पर लम्बवत् स्पर्श रेखायें खीचीं गई है। प्रदर्शित करो कि उनके प्रतिच्छेद बिन्दु का बिन्दुपथ तृतीय सहकेन्द्रीय वृत्त होगा। उस वृत की त्रिज्या भी ज्ञात कीजिए।

Ans. $x^2 + y^2 = a^2 + b^2$; $r = \sqrt{a^2 + b^2}$

Sol. Both circles are concentric at the origin (see fig.). From the note given under



$$y = mx + a\sqrt{1+m^2}$$

$$y = \frac{-1}{m}x + b\sqrt{a + \frac{1}{m^2}}$$

are tangents to the given circles and are at right angles. Therefore,

$$\left(y - mx^2\right) + \left(x + my\right)^2 = \left(a^2 + b^2\right)\left(1 + m^2\right)$$

$$\Rightarrow \left(x^2 + y^2\right)\left(1 + m^2\right) = \left(a^2 + b^2\right)\left(1 + m^2\right)$$

That is, $x^2 + y^2 = a^2 + b^2$ is the required locus.

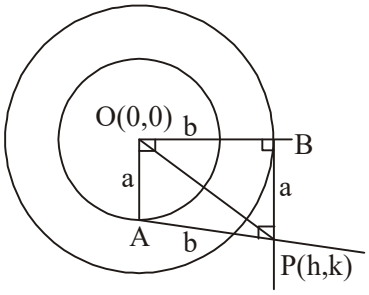
Note : in the problem if b = a, then the equation of the director circle of $x^2 + y^2 = a^2$ is $x^2 + y^2 = 2a^2$.

Alter :

$$OP = \sqrt{a^2 + b^2}$$

$$\sqrt{h^2 + k^2} = \sqrt{a^2 + b^2}$$

$x^2 + y^2 = a^2 + b^2$



18 If the variable line $3x - 4y + k = 0$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle, then the range of k is (a, b) where $a, b \in I$. Find the value of $(b - a)$.

यदि चर रेखा $3x - 4y + k = 0$ वृत्तों $x^2 + y^2 - 2x - 2y + 1 = 0$ तथा $x^2 + y^2 - 16x - 2y + 61 = 0$ के बीच में बिना किसी वृत को स्पर्श या प्रतिच्छेद करते हुये स्थित हो, तो यदि k का परिसर (a, b) है जहाँ $a, b \in I$ तो $(b - a)$ का मान ज्ञात कीजिए।

Ans. 6

Sol. The given circles :

$c_1(x - 1)^2 + (y - 1)^2 = 1$

$c_2(x - 8)^2 + (y - 1)^2 = 4$

$3x - 4y + k = 0$

$(3\cdot1 - 4\cdot1 + k)(3\cdot8 - 4\cdot1 + k) < 0 \Rightarrow (k - 1)(k + 20) < 0 \Rightarrow k \in (-20, 1)$

For C_1

$$\frac{|3\cdot1 - 4\cdot1 + k|}{5} > 1 \Rightarrow \frac{|k - 1|}{5} > 1$$

$\Rightarrow k - 1 > 5$ or $k - 1 < 5$

$\Rightarrow k > 6$ or $k < -4$

For C_2

$$\frac{|3\cdot 8 - 4\cdot 1 + k|}{5} > 2 \Rightarrow \frac{|k+20|}{5} > 2$$

$\Rightarrow k + 20 > 10$ or $k + 20 < -10$

$\Rightarrow k > -10$ or $k < -30$

$\Rightarrow k \Rightarrow (-10,-4) \Rightarrow a = -10, b = -4$

$\Rightarrow b - a = -4 + 10 = 6$

19 A circle with center in the first quadrant is tangent to $y = x + 10$, $y = x - 6$, and the y-axis. Let (h, k) be the center of the circle. If the value of $(h + k) = a + b\sqrt{a}$, find the value of a + b.

एक वृत्त का केन्द्र प्रथम चतुर्थांश में है तथा यह वृत्त रेखाओं $y = x + 10$, $y = x - 6$ तथा y-अक्ष को स्पर्श करता है। यदि (h, k) वृत्त का केन्द्र है तथा (h + k) का मान $a + b\sqrt{a}$ है, a + b का मान ज्ञात कीजिए।

Ans. 10

Sol. $y = x + 10$ and $y = x - 6$ have same slope, i.e.m for these lines to be the tangent to the circle, the perpendicular distance between them should be equal to the diameter of the circle.

i.e., $y = x + 2$ passes through the centre of the circle.

$\Rightarrow k = h + 2$

So, radius of the circle $= \frac{1}{2}\left|\frac{16}{\sqrt{2}}\right| = 4\sqrt{2}$

As y-axis is a tangent to the circle, $h = 4\sqrt{2}$, i.e. $k = 2 + 4\sqrt{2}$ So, $a + b = 2 + 8 = 10$

POWER OF POINT

20. A point moving around circle $(x + 4)^2 + (y + 2)^2 = 25$ with centre C broke away from it either at the point A or point B on the circle and moved along a tangent to the circle passing through the point D (3, – 3). Find the following.

(i) Equation of the tangents at A and B.

(ii) Coordinates of the points A and B.

(iii) Angle ADB and the maximum and minimum distances of the point D from the circle.

(iv) Area of quadrilateral ADBC and the ΔDAB.

(v) Equation of the circle circumscribing the ΔDAB and also the intercepts made by this circle on the coordinate axes.

एक बिन्दु जो वृत्त $(x + 4)^2 + (y + 2)^2 = 25$ पर विचरण कर रहा है, वह वृत से बिन्दु A या बिन्दु B पर अलग हो जाता है तथा स्पर्श रेखा पर विचरण करता हुआ बिन्दु D (3, – 3) से गुजरता है। निम्न ज्ञात कीजिए।

(i) A व B पर स्पर्श रेखाओं का समीकरण

(ii) A व B के निर्देशांक

(iii) कोण ADB व D की वृत से अधिकतम व न्यूनतम दूरी

(iv) चतुर्भुज ADBC व त्रिभुज ΔDAB का क्षेत्रफल

(v) त्रिभुज ΔDAB के परिवृत का समीकरण तथा इस वृत के द्वारा अक्षों पर काटे गये अन्तः खण्ड

Ans. (i) $3x - 4y = 21;\ 4x + 3y = 3$ (ii) A(0, 1) and B (–1, – 6) (iii) 90°, $5\left(\sqrt{2} \pm 1\right)$ units

(iv) 25 sq. units, 12.5 sq. units (v) $x^2 + y^2 + x + 5y - 6 = 0$, x intercept 5; y intercept 7

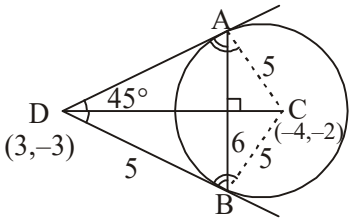
Sol. (i) $CD = \sqrt{45+1} = \sqrt{50}$

$AD = \sqrt{CD^2 + AC^2}$

$= \sqrt{50 - 25} = 5$

Hence $AC = BC = BD = AD = 5$

$\angle ADC = \angle BDC = 45^\circ$



Let

Slope of $AD = m_1$

Slope of $BD = m_2$

Slope of $CD = \dfrac{-2+3}{-4-3} = \dfrac{-1}{7}$

$$\frac{\pi}{4} = \left|\frac{m + \frac{1}{7}}{1 - \frac{m}{7}}\right|$$

$$\frac{\pi}{4} = \left|\frac{m + \frac{1}{7}}{1 - \frac{m}{7}}\right|$$

Taking +ve

$$\left(1 - \frac{m}{7}\right) = \left(\frac{1}{7} + m\right)$$

Taking –ve

$$-1 + \frac{m}{7} = m + \frac{1}{7}$$

$m = \frac{3}{4}$ $m = \frac{-4}{3}$

Equation of line Equation of line

$y + 3 = \frac{3}{4}(x - 3)$ $y + 3 = \frac{-4}{3}(x - 3)$

$4y - 3x + 21 = 0$ $3y + 4x - 3 = 0$

(ii) Slope of $DC = \frac{-1}{\text{slope of AD}}$ Slope of BC =

$= \frac{-4}{3}$ $= \frac{3}{4}$

Equation of AC

$y + 2^{o} = \frac{-4}{3}(x + 4)$

$3y + 6 = -4x - 16$

$4x + 3y + 22 = 0$

For point A, intersection of AC and AD

For point B, intersection of BC and BD

Point A = (0,1)

B = (–1,–6)

(iii) Minimum distance = $5\sqrt{2} - 5$

Maximum distance = $5\sqrt{2} + 5$

(iv) Area of quadrilateral ABBC = $5^2 = 25$

Area of ODAB = $\frac{1}{2} \times 25 = 12.5$

(v) Mid point of $CD = \left(\frac{-1}{2}, \frac{-5}{2}\right)$

Equation of circle,

$\left(x + \frac{1}{2}\right)^2 + \left(y + \frac{5}{2}\right)^2 = \left(\frac{5\sqrt{2}}{2}\right)^2$

$x^2 + y^2 + x + 5y - 6 = 0$

21. Through a given point P(5, 2), secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1)$, $A_2(B_2)$, $A_3(B_3)$, $A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$.

Find the value of $\sum_{i=1}^{5} PA_i^2 + \sum_{i=1}^{5} PB_i^2$. [Note : $A_r(B_r)$ denotes that the line passing through P(5, 2) meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r.]

यदि दिये गये एक बिन्दु P(5, 2) से वृत $x^2 + y^2 = 25$ पर डाली गयी छेदक रेखायें वृत को बिन्दुओं $A_1(B_1)$, $A_2(B_2)$, $A_3(B_3)$, $A_4(B_4)$ एवं $A_5(B_5)$ पर प्रतिच्छेद करती है कि $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ तथा $PA_5 + PB_5 = 9$ है। $\sum_{i=1}^{5} PA_i^2 + \sum_{i=1}^{5} PB_i^2$ का मान ज्ञात कीजिए। [नोट : $A_r(B_r)$ का तात्पर्य वह रेखा है जो बिन्दु P(5, 2) से गुजरती है तथा वृत $x^2 + y^2 = 25$ को दो बिन्दुओं A_r तथा B_r पर काटती है।]

Ans. 215

Sol. $A^2 + B^2 = (A + B)^2 - 2AB$

$\Rightarrow PA_1^2 + PA_2^2 + PA_3^2 + PA_4^2 + PA_5^2 + PB_1^2 + PB_2^2 + PB_3^2 + PB_4^2 + PB_5^2$

$\Rightarrow (PA_1 + PB_1)^2 - 2PA_1PB_1 + (PA_2 + PB_2)^2 - 2PA_2PB_2 + (PA_3 + PB_3)^2 - 2PA_3PB_3 + (PA_4 + PB_4)^2 - 2PA_4PB_4 + (PA_5 + PB_5)^2 - 2PA_5PB_5$

$\Rightarrow 25 + 36 + 49 + 64 + 81 - (2 \times 5\ PT^2) \Rightarrow$ Since we know $[PA \cdot PB = PT^2]$

$\Rightarrow 255 - 10PT^2$ ....(1)

Also, using distance formula $\Rightarrow PT = \sqrt{(5-5)^2 + (2-0)^2}$

$\Rightarrow PT = 2$

$\Rightarrow PT^2 = 4$ ....(2)

Using (2) and (1)

$\Rightarrow 255 - 10 \times 4 = 215$

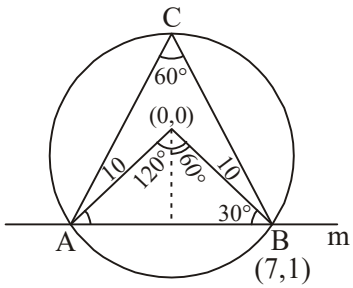
DIRECTOR CIRCLE

22. Chord AB of the circle $x^2 + y^2 = 100$ passes through the point (7, 1) and subtends an angle of 60° at the circumference of the circle. If m_1 and m_2 are the slopes of two such chords then the value of m_1m_2, is :

वृत $x^2 + y^2 = 100$ की जीवा AB बिन्दु (7, 1) से जाती है तथा परिधि के साथ 60° का कोण बनाती है। यदि m_1 तथा m_2 ऐसी जीवा की प्रवणताऐं है तो m_1m_2, का मान है :

Ans. –1

Sol.



CHORD WHOSE MID POINT IS GIVEN

23. Find the locus of the mid point of the chord of a circle $x^2 + y^2 = 4$ such that the segment intercepted by the chord on the curve $x^2 - 2x - 2y = 0$ subtends a right angle at the origin.

वृत्त $x^2 + y^2 = 4$ की उस जीवा के मध्य बिन्दु का बिन्दुपथ ज्ञात करें जिस जीवा के द्वारा वक्र $x^2 - 2x - 2y = 0$ पर काटा गया अन्तःखण्ड मूल बिन्दु पर समकोण बनाता है।

Ans. $x^2 + y^2 - 2x - 2y = 0$

Sol. Let the midpoint be (h, k)

Equation of chrod T = s

$\Rightarrow hx + ky = h^2 + k^2$

Let the above line intersects curve $x^2 - 2x - 2y = 0$ at A and B.

Let O be the origin.

For finding OA × OB, we will use homogenization

$$x^2 - 2x\left(\frac{(hx+ky)}{h^2+k^2}\right) - 2y\left(\frac{hx+ky}{h^2+k^2}\right) = 0$$

$$x^2\left(h^2+k^2\right) - x\left(hx+ky\right) - 2y\left(hx+ky\right) = 0$$

$$\Rightarrow x^2\left(h^2+k^2\right) + x^2\left(-2h\right) - 2kxy - 2hxy - 2ky^2 = 0$$

$$\Rightarrow x^2\left(h^2+k^2-2h\right) + y^2\left(-2k\right) + y\left(-2hx-2kx\right) = 0$$

$\because$ OA and OB are perpendicular.

$$\Rightarrow h^2 + k^2 - 2h - 2k = 0$$

$\Rightarrow h^2 + k^2 = 2(h + k)$

$\therefore$ Locus is $x^2 + y^2 = 2(x + y)$

$\Rightarrow x^2 + y^2 - 2x - 2y = 0$

24. Find the locus of the mid point of all chords of the circle $x^2 + y^2 - 2x - 2y = 0$ such that the pair of lines joining (0, 0) and the point of intersection of the chords with the circles make equal angle with axis of x.

वृत्त $x^2 + y^2 - 2x - 2y = 0$ की उन जीवाओं के मध्य बिन्दु का बिन्दुपथ ज्ञात करें जिनके वृत्त के साथ प्रतिच्छेद बिन्दुओं को (0, 0) से मिलाने वाली रेखा युग्म x अक्ष के साथ समान कोण बनाता है।

Ans. $x + y = 2$

Sol. Let the midpoint of chord be (h, k).

Equation of chord $T = S_1$

$\Rightarrow hx + ky - (x + h) - (y + k) = h^2 + k^2 - 2h - 2k$

Homogenising equation of circle with the above chord equation gives OA × OB equation.

$$x^2 + y^2 - 2(x + y)\left(\frac{hx + ky - x - y}{h^2 + k^2 - h - k}\right) = 0$$

$\Rightarrow x^2 (h^2 + k^2 - h - k) + y^2 (h^2 + k^2 - h - k) - 2(hx^2 + kxy - x^2 - xy + hxy + ky^2 - xy - y^2) = 0$

$\Rightarrow x^2 (h^2 + k^2 - h - k - 2h + 2) + y^2 (h^2 + k^2 - h - k - 2k + 2) + xy (-2k + 2 - 2h + 2) = 0$

$\Rightarrow x^2 (h^2 + k^2 - h - k - 2h + 2) + y^2 (h^2 + k^2 - h - k - 2k + 2) + 2xy (-k - h + 2) = 0$

$\Rightarrow$ It represents the form $ax^2 + by^2 + 2hxy$ and $m_1 + m_2 = \frac{-2h}{b}$

Where m_1 and m_2 are slopes of two lines represented by the equation.

$\because$ OA and OB make equal angle with x-axis.

$$\Rightarrow \frac{-2h}{b} = 0$$

$$\Rightarrow \frac{-2(2 - k - h)}{h^2 + k^2 + 2 - h - 3k} = 0$$

$\Rightarrow 2 - k - h = 0$

$\Rightarrow h + k = 2$

$\therefore$ Locus : $x + y = 2$

CHORD OF CONTACT

25. Find the equation of the circle passing through the three points (4, 7), (5, 6) and (1, 8). Also find the coordinates of the point of intersection of the tangents to the circle at the points where it is cut by the straight line $5x + y + 17 = 0$.

तीन बिन्दुओं (4, 7), (5, 6) व (1, 8) से गुजरने वाले वृत का समीकरण ज्ञात करें। उन बिन्दुओं पर खीची गई स्पर्श रेखाओं का प्रतिच्छेद बिन्दु ज्ञात करें जहाँ $5x + y + 17 = 0$ वृत को प्रतिच्छेद करती है।

Ans. $(-4, 2)$, $x^2 + y^2 - 2x - 6y - 15 = 0$

Sol. Equation : $(x-h)^2 + (y-k)^2 = r^2$

A : $(4-h)^2 + (7-k)^2 = r^2$

$\Rightarrow 16 + h^2 - 8h + 49 + k^2 - 14 = r^2$

$\Rightarrow h^2 + k^2 - 8h - 14k + 65 = r^2$ _____(1)

B : $(5-h)^2 + (6-k)^2 = r^2$

$\Rightarrow h^2 + k^2 - 10h - 12k + 61 = r^2$ _____(2)

C : $(1-h)^2 + (8-k)^2 = r^2$

$h^2 + k^2 - 2h - 16k + 65 = r^2$ _____(3)

Substracting equation (1) and (2)

$h^2 + k^2 - 8h - 14k + 65 = r^2$

$h^2 + k^2 - 2h - 16k + 65 = r^2$

$-6k + 2k = 0 \ldots.(4) \times 2$

$64 = 2k$

$-12h + hk = 0$ _____(5)

Subtracting equation (2) and (3)

$h^2 + k^2 - 2h - 16k + 65 = r^2$

$h^2 + k^2 - 10h - 12k + 61 = r^2$

$8h - 4k + 4 = 0$ _____(6)

Adding equation (5) and (6)

$-12h + 4k = 0$

$8h - 4k + 4 = 0$

$-4h + 4 = 0$

$h = 1$

$-12 + 4k = 0$

$4k = 12$

$k = 3$

$\therefore$ center $0 \equiv (h, k) = (1,3)$

Radius : $OA = \sqrt{(1-4)^2 + (3-7)^2} = \sqrt{3^2 + 4^2}$

$\sqrt{9+16} = \sqrt{25} = 5$ cm

Equation of circle : $(x-h)^2 + (y-k)^2 = r^2$

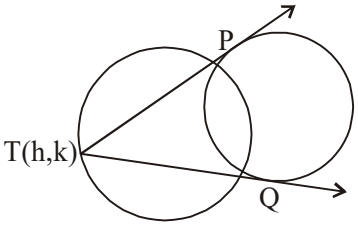
$\Rightarrow (x-1)^2 + (y-3)^2 = 5^2$

26. Tangents TP and TQ are drawn from a point T to the circle $x^2 + y^2 = a^2$. If the point T lies on the line $px + qy = r$, find the locus of centre of the circumcircle of triangle TPQ.

बिन्दु T से वृत्त $x^2 + y^2 = a^2$ पर TP व TQ स्पर्श रेखायें खीचीं गई है। यदि T रेखा $px + qy = r$ पर स्थित है, तो त्रिभुज TPQ के परिवृत्त के केन्द्र का बिन्दुपथ ज्ञात कीजिए।

Ans. $2px + 2qy = r$

Sol.



Circle TPQ is $S + \lambda P = 0$ where P is chord of contact of T(h,k).

Circumcircle is $(x^2 + y^2 - a^2) + \lambda(hx + ky - a^2)$

It passes through T(h, k)

$\therefore (h^2 + k^2 - a^2) + \lambda(h^2 + k^2 - a^2) = 0$

or $\lambda = -1$

Circle is $x^2 + y^2 - 4x - ky = 0$

Its centre $x = \frac{h}{2}$, $y = \frac{k}{2}$. But (h,k) lies on $px + qy + r = 0$, $\therefore ph + qk + r = 0$.

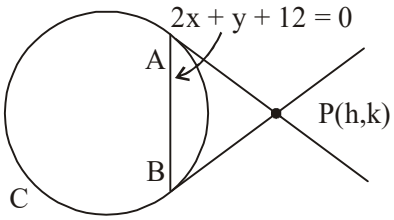
Eliminating h, k we get the locus as $p \cdot 2x + q \cdot 2y + r = 0$.

27. Find the point of intersection of the tangents drawn to the circle $x^2 + y^2 - 4x + 3y - 1 = 0$ at the points where the line $2x + y + 12 = 0$ meets it.

रेखा $2x + y + 12 = 0$ द्वारा वृत $x^2 + y^2 - 4x + 3y - 1 = 0$ काटे गए बिन्दुओं पर स्पर्श रेखाओं का प्रतिच्छेदन बिन्दु ज्ञात करें।

Ans. $(1,-2)$

Sol.



$S : x^2 + y^2 - 4x + 3y - 1 = 0.$

Let the point of intersection of tangents by p(h,k).

Line aB $2x + y + 12 = 0$ is chord of contact.

Equation of chord of contact is given by :

$$x(h) + y(k) - 4\left(\frac{x+h}{2}\right) + 3\left(\frac{y+k}{2}\right) - 1 = 0$$

$$\Rightarrow xh + yk - 2x - 2h + \frac{3y}{2} + \frac{3k}{2} - 1 = 0$$

$$\Rightarrow x(h-2) + y\left(k + \frac{3}{2}\right) - \left(2h - \frac{3k}{2} + 1\right) = 0$$

Comparing this with given equation $2x + y + 12 = 0$

$$\frac{h-2}{2} = \frac{2k+3}{2} = -\left(\frac{4h - 3k + 2}{24}\right)$$

Solving, we get

$h = 1, k = -2$

$(1, -2)$.

POLE AND POLAR

28. The chords of contact of the pair of tangents to the circle $x^2 + y^2 = 1$ drawn from any point on the line $2x + y = 4$ pass through the point (α, β), then find $\alpha^2 + \beta^2$.

रेखा $2x + y = 4$ के किसी बिन्दु से वृत $x^2 + y^2 = 1$ के सापेक्ष सभी स्पर्श रेखाऐं बिन्दु (α, β) से गुजरती है, तो $\alpha^2 + \beta^2$ का मान ज्ञात कीजिए।

Ans. $\frac{5}{16}$

Sol. Let (h,k) be any point on the given line

$\therefore$ 2h + k = 4 and C.C. is hx + ky = 1

or hx + (4 – 2h)y = 1 $\quad \because$ k = 4 – 2h

or (4y – 1) + h(x + 2y) = 0

It is of the form P + λQ = 0. It passes through the intersection of P \ 0 and Q = 0 or $\left(\frac{1}{2}, \frac{1}{4}\right)$.

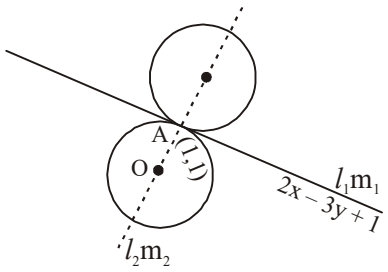
FAMILY OF CIRCLES

29. The line 2x – 3y + 1 = 0 is tangent to a circle S = 0 at (1, 1). If the radius of the circle is $\sqrt{13}$. Find the equation of the circle S.

रेखा 2x – 3y + 1 = 0 वृत S = 0 को (1, 1) पर स्पर्श करती है। यदि वृत की त्रिज्या $\sqrt{13}$ है तो वृत का समीकरण ज्ञात कीजिए।

Ans. $x^2 + y^2 – 6x + 4y=0$ OR $x^2 + y^2 + 2x – 8y + 4=0$

Sol.



Slope of line $\ell_2 = \frac{-1}{\text{slope of line } \ell_1}$

$= -\frac{1}{\frac{2}{3}}$

$= -\frac{3}{2}$

Equation of line $\ell_2 = y - 1 = -\frac{3}{2}(x - 1)$

$y = -\frac{3}{2}x + \frac{5}{2}$

Let coordinate of centre $\equiv \left(a, -\frac{3a}{2} + \frac{5}{2}\right)$

Thus $OA = \sqrt{13}$

$OA^2 = 13$

$$(a-1)^2 + \left(-\frac{3a}{2} + \frac{5}{2} - 1\right)^2 = 13$$

$$(a-1)^2 + \frac{9}{4}(a-1)^2 = 13$$

$$(a-1)^2 \times \frac{13}{4} = 13$$

$(a-1)^2 = 4$

$a = 3, -1$

Thus coordinates of centre are $(3, -2)$ & $(-1,4)$

Equation of circles

$(x-3)^2 + (y+2)^2 = 13$ & $(x+1)^2 + (y-4)^2 = 13$

Note : (1) Two circles come from the solution which is true because one circle on each side of line is possible.

(2) Distance Formula = $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ between two points (x_1, y_1) & (x_2, y_2).

(3) Putting x-coordinate which is a in line L_2 we get y-coordinate of centre.

30. Find the equation of the circle which passes through the point (1, 1) & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point (2, 3) on it.

उस वृत का समीकरण ज्ञात करें जो (1, 1) से गुजरता है तथा वृत $x^2 + y^2 + 4x - 6y - 3 = 0$ को (2, 3) पर स्पर्श करता है।

Ans. $x^2 + y^2 + x - 6y + 3 = 0$

Sol. Equation of tangent to circle

$x^2 + y^2 + 4x - 6y - 3 = 0$ at $(2,3) \Rightarrow 2x + 3y + 2(x+2) - 3(y+3) - 3 = 0$

$4x - 8 = 0 \quad \Rightarrow \quad x - 2 = 0$

family of cirlce $\Rightarrow \quad S + \lambda L = 0$

$x^2 + y^2 + 4x - 6y - 3 + \lambda (x - 2) = 0$ .....(1)

Passes through (1, 1)

$-3 - \lambda = 0 \quad \Rightarrow \quad \lambda = -3$

Putting in (1)

$x^2 + y^2 + x - 6y + 3 = 0$

31. Find the equation of the circle whose radius is 3 and which touches the circle $x^2 + y^2 - 4x - 6y - 12=0$ internally at the point $(-1, -1)$.

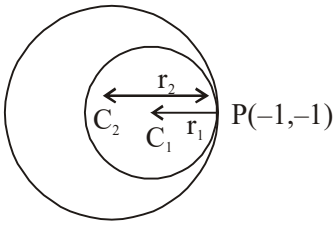
उस वृत्त का समीकरण ज्ञात करें जिसकी त्रिज्या 3 है तथा जो वृत $x^2 + y^2 - 4x - 6y - 12=0$ को बिन्दु $(-1, -1)$ पर आन्तरिक स्पर्श करता है।

Ans. $5x^2 + 5y^2 - 8x - 14y - 32 = 0$

Sol. $5x^2 + 5y^2 - 8x - 14y - 32 = 0$

$C_1 : (x_1, y_1), r_1 = 3$

$C_2 : (2,3), r_2 = 5$



Point P divides C_2C_1 externally in the ratio of their radius.

$C_2P : C_1P = -5 : 3$

or $C_1C_2 : C_1P = 2 : 3$

co-ordinates of $C_1 = \left(\frac{4}{5}, \frac{7}{5}\right)$

Equation of circle is

$$\left(x - \frac{4}{5}\right)^2 + \left(y - \frac{7}{5}\right)^2 = 9$$

$\Rightarrow 5x^2 + 5y^2 - 8x - 14y - 32 = 0$

32. Find the equation of the circle passing through the points of intersection of the circles $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 - 10x - 12y + 40 = 0$ and whose radius is 4.

वृतों $x^2 + y^2 - 2x - 4y - 4 = 0$ एवं $x^2 + y^2 - 10x - 12y + 40 = 0$ के प्रतिच्छेदन बिन्दुओं से गुजरने वाले वृत का समीकरण ज्ञात करें जिसकी त्रिज्या 4 है।

Ans. $2x^2 + 2y^2 - 18x - 22y + 69 = 0$ and $x^2 + y^2 - 2y - 15 = 0$

Sol. Equation of the any circle through the points of intersection of given circles is

$(x^2 + y^2 - 2x - 3y - 4) + \lambda (x^2 + y^2 - 10x - 12y + 40) = 0$

$\Rightarrow x^2(1 + \lambda) + y^2 (1 + \lambda) - 2x (1 + 5\lambda) - 2y (2 + 6\lambda) - 4 + 40\lambda = 0$

or $x^2 + y^2 - 2x\frac{(1+5\lambda)}{(1+\lambda)} - 2y\frac{(2+6\lambda)}{(1+\lambda)} + \frac{(40\lambda-4)}{(1+\lambda)} = 0$ _____(1)

Its radius $\sqrt{\left(\frac{1+5\lambda}{1+\lambda}\right)^2 + \left(\frac{2+6\lambda}{1+\lambda}\right)^2 - \left(\frac{40\lambda-4}{1+\lambda}\right)} = 4$ (given)

$\Rightarrow \frac{(1+5\lambda)^2 + (2+6\lambda)^2 - (40\lambda-4)(1+\lambda)}{(1+\lambda)^2} = 16$

$\Rightarrow 5\lambda^2 + 34\lambda - 7 = 0$

or $(\lambda - 7)(5\lambda + 1) = 0$

$\therefore \lambda = 7$ or $\lambda = \frac{1}{5}$

Substituting the values of λ in (1), the required circles are

$2x^2 + 2y^2 - 18x - 22y + 69 = 0$ and $x^2 + y^2 - 2y - 15 = 0$ Ans.

33. Find the equation to the circle which passes through the points $(1, -2)$ and $(4, -3)$ and which has its centre on the straight line $3x + 4y = 7$.

बिन्दु $(1, -2)$ एवं $(4, -3)$ से गुजरने वाले वृत्त का समीकरण ज्ञात कीजिये, जिसका केन्द्र सरल रेखा $3x + 4y = 7$ पर स्थित है।

Ans. $15x^2 + 15y^2 - 94x + 18y + 55 = 0$

Sol. Let the equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

Centre of circle is $(-g, -f)$.

Centre lies on the line $3x + 4y = 7$.

Therefore,

$-3g - 4f = 7$

$3g + 4f + 7 = 0$ _____(1)

Circle passes through the points $(1, -2)$ and $(4, -3)$.

By putting these values in the equation of circle,

$2g - 4f + c + 5 = 0$ and $8g - 6f + c + 25 = 0$

Solving these two equation we get,

$6g - 2f + 20 = 0$ _____(2)

Now, solving equation (1) & (2), we get,

$f = \frac{3}{5}$ and $g = \frac{-47}{15}$

Also, $c = \frac{11}{3}$

Hence, the equation of circle is,

$$x^2 + y^2 + 2x\left(\frac{-47}{15}\right) + 2y\left(\frac{3}{5}\right) + \frac{11}{3}$$

$15(x^2 + y^2) - 94x + 18y + 55 = 0$

POSITION OF TWO CIRCLES AND COMMON TANGENTS BETWEEN THEM

34. Let P and Q be 2 circles externally tangent at point X. A straight line is tangent to P at point A and is tangent to Q at point B($A \neq B$). The line tangent to P and Q at X intersects line AB at a point Y. If AY = 10 and the radius of P is 9. If the radius of the circle Q can be expressed as a rational number a/b in the lowest form, find the value of (a + b).

माना P तथा Q, दो वृत जो बिन्दु X पर बाह्य स्पर्श करते है। एक सरल रेखा P को बिन्दु A पर तथा Q को बिन्दु B पर ($A \neq B$) स्पर्श करती है। बिन्दु X पर खींची गई स्पर्श रेखा AB को Y पर प्रतिच्छेद करती है। AY = 10 है तथा P की त्रिज्या 9 है। यदि वृत Q की त्रिज्या को परिमेय संख्या a/b के न्यूनतम समकक्ष रूप में प्रदर्शित किया जा सकता है तो (a + b) का मान ज्ञात कीजिये।

Ans. 109

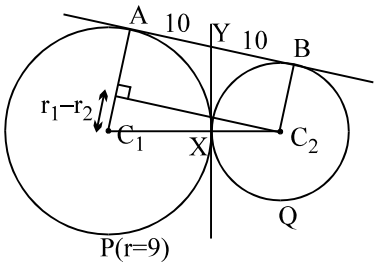
Sol. AY = BY ($\because$ Y lies on radical axis)

$(AB)^2 = (r_1 + r_2)^2 - (r_1 - r_2)^2 = 4r_1r_2$

$AB = 2\sqrt{r_1r_2}$

$AB = 20 = 2\sqrt{9r_2}$

$r_2 = \frac{100}{9}$ Ans.



35. Find the sum of all integral value(s) of 'r'($r > 0$) for which the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42 = 0$ intersect in two real distinct points.

यदि वृत्त $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ तथा $x^2 + y^2 + 6x - 14y + 42 = 0$ एक दूसरे को यदि दो भिन्न–भिन्न बिन्दुओं पर काटते है 'r' के सभी सम्भावित ($r > 0$) पूर्णांक मानों का योग क्या होगा।

Ans. 119

Sol. We have $(x - 5)^2 + (y + 8)^2 = 25 + 64 + r^2 - 89$

and $(x + 3)^2 + (y - 7)^2 = 49 + 9 - 42 = 16$

$\Rightarrow$ $(x - 5)^2 + (y + 8)^2 = r^2$

and $(x + 3)^2 + (y - 7)^2 = (4)^2$

$\underline{(5,-8) \qquad (-3,7)}$ $\Rightarrow$ $\sqrt{64+225} = \sqrt{289} = 17$ = distance between their centres

Now, $|r-4| < 17 < r+4 \Rightarrow r+4 > 17 \Rightarrow r > 13$

and $-17 < r-4 < 17 \Rightarrow -13 < r < 21$

Hence $13 < r < 21$

$\therefore$ Possible values of 'r' can be 14, 15, 16, 17, 18, 19, 20

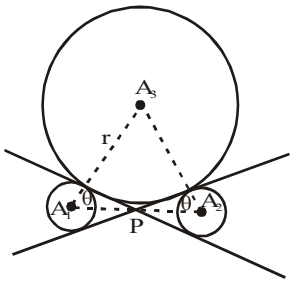
Hence required sum = 119

36. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid point of the line segment joining the centres of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C, then the radius of the circle C is. **[JEE-2009]**

दो वृत्त जिनमें से प्रत्येक की त्रिज्या इकाई है, के केन्द्र परस्पर 6 इकाई दूर है। माना कि C_1 व C_2 को जोड़ने वाले रेखा खण्ड का मध्य बिन्दु P है तथा C_1 व C_2 की बाह्य स्पर्श करने वाला एक वृत्त C है। यदि C_1 व C_2 की एक उभयनिष्ठ स्पर्श रेखा जो P से गुजरती है, C_2 व C की भी उभयनिष्ठ स्पर्श रेखा है तो वृत्त C की त्रिज्या ज्ञात कीजिए। **[JEE-2009]**

Ans. 8

Sol.



In triangle $A_1A_2A_3$

$A_1A_3 = A_3A_2$

Let angle $A_3A_1A_2 = \theta$, $\cos\theta = \frac{1}{3}$, $\sin\theta = \frac{2\sqrt{2}}{3}$

Apply sine rule in triangle $A_1A_2A_3$

$$\frac{6}{\sin(\pi-2\theta)} = \frac{r+1}{\sin\theta}$$

$\Rightarrow r = 8$

RADICAL AXIS AND ITS PROPERTIES

37. Show that the equation of a straight line meeting the circle $x^2 + y^2 = a^2$ in two points at equal distances 'd' from a point (x_1, y_1) on its circumference is $xx_1 + yy_1 - a^2 + \left(\frac{d^2}{2}\right) = 0$.

प्रदर्शित किजिये कि वृत्त $x^2 + y^2 = a^2$ पर स्थित बिन्दु (x_1, y_1) से समान दूरी 'd' पर स्थित दो बिन्दु, जो वृत $x^2 + y^2 = a^2$ पर स्थित है, से गुजरने वाली सरल रेखा का समीकरण $xx_1 + yy_1 - a^2 + \left(\frac{d^2}{2}\right) = 0$ है।

Ans.

Sol. The equation of the circle is $x^2 + y^2 = a^2$

Let the points the line meets circle be,

A $(a\cos\theta_1, a\sin\theta_1)$ and $(a\cos\theta_2, a\sin\theta_2)$

Given these points are at distance 'd' from (x_1, y_1)

$\Rightarrow (x_1 - a\cos\theta_1)^2 + (y_1 - a\sin\theta_1)^2 = (x_1 - a\cos\theta_2)^2 + (y_2 - a\sin\theta_2)^2 = d^2$ ...(1)

$\Rightarrow x_1^2 - 2ax_1\cos\theta_1 + a^2\cos^2\theta_1^2 + y_1^2 + a^2\sin^2\theta_1^2 - 2ay_1\sin\theta_1 = x_1^2 - 2ax_1\cos\theta_2 + a^2\cos^2\theta_2 + y_1^2$

$-2ay_1\sin\theta_2 + a^2\sin^2\theta_2$

$x_1\cos\theta_1 + y_1\sin\theta_1 = y_1\sin\theta_2 + x_1\cos\theta_2$

$\Rightarrow \frac{\sin\theta_2 - \sin\theta_1}{\cos\theta_2 - \cos\theta_1} = \frac{-x_1}{y_1}$ ...(2)

The equation of the line joining A and B is

$(y - a\sin_2) = \frac{a\sin\theta_2 - a\sin\theta_1}{a\cos\theta_2 - a\cos\theta_1}(x - a\cos\theta_2)$

$\Rightarrow (y - a\sin\theta_2) = \frac{-x_1}{y_1}(x - a\cos\theta_2)$

$\Rightarrow y_1y + xx_1 - y_1a\sin\theta_2 - ax_1\cos\theta_2 = 0$

$\Rightarrow xx_1 + yy_1 + yy_1 + \left(\frac{x_1^2 - 2ax_1\cos\theta_2 + a^2\cos^2\theta_2}{2}\right) + \left(\frac{y_1^2 - 2ay_1\sin\theta_2 + a^2\sin^2\theta_2}{2}\right)$

$-\left(\frac{x_1^2 + y_1^2}{2}\right) - \frac{a^2}{2} = 0$

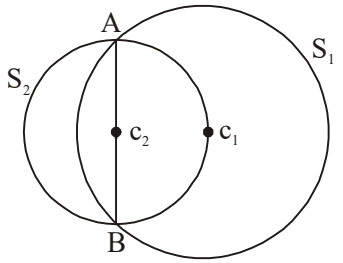
$\Rightarrow xx_1 + yy_1 + \frac{d^2}{2} - a^2 = 0$

38. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where $a, b > 0$), then find the maximum value of (ab).

यदि वृत्त $x^2 + y^2 + 4x + 22y + a = 0$, वृत्त $x^2 + y^2 - 2x + 8y - b = 0$ की परिधि को समद्विभाजित करता है (जहाँ $a, b > 0$), तो (ab) का अधिकतम मान ज्ञात कीजिये।

Ans. 625

Sol.



Given that $S_1 = x^2 + y^2 + 4x + 22y + c = 0$

$S_2 = x^2 + y^2 - 2x + 8y - d = 0$

AB is a common chord passing through C_2.

Equation of AB : $S_1 - S_2 = 0$

$\Rightarrow (x^2 + y^2 + 4x + 22y + c) - (x^2 + y^2 - 2x + 8y - d)$

$\Rightarrow 6x + 14y + c + d = 0$

The common chord passes through centre of S_2, i.e. $C_2 (1, -4)$

$\Rightarrow 6(1) + 14(-4) + c + d = 0$

$\Rightarrow c + d = 50$

Since A.M. $\geq$ G.M.

$$\frac{c+d}{2} \geq \sqrt{cd}$$

$$\Rightarrow 25 \geq \sqrt{cd}$$

$$\Rightarrow cd \leq 625$$

39. Find the equation of a circle which is co–axial with circles $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ and $x^2 + y^2 + 4x + 2y + 1 = 0$. It is given that the centre of the circle to be determined lies on the radical axis of these two circles.

उस वृत का समीकरण ज्ञात करें जो वृत $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ तथा $x^2 + y^2 + 4x + 2y + 1 = 0$ के साथ समाक्षी है। यह दिया गया है कि उस वृत का केन्द्र, जिसे ज्ञात करना है। इन दो वृतो के मूलाक्ष अक्ष पर स्थित है।

Ans. $4x^2 + 4y^2 + 6x + 10y - 1 = 0$

Sol. The two given circles in standard from we

$S_1 = 0$ or $x^2 + y^2 - x + 3y - \frac{3}{2} = 0$

$S_2 = 0$ or $x^2 + y^2 + 4x + 2y + 1 = 0$

Radical axis is given by $S_1 - S_2 = 0$

$\therefore -5x + y - \frac{5}{2} = 0$

or $10x - 2y + 5 = 0$ ...(1)

Circle coaxial with the given circles is

$S_1 + \lambda S_2 = 0$

$\left(x^2 + y^2 - x + 3y - \frac{3}{2}\right) + \lambda(x^2 + y^2 + 4x + 2y + 1) = 0$

or $(x^2 + y^2)(1 + \lambda) - (1 - 4\lambda)x + (3 + 2\lambda)y - \frac{3}{2} + \lambda = 0$

or $x^2 + y^2 - \frac{1-4\lambda}{1+\lambda}x + \frac{3+2\lambda}{1+\lambda}y + \frac{\lambda - \frac{3}{2}}{1+\lambda} = 0$

Its centre $\left(\frac{1}{2}, \frac{1-4\lambda}{1+\lambda}, -\frac{1}{2}, \frac{3+2\lambda}{1+\lambda}\right)$

i.e. $(-g, -f)$ lies on (1)

$\therefore \frac{5(1-4\lambda)}{1+\lambda} + \frac{3+2\lambda}{1+\lambda} + 5 = 0$

or $5 - 20\lambda + 3 + 2\lambda + 5 + 5\lambda = 0$

or $13 - 13\lambda = 0$ $\therefore \lambda = 1$

putting in (2), the required circle is

$x^2 + y^2 + \frac{3}{2}x + \frac{5}{2}y - \frac{1}{4} = 0$

or $4x^2 + 4y^2 + 6x + 10y - 1 = 0$

ANGLE BETWEEN TWO CIRCLES & ORTHOGONALITY OF CIRCLES

40. Find the equation of the circle through the points of intersection of circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ and cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

वृतों $x^2 + y^2 - 4x - 6y - 12 = 0$ तथा $x^2 + y^2 + 6x + 4y - 12 = 0$ के प्रतिच्छेद बिन्दुओं से जाने वाले तथा वृत $x^2 + y^2 - 2x - 4 = 0$ के लम्बकोणीय वृत का समीकरण ज्ञात कीजिए।

Ans. $x^2 + y^2 + 16x + 14y - 12 = 0$

Sol. Let, the equation of the circle through the intersection of given circle is,

$x^2 + y^2 - 4x - 6y + 12 = 0$ ...(1)

Here, the equation of radical axis for the circles

$x^2 + y^2 - 4x - 6y + 12 = 0$

and $x^2 + y^2 + 6x - + 4y - 12 = 0$ is,

$-10x - 10y = 0$

So, equation (1) can be written as,

$x^2 + y^2 - 4x - 6y - 12 + \lambda(-10x - 10y) = 0$

$\Rightarrow x^2 + y^2 - x(10\lambda + 4) - y(10\lambda + 6) - 12 = 0$

It cuts the circle v orthogonally.

$\therefore 2gg_1 + 2ff_1 = c + c_1$

$\Rightarrow (10\lambda + 4)(1) + (10\lambda + 6)(0) = -12 - 4$

$\Rightarrow \lambda = -2$

Hence, the required equation of the circle is,

$x^2 + y^2 - 4x - 6y - 12 - 2(-10x - 10y) = 0$

$\Rightarrow x^2 + y^2 + 16x + 14y - 12 = 0$

41. The centre of the circle $S = 0$ lie on the line $2x - 2y + 9 = 0$ and $S = 0$ cuts orthogonally the circle $x^2 + y^2 = 4$. Show that circle $S = 0$ passes through two fixed points & find their coordinates.

वृत $S = 0$ का केन्द्र रेखा $2x - 2y + 9 = 0$ पर स्थित है तथा $S = 0$ वृत $x^2 + y^2 = 4$ के लम्बकोणीय है। प्रदर्शित करों कि वृत $S = 0$ दो अचर बिन्दुओं से गुजरता है तथा उनके निर्देशांक ज्ञात कीजिए।

Ans. $(-4, 4)$; $(-1/2, 1/2)$

Sol. Let the circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$

Its centre (–g, –f) lies on 2x – 2y + 9 = 0.

$\therefore$ –2g + 2f + 9 = 0 ...(1)

IT cuts orthogonally the circle $x^2 + y^2 - 4 = 0$

$\therefore\ 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

or 0 + 0 = c – 4 $\therefore$ c = 4

and from (1), 2g = 9 + 2f.

Putting the values of c and g we get the equation of the circle as

$x^2 + y^2 + (9 + 2g)x + 2fy + 4 = 0$

or $(x^2 + y^2 + 9x + 4) + 2f(x + y) = 0$

Above is of the form $S + \lambda P = 0$

Above represents a family of circles which passes through the points of intersection of S = 0 and P = 0.

Solving $x^2 + y^2 + 9x + 4 = 0$ and x + y = 0 we get the fixed points as $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and (–4, 4).

42. Find the equation of a circle passing through the origin if the line pair, xy – 3x + 2y – 6 = 0 is orthogonal to it. If this circle is orthogonal to the circle $x^2 + y^2 - kx + 2ky - 8 = 0$ then find the value of k.

मूल बिन्दु से गुजरने वाले वृत का समीकरण ज्ञात करों, यदि रेखा युग्म xy – 3x + 2y – 6 = 0 वृत्त के लम्बकोणीय हो। यदि यह वृत, वृत $x^2 + y^2 - kx + 2ky - 8 = 0$ के लम्बकोणीय है तो k का मान ज्ञात कीजिए।

Ans. $x^2 + y^2 + 4x - 6y = 0$; k = 1

Sol. Let the equation of circle be

$x^2 + y^2 + 2gx + 2fy = 0$

it passes through origin)

The line pair is

xy – 3x + 2y – 6 = 0

x(y – 3) + 2 (y – 3) = 0

(x + 2) (y – 3) = 0

The centre is point of intersection of these two lines.

c ≡ (–2,3)

g = 2 and f = –3

$s \equiv x^2 + y^2 + 4x - 6y = 0$

$s_1 = x^2 + y^2 - kx + 2ky - 8 = 0$

Since s & s_1 are orthogonal

$\therefore 2gg_1 + 2ff_1 = 0 - 8$

$\Rightarrow 2(-k) + (-3) \times 2k = 0 - 8$

$\therefore k = 1$

43. Find the equation of the circle which cuts the circle $x^2 + y^2 - 14x - 8y + 64 = 0$ and the coordinate axes orthogonally.
उस वृत का समीकरण ज्ञात करें जो वृत $x^2 + y^2 - 14x - 8y + 64 = 0$ तथा निर्देशांक अक्षो को लम्बवत् प्रतिच्छेद करता है।

Ans. $x^2 + y^2 = 64$

Sol. $x^2 + y^2 - 14x - 8y + 64 = 0$

$(g, f) = (7,4); c_1 = 64$

Let $x^2 + y^2 = a^2$

$\Rightarrow x^2 + y^2 - a^2 = 0$

$(g, f) = (0,0); c_2 = -a^2$

$2g_1g_2 + 2f_1f_2 = c_1 + c_2$

$0 + 0 = 64 + (-a^2)$

$a^2 = 64$

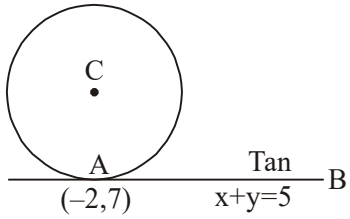
i.e., $x^2 + y^2 = 64$

44. Find the equation of a circle which touches the line $x + y = 5$ at the point $(-2, 7)$ and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally.

रेखा $x + y = 5$ को बिन्दु $(-2, 7)$ पर स्पर्श करने वाले तथा वृत $x^2 + y^2 + 4x - 6y + 9 = 0$ को लम्बकोणीय प्रतिच्छेद करने वाले वृत का समीकरण ज्ञात कीजिए।

Ans. $x^2 + y^2 + 7x - 11y + 38 = 0$

Sol.



$(x + 2)^2 + (y - 7)^2 + \lambda(x + y - 5) = 0$

or $x^2 + y^2 + x(4 + \lambda) + y(-14 + \lambda) + (53 + 5\lambda) = 0$

It cuts orthogonally the circle

$$2\frac{4+\lambda}{2}\cdot 2+2\frac{(-14+\lambda)}{2}(-3)=9+53-5\lambda$$

or $\lambda(2-3+5)=62-42-8$

or $4\lambda=12\Rightarrow\lambda=3$

Hence the required circle is

$x^2+y^2+7x-11y+38=0$

45. Find the equation of the circle passing through the point (–6 , 0) if the power of the point (1, 1) w.r.t. the circle is 5 and it cuts the circle $x^2+y^2-4x-6y-3=0$ orthogonally.

बिन्दु (–6 , 0) से गुजरने वाले वृत का समीकरण ज्ञात करें, यदि बिन्दु (1, 1) का वृत के सापेक्ष शक्ति 5 है तथा यह वृत, वृत $x^2+y^2-4x-6y-3=0$ के लम्बकोणीय प्रतिच्छेद करता है।

Ans. $x^2+y^2+6x-3y=0$

Sol. Let the circle be $x^2+y^2+2gx+2fy+c=0$

(–6,0) lies on the circle.

$\therefore 36-12g+c=0$ _____(i)

The power of (i,i) is 5.

$\Rightarrow 1+1+2g+2f+c=5$

$\Rightarrow 2g+2f+c=3$ _____(ii)

S is orthogonal to

$x^2+y^2-4x-6y-3=0$

$\Rightarrow 2g(-2)+2f(-3)=c-3$

$\therefore 4g+6f+c-3=0$ _____(iii)

From (ii) and (iii)

$2g+2c=6\Rightarrow g+c=3$ _____(iv)

From (i) and (iv) $g=3$

$\Rightarrow c=0$ & $f=\frac{-3}{2}$

$s\equiv x^2+y^2+6x-3y=0$

46. Find the equation of the circle which passes through the points of intersection of circles $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 2x - 6y + 6 = 0$ and intersects the circle $x^2 + y^2 + 4x + 6y + 4 = 0$ orthogonally.

वृत्तों $x^2 + y^2 - 2x - 6y + 6 = 0$ व $x^2 + y^2 + 2x - 6y + 6 = 0$ के प्रतिच्छेद बिन्दुओं से गुजरने वाले उस वृत्त समीकरण ज्ञात करें जो वृत $x^2 + y^2 + 4x + 6y + 4 = 0$ को लम्बकोणीय प्रतिच्छेद करता है।

Ans. $x^2 + y^2 + 14x - 6y + 6 = 0$

Sol. By $S_1 + \lambda S_2$ the required circle is

$(x^2 + y^2 - 2x + 6y6) + \lambda(x^2 + y^2 + 2x - 6y + 6) = 0$

or $(x^2 + y^2 - 6y + 6)(1 + \lambda) + 2x(\lambda - 1) = 0$

or $x^2 + y^2 + 2x\dfrac{\lambda - 1}{\lambda + 1} - 6y + 6 = 0$ _____(1)

It cuts $x^2 + y^2 + 4x + 6y + 4 = 0$ orthogonally.

Hence, $2g_1g_2 + 2f_1f_2 = c_1 + c_2$

i.e. $2\dfrac{\lambda - 1}{\lambda + 1} \cdot 2 - 2 \cdot 3 \cdot 3 = 6 + 4$

or $4(\lambda - 1) = 28(\lambda + 1)$

$\lambda - 1 = 7\lambda + 7 \qquad \therefore \lambda = \dfrac{-4}{3}$

Hence the required circle by (1) is

$x^2 + y^2 + 14x - 6y + 6 = 0$

47. Line $2x + 3y + 1 = 0$ is a tangent to a circle at $(1, -1)$. This circle is orthogonal to a circle which is drawn having diameter as a line segment with end points $(0, -1)$ and $(-2, 3)$. Find equation of circle.

रेखा $2x + 3y + 1 = 0$ एक वृत्त को $(1, -1)$ पर स्पर्श करती है। यह वृत एक ऐसे वृत्त को लम्बकोणीय प्रतिच्छेद करता है जो कि रेखाखण्ड जिसके सिरे $(0, -1)$ व $(-2, 3)$ को व्यास लेकर खींचा गया है। वृत्त का समीकरण ज्ञात कीजिए।

Ans. $2x^2 + 2y^2 - 10x - 5y + 1 = 0$

Sol. The required circle is $(x - 1)^2 + (y + 1)^2 + \lambda(2x + 3y + 1) = 0$

or $x2 + y2 + 2(\lambda - 1)x + (3\lambda + 2)y + \lambda + 2 = 0$ _____(1)

The circle having the line joining $(0,-1)$ and $(-2,3)$ as diameter is

$x(x + 2) + (y + 1)(y - 3) = 0$

or $x^2 + y^2 + 2x - 2y - 3 = 0$ _____(2)

Equations (1) and (2) are orthogonal.

$\therefore\ 2(\lambda - 1) - (3\lambda + 2) = \lambda + 2 - 3$

On simplifying,

$$\Rightarrow \lambda = -\frac{3}{2}$$

Hence euqation (1) becomes

$$x^2 + y^2 + 2\left(-\frac{3}{2} - 1\right)x + \left(3 \times -\frac{3}{2} + 2\right)y - \frac{3}{2} + 2 = 0$$

$$\Rightarrow x^2 + y^2 - 5x - \frac{5}{2}y + \frac{1}{2} = 0$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 48 to 50

Two fixed points A and B are 4 units apart, and are on the same side of a moving line L. If perpendicular distances of A and B say p_1 and p_2 from the line L are such that $p_1 + 3p_2 = k$, k being a constant, then the line L always touches a fixed circle C.

दो स्थिर बिन्दु A एवं B के बीच की दूरी 4 है तथा यह एक गतिशील रेखा L के एक ही तरफ है यदि A एवं B की रेखा L से दूरी p_1 एवं p_2 है जहाँ $p_1 + 3p_2 = k$ (स्थिरांक) तो रेखा L हमेशा एक स्थिर वृत C को स्पर्श करेगी।

48. The centre of the circle C lies on

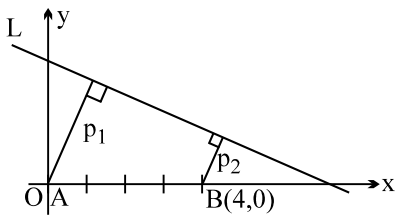
(A) line segment joining AB
(B) perpendicular bisector of AB
(C) one of A or B
(D) nothing definite can be said

C का केन्द्र होगा :

(A) AB रेखाखण्ड पर
(B) AB के लम्बार्धक पर
(C) या तो A या B पर
(D) कुछ निश्चित नही कहा जा सकता

Ans. A

Sol. Let A = (0, 0) and B = (4, 0)
and the line be $ax + by = 1$



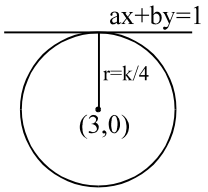
$$p_1 = \left|\frac{1}{\sqrt{a^2 + b^2}}\right| \quad ; \quad p_2 = \left|\frac{4a - 1}{\sqrt{a^2 + b^2}}\right|$$

$p_1 + 3p_2 = k$

$$\left|\frac{1}{\sqrt{a^2 + b^2}}\right| + 3\left|\frac{4a - 1}{\sqrt{a^2 + b^2}}\right| = k$$

now (0, 0) and (4, 0) must give the same sign i.e. – ve with the line L $(4a - 1 < 0)$

$$\therefore \quad \frac{1}{\sqrt{a^2+b^2}} + \frac{3(1-4a)}{\sqrt{a^2+b^2}} = k$$



$$\left|\frac{4(1-3a)}{\sqrt{a^2+b^2}}\right| = k \; ; \qquad \left|\frac{3a-1}{\sqrt{a^2+b^2}}\right| = \frac{k}{4}$$

hence centre of the fixed circle is (3, 0) which lies on the line segment AB $\Rightarrow$ (A)

49. If k = 4 then the radius of the circle is :

यदि k = 4 तो वृत C की त्रिज्या होगी :

(A) 1 (B) 2 (C) 4 (D) 8

Ans. A

Sol. If $k = 4 \quad \Rightarrow \quad r = 1$ Ans.

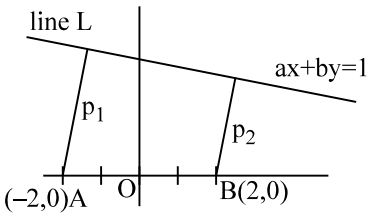
50. If A and B are (–2, 0) and (2, 0) respectively, then the centre of the circle C is :

यदि A (–2, 0) तथा B (2, 0) तो C की त्रिज्या होगी :

(A) (0, 1) (B) (1, 0) (C) (3/2, 0) (D) can not be found

Ans. B

Sol. $p_1 = \left|\frac{-2a-1}{\sqrt{a^2+b^2}}\right|; \qquad p_2 = \left|\frac{2a-1}{\sqrt{a^2+b^2}}\right|$



hence, with the same argument, $2a - 1 < 0$

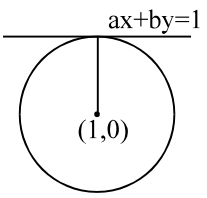
and $-2a - 1 < 0$

$\therefore \quad p_1 + 3p_2 = k$

$$\frac{1+2a}{\sqrt{a^2+b^2}} + \frac{3(1-2a)}{\sqrt{a^2+b^2}} = k$$

$$\frac{4(1-a)}{\sqrt{a^2+b^2}} = k$$

$$\frac{|a-1|}{\sqrt{a^2+b^2}} = \frac{k}{4}$$



hence centre is (1, 0) Ans.

Paragraph for Question Nos. 51 to 53

A circle C of radius 1 is inscribed in an equilateral triangle PQR. The points of contact of C with the sides PQ, QR, RP are D, E, F respectively. The line PQ is given by the equation $\sqrt{3}\,x + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$. Further, it is given that the origin and the centre of C are on the same side of the line PQ.

समबाहु त्रिभुज PQR के अन्दर एक वृत्त स्थित है जिसकी त्रिज्या 1 है। वृत्त C के त्रिभुज की भुजाओं PQ, QR, RP के साथ स्पर्श बिन्दु क्रमशः D, E, F है। रेखा PQ का समीकरण $\sqrt{3}\,x + y - 6 = 0$ तथा बिन्दु D $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$ है। यह भी दिया है कि मूलबिन्दु व C का केन्द्र रेखा PQ के एक तरफ ही स्थित है।

51. The equation of circle C is : **[JEE-2008]**

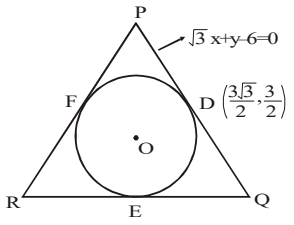
वृत्त C का समीकरण है : **[JEE-2008]**

(A) $(x - 2\sqrt{3})^2 + (y - 1)^2 = 1$ (B) $(x - 2\sqrt{3})^2 + (y + \frac{1}{2})^2 = 1$

(C) $(x - \sqrt{3})^2 + (y + 1)^2 = 1$ (D) $(x - \sqrt{3})^2 + (y - 1)^2 = 1$

Ans. D

Sol. $m_{PQ} = -\sqrt{3}$



so slope of OD $= \frac{1}{\sqrt{3}}$

$\tan\theta = \frac{1}{\sqrt{3}}$

$\therefore \frac{x - \frac{3\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}} = \frac{y - \frac{3}{2}}{\frac{1}{2}} = \pm 1$

$\left(2\sqrt{3}, 2\right)$(not possible) and $\left(\sqrt{3}, 1\right)$

hence circle is $\left(x - \sqrt{3}\right)^2 + (y - 1)^2 = 1$

52. Points E and F are given by : **[JEE-2008]**

बिन्दु E व F होगें : **[JEE-2008]**

(A) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), \left(\sqrt{3}, 0\right)$ (B) $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right), \left(\sqrt{3}, 0\right)$

(C) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

Ans. A

Sol. For point E $\frac{x-\sqrt{3}}{-\frac{\sqrt{3}}{2}} = \frac{y-1}{\frac{1}{2}} = 1 \quad \left[\therefore \ E\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right)\right]$

For point F $\frac{x-\sqrt{3}}{0} = \frac{y-1}{-1} = 1 \qquad \left[\therefore \ F\left(\sqrt{3}, 0\right)\right]$

53. Equations of the sides RP, RQ are : **[JEE-2008]**

भुजाओं RP व RQ के समीकरण होगें : **[JEE-2008]**

(A) $y = \frac{2}{\sqrt{3}}x + 1, \ y = -\frac{2}{\sqrt{3}}x - 1$ (B) $y = \frac{1}{\sqrt{3}}x, \ y = 0$

(C) $y = \frac{\sqrt{3}}{2}x + 1, \ y = -\frac{\sqrt{3}}{2}x - 1$ (D) $y = \sqrt{3}x, \ y = 0$

Ans. D

Sol. Equation of line RP $y = 0$

Equation of line QR $y - \frac{3}{2} = \sqrt{3}\left(x - \frac{\sqrt{3}}{2}\right)$

$y = \sqrt{3}x$

Paragraph for Question Nos. 54 to 56

Let S_1, S_2 and S_3 be the circles $x^2 + y^2 = 4$, $x^2 + y^2 - 6x + 5 = 0$ and $x^2 + y^2 - 4y - 5 = 0$ respectively, then

मानाकि वृत्त S_1, S_2 तथा S_3 क्रमशः $x^2 + y^2 = 4$, $x^2 + y^2 - 6x + 5 = 0$ एवं $x^2 + y^2 - 4y - 5 = 0$ है, तो

54. Point (2, 2) lies :

(A) in the circle S_1 only (B) in the circle S_2 only

(C) in the circle S_3 only (D) in the circles S_1 and S_2 but not in S_3

बिन्दु (2, 2) स्थित है :

(A) केवल वृत्त S_1 के अन्दर
(B) केवल वृत्त S_2 के अन्दर
(C) केवल वृत्त S_3 के अन्दर
(D) वृत्त S_1 तथा S_2 के अन्दर लेकिन S_3 के अन्दर नहीं

Ans. C

Sol. $S_3 < 0$

55. The point (1, 2) lies :

(A) in circles S_2 and S_3 but not in circle S_1
(B) in circles S_2 and S_1 but not in circle S_3
(C) in circle S_3 but not in circles S_1 and S_2
(D) in circles S_1, S_2 and S_3

बिन्दु (1, 2) स्थित है :

(A) वृत्त S_2 तथा S_3 के अन्दर लेकिन S_1 के अन्दर नहीं
(B) वृत्त S_2 तथा S_1 के अन्दर लेकिन S_3 के अन्दर नहीं
(C) वृत्त S_3 के अन्दर लेकिन S_2 तथा S_1 के अन्दर नहीं
(D) वृत्त S_1, S_2 तथा S_3 के अन्दर

Ans. C

Sol. $S_3 < 0$

56. The line $x - y + 2\sqrt{2} = 0$ is :

(A) tangent to S_1 and does not meet S_2 and S_3
(B) tangent to S_1, secant of S_3 and does not meet S_2
(C) tangent to S_1 and secant of S_2 and S_3
(D) tangent to S_3 and secant of S_1 and S_2

रेखा $x - y + 2\sqrt{2} = 0$:

(A) S_1 की स्पर्श रेखा है तथा S_2 एवं S_3 को नहीं मिलती है
(B) S_1 की स्पर्श रेखा है, S_3 की जीवा है तथा S_2 को नहीं मिलती है
(C) S_1 की स्पर्श रेखा है तथा S_2 एवं S_3 की जीवा है
(D) S_3 की स्पर्श रेखा है तथा S_1 एवं S_2 की जीवा है

Ans. B

Sol. For $S_1 \rightarrow P = r$, $S_2 \rightarrow P > r$, $S_3 \rightarrow P < r$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

57. A circle passes through the points $(-1, 1)$, $(0, 6)$ and $(5, 5)$. The point(s) on this circle, the tangent(s) at which is/are parallel to the straight line joining the origin to its centre is/are :

एक वृत बिन्दुओं $(-1, 1)$, $(0, 6)$ तथा $(5, 5)$ से गुजरता है। वृत पर स्थित बिन्दु जहाँ पर स्पर्श रेखा मूल बिन्दु को केन्द्र से जोड़ने वाली रेखा के समानान्तर है :

(A) $(1, -5)$ (B) $(5, 1)$ (C) $(-5, -1)$ (D) $(-1, 5)$

Ans. BD

Sol. Note that Δ is right angled at $(0, 6)$. Centre of the circle is $(2, 3)$. Slope of the line joining origin to the centre is $3/2$. Take parametric equation of a line through $(2, 3)$ with

$$\tan\theta = -\frac{2}{3} \text{ as } \frac{x-2}{\cos\theta} = \frac{y-3}{\sin\theta} = \pm r \text{ where } r = \sqrt{13}.$$

Get the co–ordinates on the circle

58. Three distinct lines are drawn in a plane. Suppose there exist exactly n circles in the plane tangent to all the three lines, then the possible values of n is/are :

किसी तल में तीन भिन्न–भिन्न रेखाऐं है माना कि उस तल में कुल n वृत है जो कि तीनो रेखाओं को स्पर्श करती है तो n का सम्भावित मान होगा :

(A) 0 (B) 1 (C) 2 (D) 4

Ans. ACD

Sol. Case-1: If lines form a triangle then $n = 4$
i.e. 3 excircles and 1 incircle
Case-2: If lines are concurrent
or all 3 parallel then $n = 0$
Case-3: If two are parallel
and third cuts then $n = 2$
hence (A), (C), (D)

59. Equations of circles which pass through the points $(1, -2)$ and $(3, -4)$ and touch the x–axis is :

(A) $x^2 + y^2 + 6x + 2y + 9 = 0$ (B) $x^2 + y^2 + 10x + 20y + 25 = 0$
(C) $x^2 + y^2 - 6x + 4y + 9 = 0$ (D) none

बिन्दुओं $(1, -2)$ एवं $(3, -4)$ से गुजरने वाले एवं x–अक्ष को स्पर्श करने वाले वृत्त का समीकरण है :

(A) $x^2 + y^2 + 6x + 2y + 9 = 0$ (B) $x^2 + y^2 + 10x + 20y + 25 = 0$
(C) $x^2 + y^2 - 6x + 4y + 9 = 0$ (D) इनमें से कोई नहीं

Ans. BC

Sol. Let equation of required circle is
$x^2 + y^2 + 2gx + 2fy + c = 0$
it passes through $(1, -2)$ & $(3, -4)$
$2g - 4f + c = -5$

$6g - 8f + c = -25$

$4g - 8f + 2c = -10$

$6g - 8f + c = -25$

$-2g + c = 15$

circle touches x-axis $g^2 = c \Rightarrow$

$g^2 - 2g - 15 = 0$

$g = 5, -3$

$g = 5,\ c = 25,\ f = 10 \Rightarrow x^2 + y^2 + 10x + 20y + 25 = 0$

$g = -3,\ c = 9,\ f = 2 \Rightarrow x^2 + y^2 - 6x + 4y + 9 = 0$

60. The circle $x^2 + y^2 - 2x - 3ky - 2 = 0$ passes through two fixed points :

वृत्त $x^2 + y^2 - 2x - 3ky - 2 = 0$ दो स्थिर बिन्दुओं से गुजरता है, वे है :

(A) $\left(1+\sqrt{3}, 0\right)$ (B) $\left(-1+\sqrt{3}, 0\right)$ (C) $\left(-\sqrt{3}-1, 0\right)$ (D) $\left(1-\sqrt{3}, 0\right)$

Ans. AD

Sol. Two fixed pts. are point of intersection of

$x^2 + y^2 - 2x - 2 = 0$ & $y = 0$

Point $x^2 - 2x - 2 = 0$

$(x-1)^2 - 3 = 0 \Rightarrow x - 1 = \sqrt{3},\ x - 1 = -\sqrt{3} \Rightarrow \left(1+\sqrt{3}, 0\right)\left(1-\sqrt{3}, 0\right)$

61. The equation of the circle which touches both the axes and the line $\frac{x}{3} + \frac{y}{4} = 1$ and lies in the first quadrant is $(x-c)^2 + (y-c)^2 = c^2$ where c is :

प्रथम चतुर्थांश में दोनों अक्षों तथा सरल रेखा $\frac{x}{3} + \frac{y}{4} = 1$ को स्पर्श करने वाले वृत्त का समीकरण $(x-c)^2 + (y-c)^2 = c^2$ है जहाँ c का मान है :

(A) 1 (B) 2 (C) 4 (D) 6

Ans. AD

Sol. $\left|\frac{4C + 3C - 12}{5}\right| = C \Rightarrow C = 1, 6$

62. Tangents PA and PB are drawn to the circle $S \equiv x^2 + y^2 - 2y - 3 = 0$ from the point P(3, 4). Which of the following alternative(s) is/are correct?

(A) The power of point P(3, 4) with respect to circle $S = 0$ is 14

(B) The angle between tangents from P(3, 4) to the circle S = 0 is $\frac{\pi}{3}$

(C) The equation of circumcircle of ΔPAB is $x^2 + y^2 - 3x - 5y + 4 = 0$

(D) The area of quadrilateral PACB is $3\sqrt{7}$ square units where C is the centre of circle S = 0

वृत $S \equiv x^2 + y^2 - 2y - 3 = 0$ पर बिन्दु P(3, 4) से खीचीं गयी स्पर्श रेखा PA एवं PB है तो सही विकल्प चुनियें?

(A) S = 0 का बिन्दु P(3, 4) के सापेक्ष बिन्दु क्षमता 14 है

(B) PA एवं PB के बीच कोण $\frac{\pi}{3}$ है

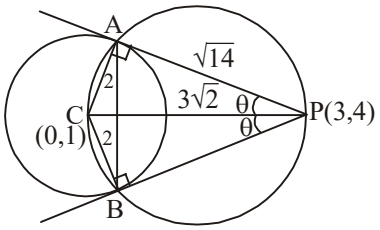
(C) ΔPAB का परिवृत $x^2 + y^2 - 3x - 5y + 4 = 0$ है

(D) चतुर्भुज PACB का क्षेत्रफल $3\sqrt{7}$ है जहाँ C वृत का केन्द्र है

Ans. AC

Sol. Clearly power of point P(3, 4) w.r.t. S = 0 is

$S_1 = 14 \Rightarrow$ option (A) is correct



Also $\sin\theta = \frac{2}{3\sqrt{2}} = \frac{\sqrt{2}}{3}$

Now angle between tangents = $2\sin^{-1}\frac{\sqrt{2}}{3}$

Also CP is diameter of circumcircle of ΔPAB, so equation of circumcircle is

$x(x-3) + (y-1)(y-4) = 0 \quad \Rightarrow \quad x^2 + y^2 - 3x - 5x + 4 = 0 \Rightarrow$ option (C) is correct

Now, area of quadrilateral PACB = $2 \times \frac{1}{2} \times 2 \times \sqrt{14} = 2\sqrt{14}$ square units. $\Rightarrow$ option (D) is incorrect

63. If the equation of circle touching the y-axis at (0, 3) and making an intercept of 8 unit on x-axis is $x^2 + y^2 + 2gx + 2fy + c = 0$, then (g + f + c) can be :

यदि वृत $x^2 + y^2 + 2gx + 2fy + c = 0$ y–अक्ष को (0, 3) पर स्पर्श कर रही है तथा x–अक्ष पर 8 लम्बाई का अन्तःखण्ड काट रही है तो (g + f + c) होगा :

(A) 1 (B) 7 (C) 11 (D) 14

Ans. AC

Sol. The circle is $x^2 + (y-3)^2 + \lambda x = 0$

putting y = 0, we get $x^2 + \lambda x + 9 = 0$

x-intercept is $|x_1 - x_2|$

$|x_1 - x_2|^2 = (x_1 + x_2)^2 - 4x_1x_2$

$64 = \lambda^2 - 36 \quad \Rightarrow \quad \lambda^2 = 10 \Rightarrow \quad \lambda = 10 \text{ or } -10$

Solving $\lambda = \pm 10$

The circles are $x^2 + y^2 \pm 10x - 6y + 9 = 0$

$\Rightarrow \quad g = \pm 5;\ f = -3;\ c = 9 \quad \Rightarrow \quad (g + f + c) = 11$ and 1 Ans.

64. The circles $x^2 + y^2 - 2x - 4y + 1 = 0$ and $x^2 + y^2 + 4x + 4y - 1 = 0$

(A) touch internally

(B) touch externally

(C) have $3x + 4y - 1 = 0$ as the common tangent at the point of contact

(D) have $3x + 4y + 1 = 0$ as the common tangent at the point of contact

वृत्तों $x^2 + y^2 - 2x - 4y + 1 = 0$ तथा $x^2 + y^2 + 4x + 4y - 1 = 0$ के लिये सत्य होगा :

(A) आन्तरिक स्पर्श करते है

(B) बाह्य स्पर्श करते है

(C) इनकी उभयनिष्ठ स्पर्शरेखा $3x + 4y - 1 = 0$ है

(D) इनकी उभयनिष्ठ स्पर्शरेखा $3x + 4y + 1 = 0$ है

Ans. BC

Sol. Correct options are (B) and (C).

For given circles S1 : $x^2 + y^2 - 2x - 4y + 1 = 0$

and $S_2 : x^2 + y^2 + 4x + 4y - 1 = 0$

$C_1 (1,2)$, $r_1 = 2$ and $C_2 (-2,-2)$, $r_2 = 3$

Now, $r_1 + r_2 = 5$ and $C_1C_2 = 5$

Hence, circles touch externally.

Also common tangent at point of contact is

$S_1 - S_2 = 0$ or $3x + 4y - 1 = 0$

65. Which of the following is/are True? The circles $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are such that:

(A) they do not intersect

(B) they touch each other.

(C) their exterior common tangents are parallel

(D) their interior common tangents are perpendicular

वृत्त $x^2 + y^2 - 6x - 6y + 9 = 0$ तथा $x^2 + y^2 + 6x + 6y + 9 = 0$ के लिये कौनसा विकल्प सत्य है :

(A) परस्पर काटते नहीं

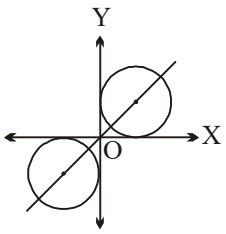
(B) परस्पर स्पर्श करते है

(C) बाह्य स्पर्श रेखायें परस्पर समान्तर है

(D) आन्तरिक स्पर्श रेखायें परस्पर लम्बवत् है

Ans. ACD

Sol. Centre (3, 3) with radius = 3 and centre (–3, 3) with radius = 3.



66. Consider the circles C_1: $x^2 + y^2 = 16$ and C_2 : $x^2 + y^2 - 12x + 32 = 0$. Which of the following statements is/are correct?

(A) Number of common tangent to these circles is 3.

(B) The point P with coordinates (4, 1) lies outside the circle C_1 and inside the circle C_2.

(C) Their direct common tangent intersect at (12, 0).

(D) Slope of their radical axis is not defined.

यदि C_1: $x^2 + y^2 = 16$ तथा C_2 : $x^2 + y^2 - 12x + 32 = 0$ तो इनमें से कौनसा विकल्प सत्य है?

(A) कुल उभयनिष्ठ स्पर्शरेखा की संख्या 3 होगी

(B) P (4, 1) वृत C_1 के बाहर तथा C_2 के अन्दर होगा

(C) उनकी उभयनिष्ठ बाह्य स्पर्श रेखा (12, 0) पर काटेगी

(D) इनके मूलाक्ष की प्रवणता अपरिभाषित होगी

Ans. ACD

Sol. C_1 : (0, 0), radius 4

C_2: (6, 0), radius 2

circle touch externally.

67. If a circle passes through P(0, 1), Q(0, 9) and touches the x – axis, then which of the following statement(s) is/are TRUE?

(A) Centres of circles are $(\pm 3, \pm 5)$.

(B) Equation of one of their direct common tangent is $y = 10$.

(C) Radii of both the circles is 5.

(D) Length of common chord of circles is 8.

बिन्दुओं P(0, 1), Q(0, 9) से जाने वाला तथा x – अक्ष को स्पर्श करने वाले वृत के लिये निम्न से कौनसा कथन सत्य है?

(A) केन्द्र $(\pm 3, \pm 5)$ होगा

(B) उनकी बाह्य उभयनिष्ठ स्पर्शरेखा में से एक रेखा $y = 10$ होगी

(C) दोनों वृत की त्रिज्या 5 हो

(D) उभयनिष्ठ जीवा की लम्बाई 8 हो

Ans. BCD

Sol. Equation of any circle through P(0, 1) and Q(0, 9) is $x^2 + (y - 1)(y - 9) + \lambda x = 0$

$\Rightarrow x^2 + y^2 - 10y + 9 + \lambda x = 0$

As above circle touches x - axis, so put $y = 0$

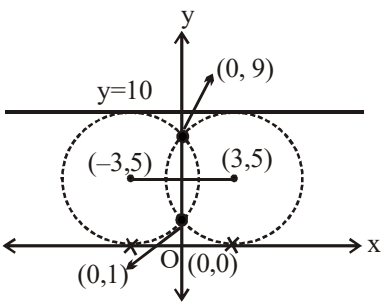
$\Rightarrow x^2 + \lambda x + 9 = 0$

should have equal roots. So, Discriminant = 0

$\Rightarrow \lambda^2 = 36 \quad \therefore \lambda = \pm 6$

Hence required circles are $x^2 + y^2 + 6x - 10y + 9 = 0$

and $x^2 + y^2 - 6x - 10y + 9 = 0$



68. A family of linear functions is given by $f(x) = 1 + c(x + 3)$ where $c \in R$. If a member of this family meets a unit circle centred at origin in two coincident points then 'c' can be equal to :

एक रेखीय फलनों का समूह $f(x) = 1 + c(x + 3)$ द्वारा दिया जाता है। यदि समूह का एक सदस्य इकाई त्रिज्या तथा मूल बिन्दु केन्द्र वाले वृत को दो सम्पाती बिन्दुओं पर मिलता है तो 'c' बराबर होगा :

(A) – 3/4 (B) 0 (C) 3/4 (D) 1

Ans. AB

Sol. Line: $y = 1 + c(x + 3)$ is tangent to circle $x^2 + y^2 = 1$

$\therefore \quad p = r$

$\therefore \quad \left|\frac{1+3c}{\sqrt{1+c^2}}\right| = 1$

$(1 + 3c)^2 = 1 + c^2$

$9c^2 + 6c = c^2$

$8c^2 = -6c$

$\therefore \quad c = 0 \quad \text{or} \quad c = -3/4 \quad \Rightarrow$ (A), (B)

69. Which of the following lines have the intercepts of equal lengths on the circle, $x^2 + y^2 - 2x + 4y = 0$?

निम्न में से कौनसी रेखायें वृत $x^2 + y^2 - 2x + 4y = 0$ पर बराबर लम्बाई के अन्तः कर्ण काटती है?

(A) $3x - y = 0$ (B) $x + 3y = 0$ (C) $x + 3y + 10 = 0$ (D) $3x - y - 10 = 0$

Ans. ABCD

Sol. Chords equidistance from the centre are equal

70. The equations of the tangents drawn from the origin to the circle, $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are :

वृत $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ पर मूल बिन्दु से डाली गई स्पर्श रेखाऐं है :

(A) $x = 0$ (B) $y = 0$

(C) $(h^2 - r^2)x - 2rhy = 0$ (D) $(h^2 - r^2)x + 2rhy = 0$

Ans. AC

Sol. $x^2 + y^2 - 2rx$ " $2hy + h^2 = 0$

$x^2 + y^2 - 2rx - 2hy + h^2 + r^2 = r^2$

$(x - r)^2 + (y - h)^2 = r^2$

(r, h) is centre, r is radius

It is touching $x = 0$ because radius Δx.

coordinate of centre are equal

$x = 0$ is a tangent

Let $y = mx$ be another tangent $[y - mx = 0]$

1 distance from (r,h) = r.

$$\left|\frac{r - mh}{\sqrt{1+m^2}}\right| = r \qquad m = 0, \qquad \frac{h^2 - r^2}{2rh}$$

Another tangent will be $(h^2 - r^2)x - 2rhy = 0$

B is correct.

MATCH THE COLUMN TYPE QUESTION(S)

71. Match the following column :

Column – I	**Column – II**
(A) Number of values of a for which the common chord of the circles $x^2 + y^2 = 8$ and $(x - a)^2 + y^2 = 8$ subtends a right angle at the origin is	(P) 4
(B) A chord of the circle $(x - 1)^2 + y^2 = 4$ lies along the line $y = 22\sqrt{3}(x - 1)$. The length of the chord is equal to	(Q) 2
(C) The number of circles touching all the three lines $3x + 7y = 2$, $21x + 49y = 5$ and $9x + 21y = 0$ are	(R) 0
(D) If radii of the smallest and largest circle passing through the point $\left(\sqrt{3}, \sqrt{2}\right)$ and touching the circle $x^2 + y^2 - 2\sqrt{2}y - 2 = 0$ are r_1 and r_2 respectively, then the mean of r_1 and r_2 is	(S) 1

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) a के उन मानों की संख्या जिनके लिए वृत्तों $x^2 + y^2 = 8$ तथा $(x - a)^2 + y^2 = 8$ की उभयनिष्ट जीवा मूल बिन्दु पर समकोण अन्तरित करती है	(P) 4
(B) यदि वृत्त $(x - 1)^2 + y^2 = 4$ की एक जीवा रेखा $y = 22\sqrt{3}(x - 1)$ के अनुदिश हो, तो जीवा की लम्बाई है	(Q) 2

(C) सभी तीन रेखाओं $3x + 7y = 2$, $21x + 49y = 5$ तथा $9x + 21y = 0$ को स्पर्श करने वाले वृत्तों की संख्या है (R) 0

(D) यदि बिन्दु $\left(\sqrt{3}, \sqrt{2}\right)$ से गुजरने वाले तथा वृत्त $x^2 + y^2 - 2\sqrt{2}y - 2 = 0$ को स्पर्श करने वाले सबसे छोटे तथा सबसे बड़े वृत्त की त्रिज्याएँ क्रमश : r_1 तथा r_2 हो, तो r_1 तथा r_2 का माध्य है (S) 1

Ans. (A) → Q; (B) → P; (C) → R; (D) → S

Sol. (A) $S_1 - S_2 = 0$ is the required common chord i.e $2x = a$

Make homogeneous, we get $x^2 + y^2 - 8.4\frac{x^2}{a^2} = 0$

As pair of lines substending angle of 90° at origin

$\therefore$ coefficient of x^2 + coefficient of $y^2 = 0$

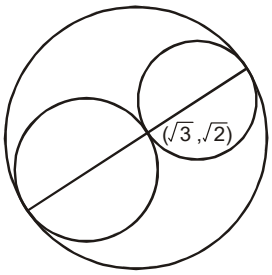
$\therefore$ $a = \pm 4$

(B) $y = 22\sqrt{3}(x - 1)$ passes through centre $(1, 0)$ of circle

(C) Three lines are parallel so not any circle is possible

(D) $2(r_1 + r_2) = 4$

$r_1 + r_2 = 2 \Rightarrow \frac{r_1 + r_2}{2} = 1$



72. Match the following column :

Column – I	**Column – II**
(A) Number of common tangents of the circles $x^2 + y^2 - 2x = 0$ and $x^2 + y^2 + 6x - 6y + 2 = 0$ is	(P) 1
(B) Number of transverse common tangents of the circles $x^2 + y^2 - 4x - 10y + 4 = 0$ & $x^2 + y^2 - 6x - 12y - 55 = 0$ is	(Q) 2
(C) Number of common tangents of the circles $x^2 + y^2 - 2x - 4y = 0$ & $x^2 + y^2 - 8y - 4 = 0$ is	(R) 3
(D) Number of direct common tangents of the circles $x^2 + y^2 + 2x - 8y + 13 = 0$ & $x^2 + y^2 - 6x - 2y + 6 = 0$ is	(S) 0

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) वृत्तो $x^2 + y^2 - 2x = 0$ तथा $x^2 + y^2 + 6x - 6y + 2 = 0$ की उभयनिष्ठ स्पर्श रेखाओं की संख्या है	(P) 1
(B) $x^2 + y^2 - 4x - 10y + 4 = 0$ तथा $x^2 + y^2 - 6x - 12y - 55 = 0$ तिर्यक उभयनिष्ठ स्पर्श रेखाओं की संख्या है	(Q) 2
(C) $x^2 + y^2 - 2x - 4y = 0$ तथा $x^2 + y^2 - 8y - 4 = 0$ की उभयनिष्ठ स्पर्श रेखाओं की संख्या है	(R) 3
(D) $x^2 + y^2 + 2x - 8y + 13 = 0$ तथा $x^2 + y^2 - 6x - 2y + 6 = 0$ की अनुक्रम उभयनिष्ठ स्पर्श रेखाओं की संख्या है	(S) 0

Ans. (A) $\rightarrow$ R; (B) $\rightarrow$ S; (C) $\rightarrow$ P; (D) $\rightarrow$ Q

Sol. (A) $C_1(1, 0)$, $r_1 = 1$ and $C_2(-3, 3)$, $r_2 = 4$

distance between centres C_1 and $C_2 = d = 5$

$d = r_1 + r_2 = 5 \quad \Rightarrow$ 3 common tangents

(B) $C_1(2, 5)$, $r_1 = 5$ and $C_2(3, 6)$, $r_2 = 10$

distance between centres C_1 and $C_2 =$

$d = \sqrt{2}$

$d < |r_1 - r_2|$

$\Rightarrow$ no common tangent

(C) $C_1(1, 2)$, $r_1 = \sqrt{5}$ and $C_2(0, 4)$, $r_2 = 2\sqrt{5}$

distance between centres C_1 and $C_2 = d = \sqrt{5}$

$|r_1 - r_2| = d$

number of common tangents is 1

(D) $C_1(-1, 4)$, $r_1 = 2$ and $C_2(3, 1)$, $r_2 = 2$

distance between centres C_1 and $C_2 = d = 5$

$d > r_1 + r_2$

$\Rightarrow$ number of direct common tangents is 2

Exercise J-M

1. The locus of the mid points of the chords of the circle $C_1 : (x-4)^2 + (y-5)^2 = 4$ which subtend an angle θ_i at the centre of the circle C_i, is a circle of radius r_i. If $\theta_1 = \frac{\pi}{3}$, $\theta_3 = \frac{2\pi}{3}$ and $r_1^2 = r_2^2 + r_3^2$, the θ_2 is equal to : **[JEE(Main)Jan.2023]**

वृत्त $C_1 : (x-4)^2 + (y-5)^2 = 4$ की उन जीवाओं, जो वृत्त C_i (जो r_i त्रिज्या का वृत्त है) के केन्द्र पर θ_i कोण बनाती है, के मध्य बिंदुओं का बिन्दुपथ होगा। यदि $\theta_1 = \frac{\pi}{3}$, $\theta_3 = \frac{2\pi}{3}$ और $r_1^2 = r_2^2 + r_3^2$ है, तब θ_2 बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. D

Sol. If a chord of circle of radius R subtends angle θ_i at the centre then locus of the midpoint of this chord is a circle of radius $r_i = R \cdot \cos\left(\frac{\theta_i}{2}\right)$

Given

$r_1^2 = r_2^2 + r_3^2$

$\Rightarrow \cos^2\frac{\theta_1}{2} = \cos^2\frac{\theta_2}{2} + \cos^2\frac{\theta_2}{2}$

$\Rightarrow \cos^2\frac{\pi}{6} = \cos^2\frac{\pi}{3} + \cos^2\frac{\theta_2}{2}$

$\Rightarrow \frac{3}{4} = \frac{1}{4} + \cos^2\frac{\theta_2}{2}$

$\Rightarrow \cos^2\frac{\theta_2}{2} = \frac{1}{2}$

$\Rightarrow \cos^{\frac{\theta_2}{2}} = \frac{1}{\sqrt{2}}$

$\Rightarrow \frac{\theta_2}{2} = \frac{\pi}{4}$

$\therefore \theta_3 = \frac{\pi}{2}$

2. The points of intersection of the line $ax + by = 0$, $(a \neq b)$ and the circle $x^2 + y^2 - 2x = 0$ are $A(\alpha, 0)$ and $B(1, \beta)$. The image of the circle with AB as a diameter in the line $x + y + 2 = 0$ is : **[JEE(Main)Jan.2023]**

रेखा $ax + by = 0$, $(a \neq b)$ तथा वृत्त $x^2 + y^2 - 2x = 0$ के प्रतिच्छेदन बिंदु $A(\alpha, 0)$ तथा $B(1, \beta)$ है। वृत्त, जिसका एक व्यास AB

है, का रेखा $x + y + 2 = 0$ में प्रतिबिंब है : **[JEE(Main)Jan.2023]**

(A) $x^2 + y^2 + 3x + 3y + 4 = 0$ (B) $x^2 + y^2 + 3x + 5y + 8 = 0$

(C) $x^2 + y^2 - 5x - 5y + 12 = 0$ (D) $x^2 + y^2 + 5x + 5y + 12 = 0$

Ans. D

Sol. If the points of intersection of the line $ax + by = 0$, $(a \neq b)$ and the circle $x^2 + y^2 - 2x = 0$ are $A(\alpha, 0)$ and

$$B(1,\beta) \quad \Rightarrow \quad \alpha = 0 \quad \beta = \frac{-a}{b} \text{ and } \beta^2 = 1$$

$$\Rightarrow a^2 = b^2 \,(a \neq b) \Rightarrow a + b = 0$$

$$\Rightarrow \beta = 1$$

So centre of circle with diameter AB is $\left(\frac{1}{2}\frac{1}{2}\right)$ and radius = $\frac{1}{\sqrt{2}}$, where A(0, 0) B (1, 1)

Now image of $\left(\frac{1}{2}\frac{1}{2}\right)$ about the line $x + y + 2 = 0$

$$\frac{x - \frac{1}{2}}{1} = \frac{y - \frac{1}{2}}{1} = -2\left(\frac{3}{2}\right)$$

$$\Rightarrow \text{Image } \left(-\frac{5}{2}, -\frac{5}{2}\right)$$

equation of Image circle

$$\left(x + \frac{5}{2}\right)^2 + \left(y + \frac{5}{2}\right)^2 = \frac{1}{2}$$

3. Let the tangents at the point $A(4, -11)$ and $B(8, -5)$ on the circle $x^2 + y^2 - 3x + 10y - 15 = 0$, intersect at the point C. Then the radius of the circle,w hose centre is C and the line joining A and B is its tangent, is equal to : **[JEE(Main)Jan.2023]**

माना वृत्त $x^2 + y^2 - 3x + 10y - 15 = 0$ के बिंदु $A(4, -11)$ व $B(8, -5)$ पर खींची गई स्पर्श रेखाएँ बिंदु C पर मिलती है। उस वृत्त, जिसका केन्द्र C है एवं A व B को मिलाने वाली रेखा जिसकी स्पर्श रेखा है, की त्रिज्या है : **[JEE(Main)Jan.2023]**

(A) $\frac{2\sqrt{13}}{3}$ (B) $2\sqrt{13}$ (C) $\frac{3\sqrt{3}}{4}$ (D) $\sqrt{13}$

Ans. A

Sol. Equation of line AB is $y + 5 = \frac{3}{2}(x - 8)$

$2y + 10 = 3x - 24$

$\Rightarrow 3x - 2y - 34 = 0$ ....(1)

Let C be (h, k) then equation of AB

$$hx + ky - \frac{3}{2}(x + h) + 5(y + k) - 15 = 0$$

$$x\left(h - \frac{3}{2}\right) + y(k + 5) - \frac{3}{2}h + 5k - 15 = 0 \qquad ...(2)$$

comparing (1) and (2)

$$\frac{h - \frac{3}{2}}{3} = \frac{k + 5}{-2} = \frac{\frac{-3}{2}h + 5k - 15}{-34}$$

$$(h, k) \equiv \left(8, \frac{-28}{3}\right)$$

$\therefore$ required radius = length of perpendicular drawn for (h, k) to line AB = $\frac{2\sqrt{13}}{3}$

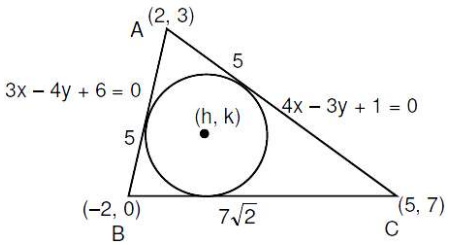
4. Let $y = x + 2$, $4y = 3x + 6$ and $3y = 4x + 1$ be three tangent lines to the circle $(x - h)^2 + (y - k)^2 = r^2$. Then $h + k$ is equal to : **[JEE(Main)Jan.2023]**

माना $y = x + 2$, $4y = 3x + 6$ तथा $3y = 4x + 1$ वृत्त $(x - h)^2 + (y - k)^2 = r^2$ की तीन स्पर्श रेखाऐं है। तो $h + k$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 5 (B) $5\sqrt{2}$ (C) 6 (D) $5(1 + \sqrt{2})$

Ans. A

Sol.



$x - y + 2 = 0$

ΔABC is isosceles (AB = AC) so angle bisector & altitude are same then

Equation of angle bisector of angle A is equal to altitude from vertex A

$x + y = \lambda$ (Passes through A)

$2 + 3 = \lambda \Rightarrow \lambda = 5$

$x + y = 5$ (Passes through center)

$h + k = 5$

5. Let a circle C_1 be obtained on rolling the circle $x^2 + y^2 - 4x - 6y + 11 = 0$ upwards 4 units on the tangent T to it at the point (3,2). Let C_2 be the image of C_1 in T. Let A and B be the centres of circles C_1 and C_2 respectively, and M and N be respectively the feet of perpendiculars drawn from A and B on the x-axis. Then the area of the trapezium AMNB is : **[JEE(Main)Jan.2023]**

माना वृत्त $x^2 + y^2 - 4x - 6y + 11 = 0$ को इसके बिंदु (3,2) पर स्पर्श रेखा T पर ऊपर की ओर 4 इकाई तक घुमाने पर वृत्त C_1 प्राप्त होता है। माना T में C_1 का प्रतिबिंब C_2 है। माना वृत्तों C_1 तथा C_2 के केन्द्र क्रमशः A तथा B है, और A तथा B से x-अक्ष पर डाले गए लंबों के पाद क्रमशः M तथा N है। तो समलंब AMNB का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

(A) $2\left(2+\sqrt{2}\right)$ (B) $3+2\sqrt{2}$ (C) $4\left(1+\sqrt{2}\right)$ (D) $2\left(1+\sqrt{2}\right)$

Ans. C

Sol. $C = (2, 3),\ r = \sqrt{2}$

Centre of $G = A = 2 + 4\dfrac{1}{\sqrt{2}}$,

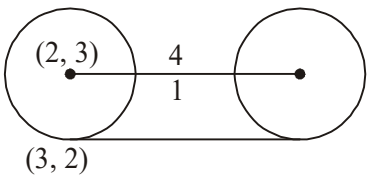
$3+\dfrac{4}{\sqrt{2}} = \left(2+\sqrt{2}, 3+2\sqrt{2}\right)$

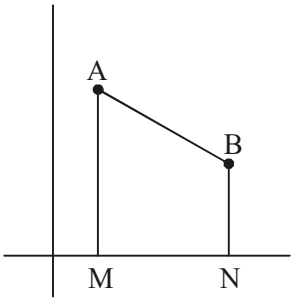
$A\left(2+2\sqrt{2}, 3+2\sqrt{2}\right)$

$B\left(4+2\sqrt{2}, 1+2\sqrt{2}\right)$

$\dfrac{x-\left(2+2\sqrt{2}\right)}{1} = \dfrac{y-\left(3+2\sqrt{2}\right)}{-1} = 2$

$\therefore$ area of trapezim : $\dfrac{1}{2}\left(4+4\sqrt{2}\right)2 = 4\left(1+\sqrt{2}\right)$





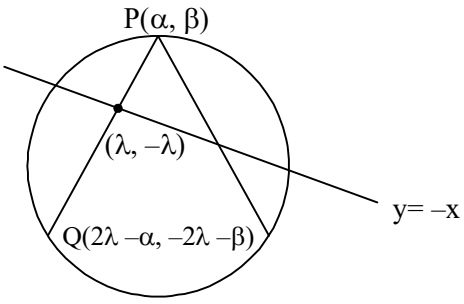
6. The set of all values of a^2 for which the line $x+y=0$ bisects two distinct chords drawn from a point $P\left(\frac{1+a}{2},\frac{1-a}{2}\right)$ on the circle $2x^2+2y^2-(1+a)x-(1-a)y=0$, is equal to : **[JEE(Main)Jan.2023]**

a^2 के सभी मानों, जिनके लिए रेखा $x+y=0$, वृत्त $2x^2+2y^2-(1+a)x-(1-a)y=0$ के बिंदु $P\left(\frac{1+a}{2},\frac{1-a}{2}\right)$ से खींची गई दो भिन्न जीवाओं को समद्विभाजित करती है, का समुच्चय बराबर है : **[JEE(Main)Jan.2023]**

(A) (0,4] (B) (8,∞) (C) (4,∞) (D) (2,12]

Ans. B

Sol.



$$x^2+y^2-\left(\frac{1+a}{2}\right)x-\left(\frac{1-a}{2}\right)y=0$$

Let $\frac{1+a}{2}=\alpha$, $\frac{1-a}{2}=\beta$

circle will be $x^2+y^2-\alpha x-\beta y=0$

Point Q will lie on circle

$$(2\lambda-\alpha)^2+(-2\lambda-\beta)^2-\alpha(2\lambda-\alpha)-\beta(-2\lambda-\beta)=0$$

$$\Rightarrow 4\lambda^2+3\lambda(\beta-\alpha)+(\alpha^2+\beta^2)=0$$

Now $D>0 \Rightarrow 9(\beta-\alpha)^2-4(4)(\alpha^2+\beta^2)>0$

$$\Rightarrow 9a^2 - 16\left(\frac{1+a^2}{2}\right) > 0$$

$$\Rightarrow a^2 > 8$$

7. Ponts P(–3, 2), Q(9,10) and R(α, 4) lie on a circle C with PR as its diameter. The tangents to C at the points Q and R intersect at the point. If S lies on the line $2x - ky = 1$, then k is equal to _____.

[JEE(Main)Jan.2023]

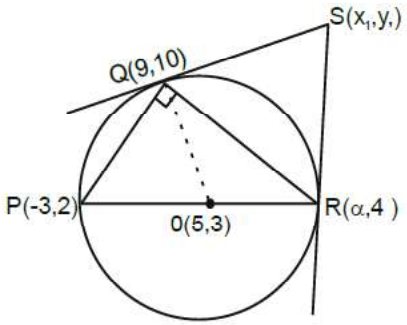
बिंदु P(–3, 2), Q(9, 10) तथा R(α, 4) एक वृत्त पर है, जिसका एक व्यास PR है। बिंदुओं Q तथा R पर वृत्त C की स्पर्श रेखाएँ बिंदु S पर मिलती है। यदि बिंदु S रेखा $2x - ky = 1$ पर है, तो k बराबर है। **[JEE(Main)Jan.2023]**

Ans. 3

Sol. $m_{PQ} \cdot m_{QR} = -1$

$$\Rightarrow \frac{10-2}{9+3} \times \frac{10-4}{9-\alpha} = -1 \Rightarrow \alpha = 13$$

$$m_{0P} \cdot m_{QS} = -1 \Rightarrow m_{QS} = -\frac{4}{7}$$



Equation of QS

$$y - 10 = -\frac{4}{7}(x - 9)$$

$\Rightarrow 4x + 7y = 106$ .....(1)

$m_{0R} - m_{RS} = -1 \Rightarrow m_{RS} = -8$

Equation of RS

$y - 4 = -8(x - 13)$

$\Rightarrow 8x + y = 108$ .....(2)

Solving eq. (1) & (2)

$$x_1 = \frac{25}{2} \quad y_1 = 8$$

$S(x_1,y_1)$ lies on $2x - ky = 1$

$25 - 8k = 1$

$\Rightarrow 8k = 24$

$\Rightarrow k = 3$

8. A circle with centre (2, 3) and radius 4 intersects the line $x + y = 3$ at the points P and Q. If the tangents at P and Q intersect at the point S(α, β), then $4\alpha - 7\beta$ is equal to _____. **[JEE(Main)Jan.2023]**

एक वृत्त, जिसका केन्द्र (2, 3) है तथा त्रिज्या 4 है, रेखा $x + y = 3$ को बिंदुओं P तथा Q पर काटता है। यदि P तथा Q पर स्पर्श रेखाएँ बिंदु S(α, β) पर मिलती है, तो $4\alpha - 7\beta$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 11

Sol. The given line is polar or P(2, β) w.r.t. given circle

$x^2 + y^2 - 4x - 6y - 3 = 0$

Chord or contact

$\alpha x + \beta y - 2(x + \alpha) - 3(y + \beta) - 3 = 0$

$\Rightarrow (\alpha - 2)x + (\beta - 3)y - (2\alpha + 3\beta + 3) = 0$ ...(i)

$\because$ But the equation of chord of contact is given

as : $x + y - 3 = 0$ ...(ii)

comparing the coefficients

$$\frac{\alpha - 2}{1} = \frac{\beta - 3}{1} = -\left(\frac{2\alpha + 3\beta + 3}{-3}\right)$$

On solving $\alpha = -6$

$\beta = -5$

Now $4\alpha - 7\beta = 11$

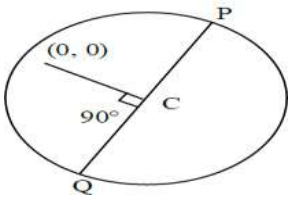
9. Let $P(a_1, b_1)$ and $Q(a_2, b_2)$ be two distinct points on a circle with center $C\left(\sqrt{2}, \sqrt{3}\right)$. Let O be the origin and OC be perpendicular to both CP and CQ. If the area of the triangle OCP is $\frac{\sqrt{35}}{2}$, then $a_1^2 + b_1^2 + a_2^2 + b_2^2$ is equal to _____. **[JEE(Main)Jan.2023]**

माना वृत्त $C\left(\sqrt{2}, \sqrt{3}\right)$ केन्द्र के एक वृत्त पर दो भिन्न बिंदु $P(a_1, b_1)$ एवं $Q(a_2, b_2)$ है। माना O मूल बिंदु है तथा OC, रेखाओं CP एवं CQ दोनों पर लम्बवत् है। यदि त्रिभुज OCP का क्षेत्रफल $\frac{\sqrt{35}}{2}$ है, तो $a_1^2 + b_1^2 + a_2^2 + b_2^2$ बराबर है।

[JEE(Main)Jan.2023]

Ans. 24

Sol. $\frac{1}{2} \times PC \times \sqrt{5} = \frac{\sqrt{35}}{2}; PC = \sqrt{7}$



$a_1^2 + b_1^2 + a_2^2 + b_2^2 = OP^2 + OQ^2$

$= 2\,(5 + 7) = 24$

10. If the tangents at the points P and Q on the circle $x^2 + y^2 - 2x + y = 5$ meet at the point $R\left(\frac{9}{4}, 2\right)$, then the area of the triangle PQR is : **[JEE(Main)April,2023]**

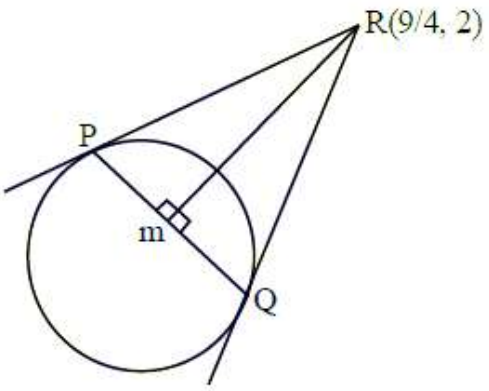
यदि वृत $x^2 + y^2 - 2x + y = 5$ के बिंदुओं P तथा Q पर स्पर्श रेखाएँ बिंदु $R\left(\frac{9}{4}, 2\right)$ पर मिलती है, तो त्रिभुज PQR का क्षेत्रफल है: **[JEE(Main)April,2023]**

(A) $\frac{5}{4}$ (B) $\frac{5}{8}$ (C) $\frac{13}{8}$ (D) $\frac{13}{4}$

Ans. B

Sol. Equation of circle is $x^2 + y^2 - 2x + y - 5 = 0$

$R = \frac{5}{2}$



Length of $PR = QR = \sqrt{S_1}$

$= \sqrt{\frac{81}{16} + 4 - \frac{2\times 9}{4} + 2 - 5} = \frac{5}{4}$

Area of triangle $PQR = \frac{RL^3}{R^2 + L^2} = \frac{\frac{5}{2}\cdot\frac{125}{64}}{\frac{25}{4} + \frac{25}{16}} = \frac{5}{8}$

11. Let O be the origin and OP and OQ be the tangents to the circle $x^2 + y^2 - 6x + 4y + 8 = 0$ at the points P and Q on it. If the circumcircle of the triangle OPQ passes through the point $\left(\alpha, \frac{1}{2}\right)$, then a value of α is :

[JEE(Main)April,2023]

माना O मूल बिंदु है तथा OP और OQ वृत्त $x^2 + y^2 - 6x + 4y + 8 = 0$ के बिंदुओं P तथा Q पर स्पर्श रेखाऐं है। यदि त्रिभुज OPQ का परिवृत्त, बिंदु $\left(\alpha, \frac{1}{2}\right)$ से होकर जाता है, तो α का एक मान है : **[JEE(Main)April,2023]**

(A) $-\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) 1

Ans. C

Sol. Equation of circumcircle whose diametric end point is (3, –2) and (0, 0)

$x(x-3) + y(y+2) = 0$

$x^2 + y^2 - 3x + 2y = 0$

pt $\left(\alpha, \frac{1}{2}\right)$ is on circle

$$\alpha^2 + \frac{1}{4} - 3\alpha + 1 = 0$$

$$\alpha^2 - 3\alpha + \frac{5}{4} = 0$$

$$4\alpha^2 - 12\alpha + 5 = 0$$

$$\alpha = \frac{10}{4}, \alpha = \frac{2}{4}$$

$$\alpha = \frac{5}{2}, \frac{1}{2}$$

12. A line segment AB of length λ moves such that the points A and B remain on the periphery of a circle of radius λ. Then the locus of the point, that divides the line segment AB in the ratio 2 : 3, is a circle of radius :

[JEE(Main)April,2023]

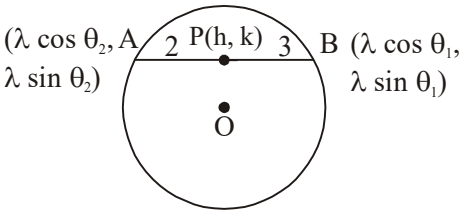
λ लंबाई का एक रेखा खंड AB इस प्रकार खिसकता है कि बिंदु A तथा B, λ त्रिज्या के एक वृत्त की परिधि पर रहते हैं। तो उस बिंदु, जो रेखाखंड AB को 2 : 3 के अनुपात में विभाजित करता है, का बिंदुपथ एक वृत्त है, जिसकी त्रिज्या है :

[JEE(Main)April,2023]

(A) $\frac{3}{5}\lambda$ (B) $\frac{\sqrt{19}}{5}\lambda$ (C) $\frac{2}{3}\lambda$ (D) $\frac{\sqrt{19}}{7}\lambda$

Ans. B

Sol.



Let O be the origin and radius of circle is λ and AB = λ

$\Rightarrow \quad 1 = 2 - 2\cos(\theta_1 - \theta_2)$

$\Rightarrow \quad \cos(\theta_1 - \theta_2) = \frac{1}{2}$

$\therefore \quad h = \frac{2\lambda\cos\theta_1 + 3\lambda\cos\theta_2}{5}$

$k = \frac{2\lambda\sin\theta_1 + 3\lambda\sin\theta_2}{5}$

$\therefore \quad h^2 + k^2 = \frac{\lambda^2}{25}\left[4 + 9 + 12\left(\cos(\theta_1 - \theta_2)\right)\right]$

$= \frac{\lambda^2}{25}\cdot 19$

$\therefore \quad$ Radius $= \frac{\lambda}{5}\sqrt{19}$

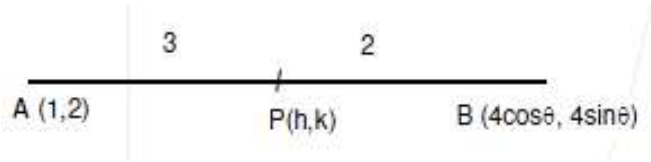
13. Let A be the point (1, 2) and B be any point on the curve $x^2 + y^2 = 16$. If the centre of the locus of the point P, which divides the line segment AB in the ratio 3 : 2 is the point C(α, β), then the length of the line segment AC is : **[JEE(Main)April,2023]**

माना A बिंदु (1, 2) है तथा वक्र $x^2 + y^2 = 16$ पर कोई बिंदु B है। यदि रेखा खंड AB को 3 : 2 के अनुपात में विभाजित करने वाले बिंदु P के बिंदुपथ का केन्द्र C(α, β) है, तो रेखाखंड AC की लंबाई है : **[JEE(Main)April,2023]**

(A) $\frac{4\sqrt{5}}{5}$ (B) $\frac{3\sqrt{5}}{5}$ (C) $\frac{6\sqrt{5}}{5}$ (D) $\frac{2\sqrt{5}}{5}$

Ans. B

Sol.



$h = \frac{12\cos\theta + 2}{5}, \quad k = \frac{12\sin\theta + 4}{5}$

$5h - 2 = 12\cos\theta, \qquad 5k - 4 = 12\sin\theta$

locus of point P is

$(5x-2)^2+(5y-4)^2=144$

its centre : C (2/5, 4/5)

$$AC=\sqrt{(1-2/5)^2+(2-4/5)^2}=\sqrt{\frac{9}{25}+\frac{36}{25}}=\sqrt{\frac{45}{25}}=\frac{3}{\sqrt{5}}$$

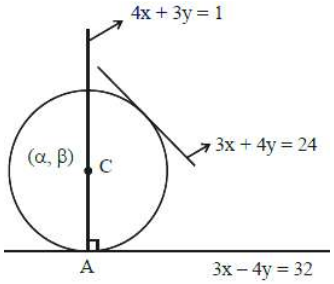
14. Let the centre of a circle be (α, β) and its radius $r < 8$. Let $3x + 4y = 24$ and $3x - 4y = 32$ be two tangents and $4x + 3y = 1$ be a normal to C. Then $(\alpha - \beta + r)$ is equal to : **[JEE(Main)April,2023]**

माना एक वृत्त C का केन्द्र (α, β) है तथा इसकी त्रिज्या $r < 8$ है। माना C की दो स्पर्श रेखाएँ $3x + 4y = 24$ तथा $3x - 4y = 32$ हैं तथा एक अभिलंब $4x + 3y = 1$ है। तो $(\alpha - \beta + r)$ बराबर है : **[JEE(Main)April,2023]**

(A) 7 (B) 5 (C) 9 (D) 6

Ans. A

Sol.



First find point A by solving $4x + 3y = 1$ and $3x - 4y = 32$

After solving, point A is $(4, -5)$

centre (α, β) lie on $4x + 3y = 1$

$$4\alpha+3\beta=1\Rightarrow\beta=\frac{1-4\alpha}{3}$$

Now distance from centre to line $3x - 4y - 32 = 0$ and $3x + 4y - 24 = 0$ are equal.

$$\left|\frac{3\alpha-4\left(\frac{1-4\alpha}{3}\right)-32}{5}\right|=\left|\frac{3\alpha+4\left(\frac{1-4\alpha}{3}\right)-24}{5}\right|$$

after solving $\alpha = 1$ and $\alpha=\frac{28}{3}$

For $\alpha = 1$, centre $(1, -1) \Rightarrow$ radius $= 5$

For $\alpha=\frac{28}{3}$, centre $\left(\frac{28}{3},\frac{-109}{2}\right)$

$\Rightarrow$ radius ≈ 49.78 (rejected)

Hence, $\alpha = 1, \beta = -1, r = 5$

$\alpha - \beta + r = 7$

15. The number of common tangents, to the circles $x^2 + y^2 - 18x - 15y + 131 = 0$ and $x^2 + y^2 - 6x - 6y - 7 = 0$:
[JEE(Main)April,2023]

वृत्तों $x^2 + y^2 - 18x - 15y + 131 = 0$ तथा $x^2 + y^2 - 6x - 6y - 7 = 0$ की उभयनिष्ठ स्पर्श रेखाओं की संख्या होगी :
[JEE(Main)April,2023]

(A) 2 (B) 4 (C) 3 (D) 1

Ans. C

Sol. $C_1\left(9, \frac{15}{2}\right)$ $r_1 = \sqrt{81 + \frac{225}{4} - 131} = \frac{5}{2}$

$C_2(3,3)$ $r_2 = 5$

$C_1C_2 = \sqrt{6^2 + \frac{81}{4}} = \frac{15}{2}$

$r_1 + r_2 = \frac{15}{2}$

$C_1C_2 = r_1 + r_2$

16. Let the point $(p, p+1)$ lie inside the region $E = \left\{(x, y) : 3 - x \le y \le \sqrt{9 - x^2}, 0 \le x \le 3\right\}$. If the set of all values of p is the interval (a, b), then $b^2 + b - a^2$ is equal to _____. **[JEE(Main)April,2023]**

माना बिंदु $(p, p+1)$ क्षेत्र $E = \left\{(x, y) : 3 - x \le y \le \sqrt{9 - x^2}, 0 \le x \le 3\right\}$ के अन्दर स्थित है। यदि p के सभी मानों का समुच्चय अन्तराल (a, b) है, तब $b^2 + b - a^2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. C

Sol. $3 - x \le y \le \sqrt{9 - x^2}$

Points $(p, p+1)$ lies on $y = x + 1$

So point of intersection between

$y = x + 1$ & $y = 3 - x$ is $x = 1, y = 2$

and point of intersection between

$x + 1 = \sqrt{9 - x^2}$ is $x = \frac{-1 + \sqrt{17}}{2}$

Hence $p \in \left(1, \frac{-1 + \sqrt{17}}{2}\right)$

Hence $b^2 + b - a^2 = 3$

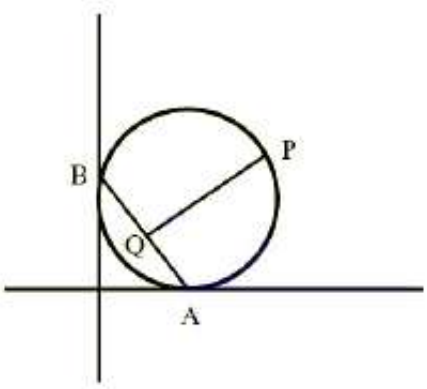
17. A circle passing through the point P(α, β) in the first quadrant touches the two coordinate axes at the points A and B. The point P is above the line AB. The point Q on the line segment AB is the foot of perpendicular from P on AB. If PQ is equal to 11 units, then the value of $\alpha\beta$ is _____. **[JEE(Main)April,2023]**

प्रथम चतुर्थांश में बिंदु P(α, β) से होकर जाने वाला एक वृत्त, निर्देशांक अक्षों को बिंदुओं A तथा B पर स्पर्श करता है। बिंदु P, रेखा AB से ऊपर है। बिंदु P से AB पर डाले गए लंब पाद रेखाखंड AB पर बिंदु Q है। यदि PQ = 11 इकाई है, तो $\alpha\beta$ का मान है _____

[JEE(Main)April,2023]

Ans. 121

Sol.



Let equation of circle is $(x-a)^2+(y-a)^2=a^2$

which is passing through P (α, β)

then $(\alpha-a)^2+(\beta-a)^2=a^2$

$\alpha^2+\beta^2-2\alpha a-2\beta a+a^2=0$

Here equation of AB is $x+y=a$

Let Q (α', β') be foot of perpendicular of P on AB

$$\frac{\alpha'-\alpha}{1}=\frac{\beta'-\beta}{1}=\frac{-(\alpha+\beta-a)}{2}$$

$$PQ^2=\left(\alpha'-\alpha\right)^2+\left(\beta'-\beta\right)^2=\frac{1}{4}(\alpha+\beta-a)^2+\frac{1}{4}(\alpha+\beta-a)^2$$

$$121=\frac{1}{2}\left(\alpha+\beta-a\right)^2$$

$242=\alpha^2+\beta^2-2\alpha a-2\beta a+a^2+2\alpha\beta$

$\Rightarrow \alpha\beta = 121$

18. Consider a circle $C_1 : x^2 + y^2 - 4x - 2y = \alpha - 5$. Let its mirror image in the line $y = 2x + 1$ be another circle $C_2 : 5x^2 + 5y^2 - 10fx - 10gy + 36 = 0$. Let r be the radius of C_2. Then $\alpha + r$ is equal to _____.

[JEE(Main)April,2023]

एक वृत $C_1 : x^2 + y^2 - 4x - 2y = \alpha - 5$ का विचार कीजिए। माना रेखा $y = 2x + 1$ में इसका दर्पण प्रतिबिंब वृत $C_2 : 5x^2 + 5y^2 - 10fx - 10gy + 36 = 0$ है। माना वृत C_2 की त्रिज्या r है। तो $\alpha + r$ बराबर है _____

[JEE(Main)April,2023]

Ans. 2

Sol. $x^2 + y^2 - 4x - 2y + 5 - \alpha = 0$

$C_1(2,1) r_1 = \sqrt{\alpha}$

$2x - y + 1 = 0$

Image of (2, 1)

$$\frac{x-2}{2} = \frac{y-1}{-1} = \frac{-2(4-1+1)}{5}$$

$$\frac{x-2}{2} = \frac{y-1}{-1} = \frac{-8}{5}$$

$$x = 2 - \frac{16}{5} = \frac{-6}{5}, y = 1 + \frac{8}{5} = \frac{13}{5}$$

$$x^2 + y^2 - 2fx - 2gy + \frac{36}{5} = 0$$

$C_2(f, g)$

$$r_2 = \sqrt{f^2 + g^2 - \frac{36}{5}}$$

$$\alpha = f^2 + g^2 - \frac{36}{5}$$

$$\therefore f = -\frac{6}{5}, g = \frac{13}{5}$$

$$\alpha = \frac{36}{25} + \frac{169}{25} - \frac{36}{5}$$

$$= \frac{36 + 169 - 180}{25} \Rightarrow \alpha = 1 \Rightarrow r = 1$$

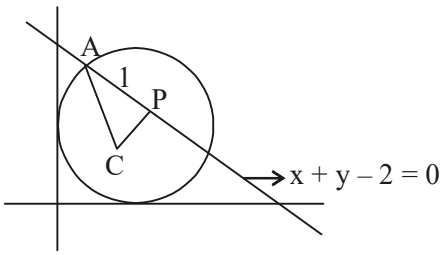
$$\therefore \alpha + r = 2$$

19. Two circles in the first quadrant of radii r_1 and r_2 touch the coordinate axes. Each of them cuts off an intercept of 2 units with the line $x + y = 2$. Then $r_1^2 + r_2^2 - r_1 r_2$ is equal to _____. **[JEE(Main)April,2023]**

प्रथम चतुर्थांश में r_1 तथा r_2 त्रिज्या के दो वृत्त निर्देशांक अक्षों को स्पर्श करते हैं। इनमें से प्रत्येक रेखा $x + y = 2$ से 2 इकाई का अंतःखंड काटता है। तो $r_1^2 + r_2^2 - r_1 r_2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 7

Sol. Let equation of circle is



$(x-r)^2+(y-r)^2=r^2$

$$CP=\frac{|r+r-2|}{\sqrt{2}}=\sqrt{r^2-1}$$

$$\left(\sqrt{2}(r-1)\right)^2=r^2-1$$

$r^2-4r+3=0$

$r_1=1, r_2=3$

$r_1^2+r_2^2-r_1r_2=1+9-3=7$

20. Four distinct points (2k, 3k), (1,0), (0,1) and (0,0) lie on a circle for k equal to : **[JEE(Main)Jan.2024]**

यदि चार भिन्न बिन्दु (2k, 3k), (1,0), (0,1) तथा (0,0) एक वृत्त पर हैं, तो k बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{3}{13}$ (B) $\frac{1}{13}$ (C) $\frac{5}{13}$ (D) $\frac{2}{13}$

Ans. C

Sol. General equation of circle is

$x^2+y^2+2gx+2fy+c=0$

As it passes through (0, 0) (0, 1) (1, 0)

$\therefore c=0$

Now $2g+1=0$

$$\Rightarrow g=\frac{-1}{2}$$

$2f+1=0$

$$f=\frac{-1}{2}$$

Hence equation of circle is

$x^2+y^2-x-y=0$

Also point (2k, 3k) lies on the circle.

$\therefore 4k^2+9k^2-5k=0$

$\Rightarrow 13k^2-5k=0$

$\Rightarrow k = 0,\ k = \frac{5}{13}$

$\therefore k = \frac{5}{13}$

21. If the circles $(x+1)^2 + (y+2)^2 = r^2$ and $x^2 + y^2 - 4x - 4y + 4 = 0$ intersect at exactly two distinct points, then:
[JEE(Main)Jan.2024]

यदि वृत्त $(x+1)^2 + (y+2)^2 = r^2$ तथा $x^2 + y^2 - 4x - 4y + 4 = 0$ एक दूसरे को ठीक दो भिन्न बिंदुओं पर काटते हैं, तो :
[JEE(Main)Jan.2024]

(A) $0 < r < 7$ (B) $\frac{1}{2} < r < 7$ (C) $5 < r < 9$ (D) $3 < r < 7$

Ans. D

Sol. for interaction

$|r_1 - r_2| < C_1C_2 < r_1 + r_2$

$|r - 2| < 5 < r + 2$

$r > 3$ ...(1)

$-5 < r - 2 < 5$

$-3 < r < 7$ ...(2)

By (1) and (2)

$3 < r < 7$

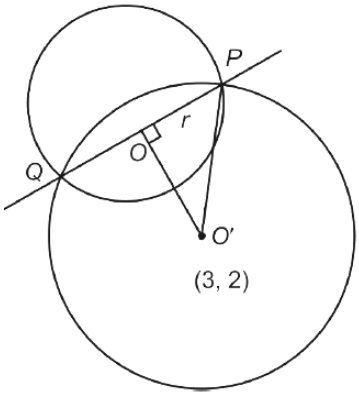
22. If one of the diameters of the circle $x^2 + y^2 - 10x + 4y + 13 = 0$ is a chord of another circle C, whose center is the point of intersection of the lines $2x + 3y = 12$ and $3x - 2y = 5$, then the radius of the circle is :
[JEE(Main)Jan.2024]

यदि वृत्त $x^2 + y^2 - 10x + 4y + 13 = 0$ का एक व्यास, एक दूसरे वृत्त C, जिसका केन्द्र, रेखाओं $2x + 3y = 12$ तथा $3x - 2y = 5$ का प्रतिच्छेदन बिन्दु हैं, की एक जीवा है, तो वृत्त C को त्रिज्या है : **[JEE(Main)Jan.2024]**

(A) 4 (B) $3\sqrt{2}$ (C) 6 (D) $\sqrt{20}$

Ans. C

Sol. Given circle $x^2 + y^2 - 10x + 4y + 13 = 0$



Centre (O) ≡ (5, – 2)

Radius (r) = $\sqrt{25+4-13}=4$

Point of intersection of 2x + 3y = 12 and 3x – 2y = 5 is (3,2)

⇒ Centre of C(O′) = (3,2)

$d = OO' = \sqrt{(5-3)^2+(-2-2)^2} = \sqrt{20}$

Radius of C $= \sqrt{r^2+d^2}$

$= \sqrt{16+20}$

= 6

23. Let a variable line passing through the centre of the circle $x^2 + y^2 - 16x - 4y = 0$, meet the positive co-ordinate axes at the points A and B. Then the minimum value of OA + OB, where O is the origin, is equal to :

[JEE(Main)Jan.2024]

माना वृत्त $x^2 + y^2 - 16x - 4y = 0$ के केन्द्र से होकर जाने वाली एक चर रेखा धनात्मक निर्देशांक अक्षों की बिन्दुओं A तथा B पर मिलती है। तो OA + OB, जहाँ O मूलबिन्दु है, का न्यूनतम मान है : **[JEE(Main)Jan.2024]**

(A) 18 (B) 12 (C) 24 (D) 20

Ans. A

Sol. Given $x^2 + y^2 - 16x - 4y = 0$

$\Rightarrow (x-4)^2 + (y-2)^2 = 20$

As variable line passes through center of circle

$\therefore (y-2) = m(x-8)$

$\therefore$ X intercept $= \left(\frac{-2}{m}+8\right)$

Y intercept $= (-8m+2)$

$$\therefore OA + OB = \frac{-2}{m} + 8 - 8m + 2$$

$$f'(m) = \frac{2}{m^2} - 8 = 0$$

$$\Rightarrow m^2 = \frac{1}{4}$$

$$\Rightarrow m = \frac{-1}{2}$$

$$\therefore f\left(\frac{-1}{2}\right) = 18$$

24. Let C : $x^2 + y^2 = 4$ and C′ : $x^2 + y^2 - 4\lambda x + 9 = 0$ be two circles. If the set of all values of λ so that the circles C and C′ intersect at two distinct points, is R – [a, b], then the point (8a + 12, 16b – 20) lies on the curve :

[JEE(Main)Jan.2024]

माना दो वृत्त C : $x^2 + y^2 = 4$ तथा C′ : $x^2 + y^2 - 4\lambda x + 9 = 0$ हैं। यदि λ के सभी मानों, जिनके लिए वृत्त C तथा C′ एक – दूसरे को दो भिन्न बिंदुओं पर काटते हैं, का समुच्चय R – [a, b] है, तो बिंदु (8a + 12, 16b – 20) किस वक्र पर है?

[JEE(Main)Jan.2024]

(A) $x^2 + 2y^2 - 5x + 6y = 3$ (B) $x^2 - 4y^2 = 7$

(C) $5x^2 - y = -11$ (D) $6x^2 + y^2 = 42$

Ans. D

Sol. $x^2 + y^2 = 4$, $c_1(0,0)$, $r_1 = 2$

$x^2 + y^2 - 4\lambda x + 9 = 0$, $C_2(2\lambda, 0)$, $r_2 = \sqrt{4\lambda^2 - 9}$

$$|r_1 - r_2| < c_1c_2 < |r_1 + r_2|$$

$$\left|2 - \sqrt{4\lambda^2 - 9}\right| < |2\lambda| < 2 + \left|2 + \sqrt{4\lambda^2 - 9}\right|$$

$$4 + 4\lambda^2 - 9 - 4\sqrt{4\lambda^2 - 9} < 4\lambda^2 < 4 + 4\lambda^2 - 9 + 4\sqrt{4\lambda^2 - 9}$$

$$-5 < 4\sqrt{4\lambda^2 - 9} \quad \text{...(A)}$$

$\lambda \in R$

and $5 < 4\sqrt{4\lambda^2 - 9}$

$$25 < 16\left(4\lambda^2 - 9\right)$$

$$\frac{169}{64} < \lambda^2$$

$$\lambda \in \left(-\infty, \frac{-13}{8}\right) \cup \left(\frac{13}{8}, \infty\right) \quad ...(1)$$

Also $4\lambda^2 - 9 \geq 0$

$$\lambda \in \left(-\infty, \frac{-3}{2}\right] \cup \left[\left(\frac{3}{2}, \infty\right)\right] \quad ...(2)$$

Hence $\lambda \in R - \left[\frac{-13}{8}, \frac{13}{8}\right]$

$(8a + 12), (16b - 20)$

$= (-1,6)$

25. Let the locus of the midpoints of the chords of the circle $x^2 + (y-1)^2 = 1$ drawn from the origin intersect the line $x + y = 1$ at P and Q. Then, the length of PQ is : **[JEE(Main)Jan.2024]**

माना मूल बिंदु से वृत्त $x^2 + (y-1)^2 = 1$ पर खींची गई जीवाओं के मध्य बिंदु का बिंदुपथ रेखा $x + y = 1$ को बिन्दुओं P तथा Q पर काटता है। तो PQ की लम्बाई है : **[JEE(Main)Jan.2024]**

(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{2}$ (D) 1

Ans. B

Sol. Let mid point be (h, k)

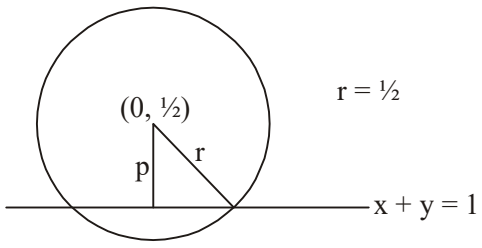
$$T = S_1$$

$hx + ky - (y + k) = h^2 + k^2 - 2k$

$(0, 0)$ lies on it

$\Rightarrow \quad h^2 + k^2 - k = 0$

$x^2 + y^2 - y = 0$



$$p = \frac{1}{2\sqrt{2}}$$

$$PQ = 2\sqrt{r^2 - p^2} = \frac{1}{\sqrt{2}}$$

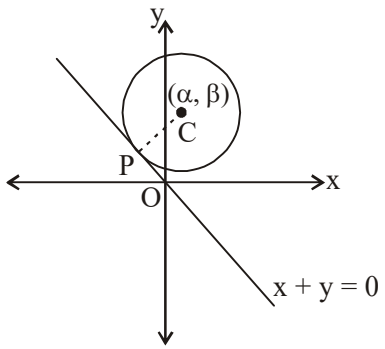
26. Consider a circle $(x-\alpha)^2+(y-\beta)^2=50$ where $\alpha, \beta > 0$. If the circle touches the line $y+x=0$ at the point P, whose distance from the origin is $4\sqrt{2}$, then $(\alpha+\beta)^2$ is equal to _____.

[JEE(Main)Jan.2024]

वृत्त $(x-\alpha)^2+(y-\beta)^2=50$, जहाँ $\alpha, \beta > 0$ हैं, का विचार कीजिए। यदि यह वृत्त रेखा $y+x=0$ को बिंदु P पर स्पर्श करता है तथा P की मूल बिंदु से दूरी $4\sqrt{2}$ है, तो $(\alpha+\beta)^2$ बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 100

Sol.



$S:(x-\alpha)^2+(y-\beta)^2=50$

$CP = r$

$\left|\frac{\alpha+\beta}{\sqrt{2}}\right| = 5\sqrt{2}$

$\Rightarrow (\alpha+\beta)^2 = 100$

27. Equation of two diameters of a circle are $2x-3y=5$ and $3x-4y=7$. The line joining the points $\left(-\frac{22}{7},-4\right)$ and $\left(-\frac{1}{7},3\right)$ intersects the circle at only one point $P(\alpha, \beta)$. Then $17\beta-\alpha$ is equal to _____.

[JEE(Main)Jan.2024]

एक वृत्त के दो व्यासों के समीकरण $2x-3y=5$ तथा $3x-4y=7$ हैं। बिन्दुओं $\left(-\frac{22}{7},-4\right)$ तथा $\left(-\frac{1}{7},3\right)$ को मिलाने वाली रेखा, वृत्त को केवल एक बिन्दु $P(\alpha, \beta)$ पर मिलती है। तो $17\beta-\alpha$ बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. B

Sol. Intersection point of diameters $2x-3y=5$ and $3x-4y=7$ is centre of circle $C(1,-1)$

Equation of line joining $\left(-\frac{22}{7},4\right)$ and $\left(-\frac{1}{7},3\right)$ is

$$y = \frac{7}{3}x + \frac{10}{3} \Rightarrow 7x - 3y + 10 = 0$$

Radius of circle is $\frac{20}{\sqrt{58}}$

$\Rightarrow$ Equation of circle is $(x-1)^2 + (y+1)^2 = \frac{400}{58}$

$$x^2 + y^2 - 2x + 2y - \frac{142}{29} = 0$$

$\Rightarrow$ Tangent P(α, β) is $\alpha x + \beta y - (\alpha + x) + (\beta + y) - \frac{142}{29} = 0$

$$\Rightarrow \frac{\alpha - 1}{7} = \frac{\beta + 1}{-3} = \frac{-\alpha + \beta - \frac{142}{29}}{10}$$

$$\Rightarrow \alpha = -\frac{41}{29},\ \beta = \frac{1}{29}$$

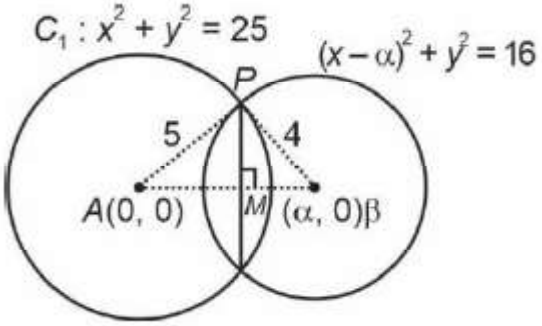
$$\Rightarrow 17\beta - \alpha = \frac{17}{29} + \frac{41}{29} = 2$$

28. Consider two circles $C_1 : x^2 + y^2 = 25$ and $C_2 : (x-\alpha)^2 + y^2 = 16$, where $\alpha \in (5, 9)$. Let the angle between the two radii (one to each circle) drawn from one of the intersection points of C_1 and C_2 be $\sin^{-1}\left(\frac{\sqrt{63}}{8}\right)$. If the length of common chord of C_1 and C_2 is β, then the value of $(\alpha\beta)^2$ equals _____. **[JEE(Main)Jan.2024]**

दो वृत्तों $C_1 : x^2 + y^2 = 25$ तथा $C_2 : (x-\alpha)^2 + y^2 = 16$, जहाँ $\alpha \in (5, 9)$ है, का विचार कीजिए। माना वृत्तों C_1 तथा C_2 के प्रतिच्छेदन बिन्दुओं में से एक से खींची गई दो त्रिज्याओं (प्रत्येक वृत्त पर एक) के बीच कोण $\sin^{-1}\left(\frac{\sqrt{63}}{8}\right)$ है। यदि C_1 तथा C_2 की उभयनिष्ठ जीवा की लंबाई β है, तो $(\alpha\beta)^2$ का मान बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 1575

Sol.



$$\cos\theta = \frac{25+16-\alpha^2}{40}, \ \because \sin\theta = \frac{\sqrt{63}}{8}$$

$$\Rightarrow \cos\theta = \frac{1}{8}$$

$$\Rightarrow \frac{1}{8} = \frac{41-\alpha^2}{40} \Rightarrow \alpha^2 = 36 \Rightarrow \alpha = 6$$

Equation of common chord is $x = \frac{15}{4}$

$$\Rightarrow AM = \frac{15}{4} \Rightarrow (MP)^2 = 25 - \left(\frac{15}{4}\right)^2$$

$$\Rightarrow \beta^2 = 4(MP)^2 = 100 - \frac{225}{4} = \frac{400-225}{4} = \frac{175}{4}$$

$$\Rightarrow \alpha^2\beta^2 = 36 \times \frac{175}{4} = 1575$$

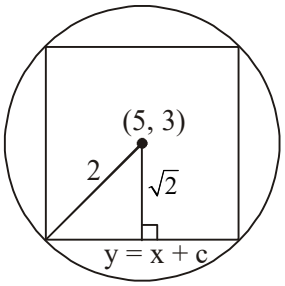
29. A square is inscribed in the circle $x^2 + y^2 - 10x - 6y + 30 = 0$. One side of this square is parallel to $y = x + 3$. If (x_i, y_i) are the vertices of the square, then $\sum\left(x_i^2 + y_i^2\right)$ is equal to : **[JEE(Main)April,2024]**

वृत्त $x^2 + y^2 - 10x - 6y + 30 = 0$ के अंतर्गत एक वर्ग है। वर्ग की एक भुजा $y = x + 3$ के समांतर है। यदि वर्ग के शीर्ष (x_i, y_i) हैं, तो $\sum\left(x_i^2 + y_i^2\right)$ बराबर है : **[JEE(Main)April,2024]**

(A) 156 (B) 148 (C) 152 (D) 160

Ans. C

Sol.



$\perp^{lr}$ from (5,3) to $x - y + c = 0$ is $\sqrt{2}$

$\therefore$ C = 0 or C = – 4.

2 Lines are y = x and y = x – 4.

Solve with circle.

$y = x$ $\qquad$ $y = x - 4$

(i) $x^2 - 8x + 15 = 0$ $\qquad$ (ii) $x^2 - 12x + 35 = 0$

x = 3, y = 3 x = 5, y = 1

x = 5, y = 5 x = 7, y = 3

$\therefore \sum x_i^2 + y_i^2 = 152$

30. Let C be a circle with radius $\sqrt{10}$ units and centre at the origin. Let the line x + y = 2 intersects the circle C at the points P and Q. Let MN be a chord of C of length 2 unit and slope –1. Then, a distance (in units) between the chord PQ and the chord MN is : **[JEE(Main)April,2024]**

माना C एक वृत्त है जिसकी त्रिज्या $\sqrt{10}$ इकाई है तथा केन्द्र मूलबिन्दु पर है। माना रेखा x + y = 2 वृत्त C को बिन्दु P तथा Q पर प्रतिच्छेद करती है। माना MN, C की एक जीवा है जिसकी लम्बाई 2 तथा प्रवणता –1 है। तो PQ तथा जीवा MN के बीच की एक दूरी (इकाई में) है : **[JEE(Main)April,2024]**

(A) $\sqrt{2}+1$ $\qquad$ (B) $\sqrt{2}-1$ $\qquad$ (C) $3-\sqrt{2}$ $\qquad$ (D) $2-\sqrt{3}$

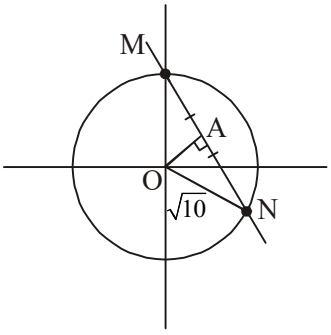
Ans. C

Sol. $C : x^2 + y^2 = 10$

$AN = \dfrac{MN}{2} = 1$

$\therefore \ell n\, \Delta OAN \rightarrow (ON)^2 = (OA)^2 + (AN)^2$

$10 = (OA)^2 + 1 \rightarrow OA = 3$



Perpendicular distance of center from $PQ = \dfrac{|0+0-2|}{\sqrt{2}} = \sqrt{2}$

Perpendicular distance between MN and $PQ = OA + \sqrt{2}$ or $\left|OA - \sqrt{2}\right|$

$= 3 + \sqrt{2}$ or $3 - \sqrt{2}$

31. Let a circle C of radius 1 and closer to the origin be such that the lines passing through the point (3, 2) and parallel to the coordinate axes touch it. Then the shortest distance of the circle C from the point (5, 5) is :

[JEE(Main)April,2024]

माना मूल बिन्दु के निकट त्रिज्या 1 का एक वृत्त C है, जिसे बिन्दु (3, 2) से होकर जाने वाली तथा निर्देशांक अक्षों के समांतर रेखाएँ स्पर्श करती हैं। तो वृत्त C की बिन्दु (5, 5) से न्यूनतम दूरी है :

[JEE(Main)April,2024]

(A) 4 (B) 5 (C) $4\sqrt{2}$ (D) $2\sqrt{2}$

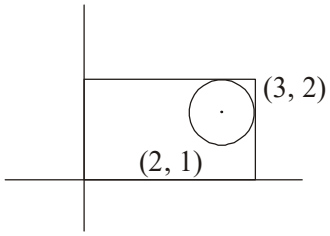
Ans. A

Sol. Centre of circle is (2, 1)

minimum distance of circle from (5, 5)

$= \sqrt{(5-2)^2+(5-1)^2}-1$

$= 5 - 1 = 4$



32. Let ABCD and AEFG be squares of side 4 and 2 units, respectively. Then point E is on the line segment AB and the point F is on the diagonal AC. Then the radius r of the circle passing through the point F and touching the line segment BC and CD satisfies :

[JEE(Main)April,2024]

माना भुजा 4 तथा 2 इकाईयों के दो वर्ग क्रमशः ABCD तथा AEFG हैं। बिन्दु E रेखाखण्ड AB पर है तथा बिन्दु F विकर्ण AC पर है। यदि बिन्दु F से होकर जाने वाले तथा रेखाखंडों BC और CD को छूने वाले वृत्त की त्रिज्या r है, तो r संतुष्ट करता है :

[JEE(Main)April,2024]

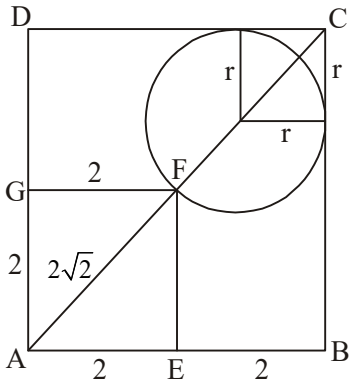
(A) $r^2 - 8r + 8 = 0$ (B) $2r^2 - 8r + 7 = 0$ (C) $2r^2 - 4r + 1 = 0$ (D) $r = 1$

Ans. A

Sol. $r + r\sqrt{2} = 2\sqrt{2}$

$r = 4 - 2\sqrt{2}$

$(r-4)^2 = 8 \Rightarrow r^2 - 8r + 8 = 0$



33. Let the circle $C_1 : x^2 + y^2 - 2(x + y) + 1 = 0$ and C_2 be a circle having centre at (–1, 0) and radius 2. If the line of common chord of C_1 and C_2 intersects the y-axis at the point P, then the square of the distance of P from the centre of C_1 is : **[JEE(Main)April,2024]**

माना वृत्त $C_1 : x^2 + y^2 - 2(x + y) + 1 = 0$ है तथा वृत्त C_2 का केन्द्र (–1, 0) और त्रिज्या 2 है। यदि C_1 तथा C_2 की उभयनिष्ठ जीवा की रेखा y-अक्ष को बिन्दु P पर काटती है, तो बिन्दु P की वृत्त C_1 के केन्द्र से दूरी का वर्ग है : **[JEE(Main)April,2024]**

(A) 6 (B) 4 (C) 2 (D) 1

Ans. C

Sol. $C_1(1,1)$, $r_1 = 1$, $(x+1)^2 + y^2 = 4$

Radical axis $4x + 2y - 4 = 0 \Rightarrow P(0,2)$

Distance = $\sqrt{(1-0)^2 + (1-2)^2}$

Square of distance = 2

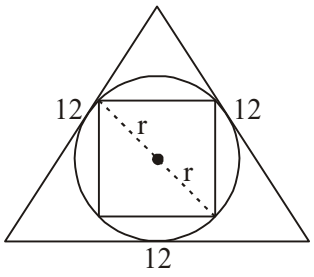
34. A circle is inscribed in an equilateral triangle of side of length 12. If the area and perimeter of any square inscribed in this circle are m and n, respectively, then $m + n^2$ is equal to : **[JEE(Main)April,2024]**

लम्बाई 12 की भुजा के किसी समबाहु त्रिभुज में एक अंतवृत्त है। यदि इस वृत्त के अंतर्गत किसी वर्ग का क्षेत्रफल व परिमाप क्रमशः m व n है, तो $m + n^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 396 (B) 408 (C) 312 (D) 414

Ans. B

Sol. Irradius of $\Delta = r = \frac{\Delta}{s} = \frac{\frac{\sqrt{3}}{4} \times 12^2}{18} = \frac{36\sqrt{3}}{18} = 2\sqrt{3}$



side length of square is a , then

$2a^2 = 4r^2$

$a^2 = 2r^2 = 24$

Now m = 24, $n = 4.\sqrt{24}$

$m + n^2 = 24 + 16 \times 24 = 17 \times 24 = 408$

35. Let the circles $C_1 : (x-\alpha)^2 + (y-\beta)^2 = r_1^2$ and $C_2 : (x-8)^2 + \left(y - \frac{15}{2}\right)^2 = r_2^2$ touch each other externally at the point (6,6). If the point (6,6) divides the line segment joining the centres of the circle C_1 and C_2 internally in the ratio 2 : 1, then $(\alpha + \beta) + 4\left(r_1^2 + r_2^2\right)$ equals : **[JEE(Main)April,2024]**

माना वृत्त $C_1:(x-\alpha)^2+(y-\beta)^2=r_1^2$ तथा $C_2:(x-8)^2+\left(y-\frac{15}{2}\right)^2=r_2^2$ एक दूसरे को बिन्दु (6,6) बाह्य स्पर्श करते है, यदि बिन्दु (6, 6) वृत्तों के केन्द्रों C_1 तथा C_2 को मिलाने वाली रेखा को 2 : 1 में अन्तः विभाजित करता है, तब $(\alpha+\beta)+4\left(r_1^2+r_2^2\right)$ बराबर होगा : **[JEE(Main)April,2024]**

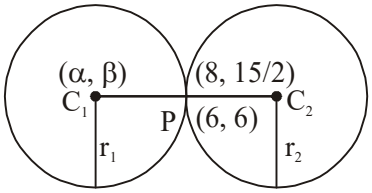
(A) 145 (B) 110 (C) 125 (D) 130

Ans. D

Sol. $C_1 \to (x-\alpha)^2+(y-\beta)^2=r_1^2$

$C_2 \to (x-8)^2+\left(y-\frac{15}{2}\right)^2=r_2^2$

Point P divide the line segment internally C_1C_2 in the ratio 2 : 1



$\frac{\alpha\times1+8\times2}{1+2}=6$, $\alpha=2$

$\frac{\beta\times1+\frac{15}{\alpha}\times2}{1+2}=6$, $\beta=3$

$r_1=\sqrt{(6-2)^2+(6-3)^2}=\sqrt{25}=5$

$r_2=\sqrt{(8-6)^2+\left(\frac{15}{\alpha}-6\right)^2}=\frac{5}{2}$

$\alpha+\beta+4\left(r_1^2+r_2^2\right)=2+3+4\left(5^2+\left(\frac{5}{2}\right)^2\right)$

$=5+4\left(\frac{125}{4}\right)$

= 130

36. If the image of the point (–4,5) in the line x + 2y = 2 lies on the circle $(x+4)^2+(y-3)^2=r^2$, then r is equal to: **[JEE(Main)April,2024]**

यदि रेखा x + 2y = 2 में (–4, 5) का प्रतिबिंब वृत्त $(x+4)^2+(y-3)^2=r^2$ पर है, तो r बराबर है :

(A) 4 (B) 1 (C) 3 (D) 2

Ans. D

Sol. $\frac{x+4}{1}=\frac{y-5}{2}=\frac{-2(-4+10-2)}{1+4}=-\frac{8}{5}$

$x+4=-\frac{8}{5}$

$y-5=-\frac{16}{5}$

$(x+4)^2+(y-3)^2=r^2$

$\frac{64}{25}+\frac{36}{25}=r^2$

$r=2$

37. Let the centre of a circle, passing through the points (0, 0), (1, 0) and touching the circle $x^2+y^2=9$, be (h, k). Then for all possible values of coordinates of the centre (h, k), $4(h^2+k^2)$ is equal to _____.

[JEE(Main)April,2024]

माना बिन्दुओं (0, 0), (1, 0) से होकर जाने वाले तथा वृत्त $x^2+y^2=9$ को स्पर्श करने वाले एक वृत्त का केन्द्र (h, k) है। तो केन्द्र (h, k) के सभी संभव निर्देशांकों के लिए $4(h^2+k^2)$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 9

Sol. Circle will touch internally

$C_1C_2=|r_1-r_2|$

$=\sqrt{h^2+k^2}=3-\sqrt{h^2+k^2}$

$\Rightarrow 2\sqrt{h^2+k^2}=3$

$\Rightarrow h^2+k^2=\frac{9}{4}$

$\therefore\ 4(h^2+k^2)=9$

Exercise J - A

Paragraph for Questions 1 and 2

A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P\left(\sqrt{3},1\right)$. A straight line L, perpendicular to PT is a tangent to the circle $(x-3)^2 + y^2 = 1$

वृत्त $x^2 + y^2 = 4$ पर बिन्दु $P\left(\sqrt{3},1\right)$ पर स्पर्श रेखा PT खींची जाती है। PT के लम्बवत् रेखा L, वृत्त $(x-3)^2 + y^2 = 1$ को स्पर्श करती है।

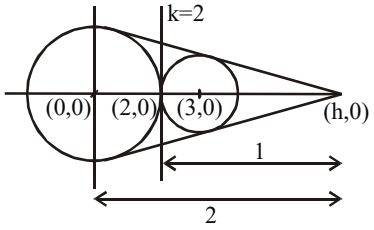
1. A possible equation of L is : **[JEE-2012]**

L का सम्भव समीकरण होगा : **[JEE-2012]**

(A) $x - \sqrt{3}y = 1$ (B) $x + \sqrt{3}y = 1$ (C) $x - \sqrt{3}y = -1$ (D) $x + \sqrt{3}y = 5$

Ans. A

Sol. $h = \dfrac{2\times 3 - 1\times 0}{2-1} = 6$



equation of tangents from (6, 0) :

$y - 0 = m(x-6) \Rightarrow \quad y - mx + 6m = 0$

use p = r

$$\left|\frac{6m}{\sqrt{1+m^2}}\right| = 2 \quad \Rightarrow 36m^2 = 4 + 4m^2$$

$$32m^2 = 4$$

$$m^2 = 1/8$$

$$\Rightarrow \quad m = \pm\frac{1}{2\sqrt{2}}$$

at $m = -\dfrac{1}{2\sqrt{2}}$

equation of tangent will be $x + 2\sqrt{2}y = 6$

2. A common tangent of the two circles is : **[JEE-2012]**

इन दो वृत्तों की उभयनिष्ठ स्पर्श रेखा होगी : **[JEE-2012]**

(A) $x = 4$ (B) $y = 2$ (C) $x + \sqrt{3}y = 4$ (D) $x + 2\sqrt{2}y = 6$

Ans. D

Sol. Equation of tangent at P will be $\sqrt{3}x + y = 4$

Slope of line L will be $\frac{1}{\sqrt{3}}$

Let equation of L be : $y = \frac{x}{\sqrt{3}} + c$

$\Rightarrow \quad x - \sqrt{3}y + \sqrt{3}c = 0$

Now this L is tangent to 2nd circle

So $\frac{3 + \sqrt{3}c}{2} = \pm 1 \quad \Rightarrow \quad c = -\frac{1}{\sqrt{3}}$

or $c = -\frac{5}{\sqrt{3}}$

using $c = -\frac{1}{\sqrt{3}}$

$y = \frac{x}{\sqrt{3}} - \frac{1}{\sqrt{3}} \Rightarrow x - \sqrt{3}y = 1$. Hence (A)

3. The locus of the mid–point of the chord of contact of tangents drawn from points lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$ is : **[JEE-2012]**

रेखा $4x - 5y = 20$ पर स्थित बिन्दुओं से वृत्त $x^2 + y^2 = 9$ के सापेक्ष स्पर्श जीवाओं के मध्य बिन्दु का बिन्दुपथ होगा : **[JEE-2012]**

(A) $20(x^2 + y^2) - 36x + 45y = 0$ (B) $20(x^2 + y^2) + 36x - 45y = 0$

(C) $36(x^2 + y^2) - 20x + 45y = 0$ (D) $36(x^2 + y^2) + 20x - 45y = 0$

Ans. A

Sol. Let mid point be (h, k),

then chord of contact :

$hx + ky = h^2 + k^2$ _____(i)

Let any point on the line $4x - 5y = 20$ be $\left(x_1, \frac{4x_1 - 20}{5}\right)$

$\therefore$ Chord of contact :

$5x_1x + (4x_1 - 20)y = 45$ _____(ii)

(i) and (ii) are same

$$\therefore \quad \frac{5x_1}{h} = \frac{4x_1 - 20}{k} = \frac{45}{h^2 + k^2}$$

$$\Rightarrow \quad x_1 = \frac{9h}{h^2 + k^2}$$

and $x_1 = \dfrac{45k + 20(h^2 + k^2)}{4(h^2 + k^2)}$

$$\Rightarrow \quad \frac{9h}{h^2 + k^2} = \frac{45k + 20(h^2 + k^2)}{4(h^2 + k^2)}$$

$\Rightarrow$ $20(h^2 + k^2) - 36h + 45k = 0$

$\therefore$ Locus is $20(x^2 + y^2) - 36x + 45y = 0$

4. Circle(s) touching x–axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ on y–axis is (are) :

[JEE(Adv.)2013]

x–अक्ष को मूलबिन्दु से 3 दूरी पर स्पर्श करने वाला (वाले) तथा y–अक्ष पर $2\sqrt{7}$ अंतःखंड बनाने वाला (वाले) वृत्त है (हैं) :

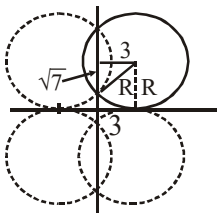
[JEE(Adv.)2013]

(A) $x^2 + y^2 - 6x + 8y + 9 = 0$ (B) $x^2 + y^2 - 6x + 7y + 9 = 0$

(C) $x^2 + y^2 - 6x - 8y + 9 = 0$ (D) $x^2 + y^2 - 6x - 7y + 9 = 0$

Ans. AC

Sol. As per figure,



$$R^2 = 3^2 + \left(\sqrt{7}\right)^2$$

$\Rightarrow R = 4$

$\therefore$ centre $\equiv (3,4)$

radius 4

$\therefore$ equation $x^2 + y^2 - 6x - 8y + 9 = 0$

such a circle can lie in all 4 quadrants as shown in figure.

$\therefore$ equation can be $x^2 + y^2 \pm 6x \pm 8y + 9 = 0$

5. A circle S passes through the point (0,1) and is orthogonal to the circles $(x-1)^2+y^2=16$ and $x^2+y^2=1$. Then:
[JEE(Adv.)2014]

(A) radius of S is 8 (B) radius of S is 7
(C) centre of S is (–7,1) (D) centre of S is (–8,1)

एक वृत्त S बिन्दु (0, 1) से गुजरता है तथा वृत्तों $(x-1)^2+y^2=16$ एवं $x^2+y^2=1$ के लम्बकोणीय (orthogonal) है। तब :
[JEE(Adv.)2014]

(A) S की त्रिज्या 8 है (B) S की त्रिज्या 7 है
(C) S का केन्द्र (–7,1) है (D) S का केन्द्र (–8,1) है

Ans. BC

Sol. Let the cirlce be

$x^2+y^2+2gx+2fy+c=0$ ...(1)

given circles

$x^2+y^2-2x-15=0$ ...(2)

$x^2+y^2-1=0$ ....(3)

(1) & (2) are orthogonal

$$\Rightarrow \quad -g+0=\frac{c-15}{2}$$

$$0+0=\frac{c-1}{2}$$

$\Rightarrow$ $c=1$ & $g=7$

so the cirle is

$x^2+y^2+14x+2fy+1=0$ it passes thrgouh

(0, 1) $\Rightarrow$ $0+1+0+2f+1=0$

$f=-1$

$\Rightarrow$ $x^2+y^2+14x-2y+1=0$

Centre (–7, 1)

radius = 7

6. Let RS be the diameter of the circle $x^2+y^2=1$, where S is the point (1, 0). Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q. The normal to the circle at P intersects a line drawn through Q parallel to RS at point E. Then the locus of E passes through the point(s) :
[JEE(Adv.)2016]

माना कि RS वृत $x^2+y^2=1$ का व्यास है, जहाँ कि S बिंदु (1, 0) है। माना कि P (R और S से भिन्न) वृत पर एक चर बिन्दु है और वृत पर बिन्दुओं S और P पर खींची गई स्पर्शरेखाएँ बिन्दु Q पर मिलती हैं। वृत के बिन्दु P पर अभिलम्ब उस रेखा को, जो Q से गुजरती है तथा RS के समानान्तर है, बिन्दु E पर प्रतिच्छेदित करता है। तब E का बिन्दुपथ निम्न बिन्दुओं से गुजरता है :
[JEE(Adv.)2016]

(A) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}}\right)$ (B) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (C) $\left(\frac{1}{3}, -\frac{1}{\sqrt{3}}\right)$ (D) $\left(\frac{1}{4}, -\frac{1}{2}\right)$

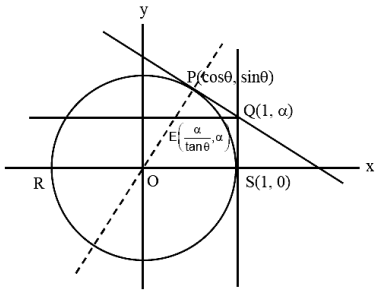
Ans. AC

Sol. $E \equiv \left(\frac{\alpha}{\tan\theta}, \alpha\right)$

$\cos\theta + \alpha \sin\theta = 1$

$\alpha = \tan\frac{\theta}{2}$

$\therefore$ locus is $y^2 = 1 - 2x$

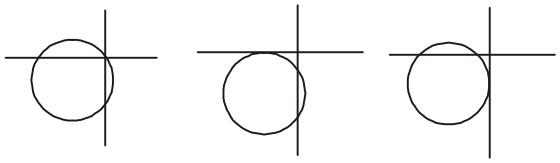


7. For how many values of p, the circle $x^2 + y^2 + 2x + 4y - p = 0$ and the coordinate axes have exactly three common points? **[JEE(Adv.)2017]**

p के कितने मानों के लिये वृत्त $x^2 + y^2 + 2x + 4y - p = 0$ तथा निर्देशांक अक्षों के केवल तीन बिन्दु उभयनिष्ठ हैं? **[JEE(Adv.)2017]**

Ans. 2

Sol.



p = 0 not possible p = –1

2 values of p.

8. Let T be the line passing through the points P(–2, 7) and Q(2, –5). Let F_1 be the set of all pairs of circles (S_1, S_2) such that T is tangent to S_1 at P, and tangent to S_2 at Q, and also such that S_1 and S_2 touch each other at a point, say, M. Let E_1 be the set representing the locus of M as the pair (S_1, S_2) varies in F_1. Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point R(1, 1) be F_2. Let E_2 be the set of the mid–points of the line segments in the set F_2. Then, which of the following statement(s) is (are) TRUE? **[JEE(Adv.)2018]**

(A) The point (–2, 7) lies in E_1 (B) The point (4/5, 7/5) does NOT lie in E_2

(C) The point (1/2, 1) lies in E_2 (D) The point (0, 3/2) does NOT lie in E_1

माना कि T, बिन्दुओं P(–2, 7) और Q(2, –5) से गुजरने वाली रेखा है। माना कि F_1 उन सभी वृत्त युग्मों (S_1, S_2) का समुच्चय है कि रेखा T, S_1 के बिन्दु P पर और S_2 के बिन्दु Q पर स्पर्शी है तथा वृत्त S_1 व S_2 एक दूसरे को बिन्दु, माना कि M, पर स्पर्श करते हैं। जब युग्म (S_1, S_2), F_1 में विचरित करता है तो माना कि समुच्चय E_1, बिन्दु M के बिन्दुपथ को दर्शाता है। माना कि F_2 उन सरल रेखा–खण्डों का समुच्चय है, जो बिन्दु R(1, 1) से गुजरती हैं तथा E_1 के दो भिन्न बिन्दुओं के युग्म को जोड़ती हैं। माना कि E_2 , समुच्चय F_2 के रेखाखण्डों के मध्य बिन्दुओं का समुच्चय है। तब निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) बिन्दु (–2, 7) समुच्चय E_1 में स्थित है

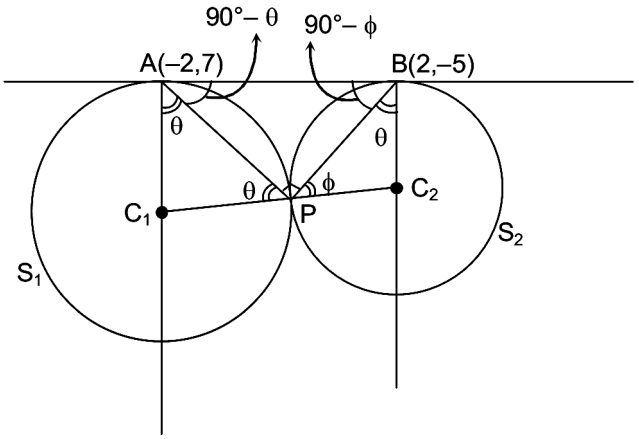
(B) बिन्दु (4/5, 7/5) समुच्चय E_2 में स्थित नहीं है

(C) बिन्दु (1/2, 1) समुच्चय E_2 में स्थित है

(D) बिन्दु (0, 3/2) समुच्चय E_1 में स्थित नहीं है

Ans. BD

Sol.



Note that $\angle APB = \frac{\pi}{2}$, hence locus of P is $(x+2)(x-2)+(y-7)(y+5)=0$

$x^2 + y^2 - 2y - 39 = 0$

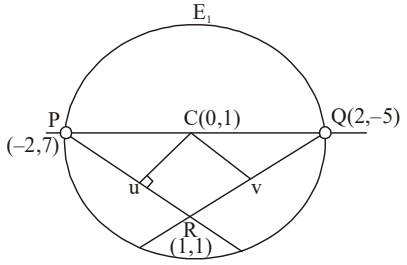
E_1 : $\{(x,y) : x^2 + y^2 - 2y - 39 = 0 \ \& \ (x, y) \neq (-2,7), (2, -5)\}$

Locus of mid-points of chords passing through (1, 1) is

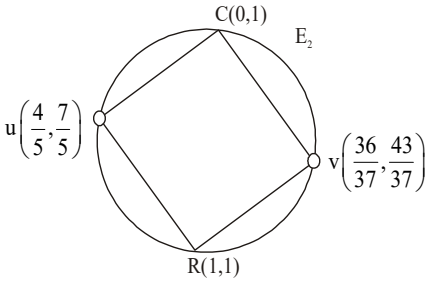
$h + K - (1 + k) = h^2 + k^2 - 2K$

$\Rightarrow \quad h^2 + K^2 - 2K - h + 1 = 0$

E_2 : $\{(x, y) : x^2 + y^2 - x - 2y + 1 = 0 \ \& \ (x,y) \neq \left(\frac{4}{5}, \frac{7}{5}\right), \left(\frac{36}{37}, \frac{43}{37}\right)\}$



$u \equiv \left(\frac{4}{5}, \frac{7}{5}\right), v\left(\frac{36}{37}, \frac{43}{37}\right)$



Paragraph for Question Nos. 9 to 10

Let S be the circle in the xy–plane defined by the equation $x^2 + y^2 = 4$.

माना कि S एक वृत्त है जो xy–समतल में समीकरण $x^2 + y^2 = 4$ के द्वारा परिभाषित है।

9. Let E_1E_2 and F_1F_2 be the chords of S passing through the point $P_0(1,1)$ and parallel to the x–axis and the y–axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slope –1. Let the tangents to S at E_1 and E_2 meet at E_3, the tangents to S at F_1 and F_2 meet at F_3, and the tangents to S at G_1 and G_2 meet at G_3. Then, the points E_3, F_3, and G_3 lie on the curve : **[JEE(Adv.)2018]**

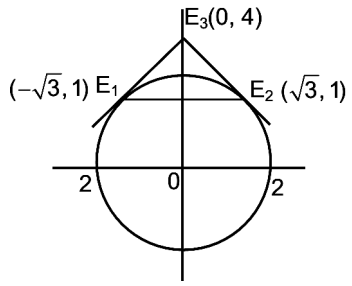
माना कि E_1E_2 और F_1F_2 वृत्त S की ऐसी जीवायें हैं जो बिन्दु $P_0(1,1)$ से गुजरती हैं और क्रमशः x–अक्ष व y–अक्ष के समान्तर हैं। माना कि G_1G_2,S की वह जीवा है जो P_0 से गुजरती है और जिसकी प्रवणता –1 है। माना कि E_1 और E_2 पर S की स्पर्शियाँ E_3 पर मिलती हैं, F_1 और F_2 पर S की स्पर्शियाँ F_3 पर मिलती हैं, तथा G_1 और G_2 पर S की स्पर्शियाँ G_3 पर मिलती हैं। तब वह वक्र जिस पर बिन्दु E_3, F_3, और G_3 स्थित हैं, है : **[JEE(Adv.)2018]**

(A) $x + y = 4$ (B) $(x-4)^2+(y-4)^2=16$ (C) $(x-4)(y-4) = 4$ (D) $xy = 4$

Ans. A

Sol. Tangent at E_1 and E_2 are $-\sqrt{3}x + y = 4$ and $\sqrt{3}x + y = 4$

They intersect at $E_3(0, 4)$



$F_1(1, \sqrt{3})$, $F_2(1, -\sqrt{3})$, $F_3(4, 0)$

$G_1(0, 2)$, $G_2(2, 0)$, $G_3(2, 2)$

E_3 F_3, G_3 lie on line $x + y = 4$

10. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N. Then, the mid–point of the line segment MN must lie on the curve :

[JEE(Adv.)2018]

माना कि P वृत्त S पर स्थित एक ऐसा बिन्दु है जिसके दोनों निर्देशांक धनात्मक हैं माना कि वृत्त S के बिन्दु P पर स्पर्शी निर्देशांक अक्षों को बिन्दुओं M और N पर प्रतिच्छेद करती है। तब वह वक्र जिस पर रेखाखण्ड MN का मध्य बिन्दु अनिवार्य रूप से स्थित है, हैं :

[JEE(Adv.)2018]

(A) $(x+y)^2 = 3xy$ (B) $x^{2/3} + y^{2/3} = 2^{4/3}$ (C) $x^2 + y^2 = 2xy$ (D) $x^2 + y^2 = x^2 y^2$

Ans. D

Sol. Let P$(2\cos\theta, 2\sin\theta)$

Tangent is $x\cos\theta + y\sin\theta = 2$

$$M\left(\frac{2}{\cos\theta}, 0\right), N\left(0, \frac{2}{\cos\theta}\right)$$

$$x = \frac{1}{\cos\theta} \text{ and } y = \frac{1}{\sin\theta} \quad \Rightarrow \quad \frac{1}{x^2} + \frac{1}{y^2} = 1 \quad \Rightarrow \quad x^2 + y^2 = x^2y^2$$

11. A line $y = mx + 1$ intersects the circle $(x-3)^2 + (y+2)^2 = 25$ at the points P and Q. If the midpoint of the line segment PQ has x–coordinate $-\frac{3}{5}$ then which one of the following options is correct? **[JEE(Adv.)2019]**

एक रेखा $y = mx + 1$ वृत्त $(x-3)^2 + (y+2)^2 = 25$ को बिन्दुओं P और Q पर प्रतिच्छेद करती है अगर रेखाखण्ड (line segment) PQ के मध्य बिन्दु का x–निर्दशांक (coordinate) $-\frac{3}{5}$ है तब निम्नलिखित में से कौन सा एक विकल्प सही है?

[JEE(Adv.)2019]

(A) $6 \le m < 8$ (B) $4 \le m < 6$ (C) $2 \le m < 4$ (D) $-3 \le m < -1$

Ans. C

Sol. Circle $(x-3)^2 + (y+2)^2 = 25$

$y = mx + 1$

Solve

$(x-3)^2 + (mx + 1 + 2)^2 = 25$

$$(1+m^2)x^2 + x(6m-6) - 7 = 0 \begin{cases} x_1 \\ x_2 \end{cases}$$

$$\frac{x_1 + x_2}{2} = \frac{-3}{5}$$

$$-\frac{(6m-6)}{2(m^2+1)} = \frac{-3}{5}$$

$5(m-1) = m^2 + 1$

$m^2 - 5m + 6 = 0$

$(m-2)(m-3) = 0$

$m = 2,3$

12. Let the point B be the reflection of the point A(2, 3) with respect to the line $8x - 6y - 23 = 0$. Let Γ_A and Γ_B be circles of radii 2 and 1 with centres A and B respectively. Let T be a common tangent to the circles Γ_A and Γ_B such that both the circles are on the same side of T. If C is the point of intersection of T and the line passing through A and B, then the length of the line segment AC is _____. **[JEE(Adv.)2019]**

माना कि बिन्दु B रेखा $8x - 6y - 23 = 0$ के सापेक्ष बिन्दु A(2,3) का प्रतिबिम्ब (reflection) है माना कि Γ_A तथा Γ_B क्रमशः त्रिज्याएँ 2 और 1 वाले वृत्त है। जिनकें केन्द्र क्रमशः A और B है माना कि वृतों Γ_A और Γ_B की एक ऐसी उभयनिष्ठ स्पर्श (common tangent) रेखा T है दोनो वृत्त जिसके एक ही तरफ है। यदि C बिन्दुओं A और B से जाने वाली रेखा T का प्रतिच्छेद बिंदु है तब रेखाखण्ड (line segment) AC की लम्बाई है_____ **[JEE(Adv.)2019]**

Ans. 10

Sol. $\frac{x-2}{8} = \frac{y-3}{-6} = -2\left(\frac{16-18-23}{100}\right)$

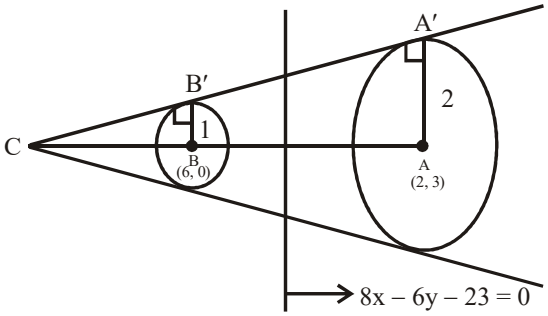
$\frac{x-2}{8} = \frac{y-3}{-6} = \frac{-2\times-25}{100} = \frac{1}{2}$

$\frac{x-2}{4} = \frac{y-3}{-3} = 1$

$x - 2 = 4 \qquad y - 3 = -3 \qquad \Rightarrow y = 0$

$x = 6$

B (6, 0)



$\frac{CB}{CA} = \frac{1}{2}$ (Δ'S CBB' and CAA' are congruent)

$\frac{CB}{CB + AB} = \frac{1}{2}$

$2CB = CB + AB$

$CB = AB = \sqrt{16+9} = 5$

$CA = 2(CB) = 10$

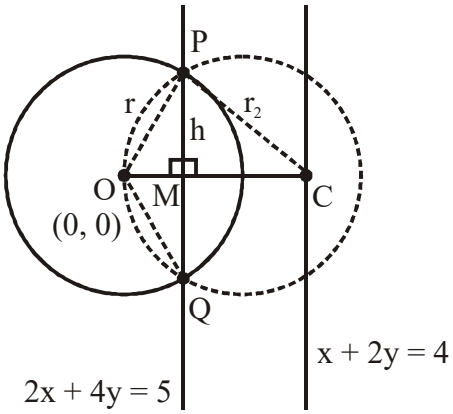
13. Let O be the centre of the circle $x^2 + y^2 = r^2$, where $r > \frac{\sqrt{5}}{2}$. Suppose PQ is a chord of this circle and the equation of the line passing through P and Q is $2x + 4y = 5$. If the centre of the circumcircle of the triangle OPQ lies on the line $x + 2y = 4$, then the value of r is _____. **[JEE(Adv.)2020]**

माना O वृत्त $x^2 + y^2 = r^2$ का केन्द्र है, जहाँ $r > \frac{\sqrt{5}}{2}$, माना PQ इस वृत्त की जीवा है तथा P व Q से गुजरने वाली रेखा का समीकरण $2x + 4y = 5$ है। यदि त्रिभुज OPQ के परिवृत्त का केन्द्र, रेखा $x + 2y = 4$ पर स्थित है, तब r का मान है _____

[JEE(Adv.)2020]

Ans. 2

Sol.



MC = distance between two lines

$$MC = \frac{3}{2\sqrt{5}}$$

$$OC = r_2 = \frac{4}{\sqrt{5}}$$

$h^2 + OM^2 = r^2$ -(1)

$h^2 + CM^2 = r_2^2$ -(2)

$$h^2 + \frac{25}{20} = r^2 \quad \text{-(1)}$$

$$h^2 + \frac{9}{20} = \frac{16}{5} \quad \text{-(2)}$$

by (1) - (2)

$r^2 = 4$

$r = 2$

14. Consider a triangle Δ whose two sides lie on the x-axis and the line $x + y + 1 = 0$. If the orthocenter of Δ is $(1, 1)$, then the equation of the circle passing through the vertices of the triangle Δ is : **[JEE(Adv.)2021]**

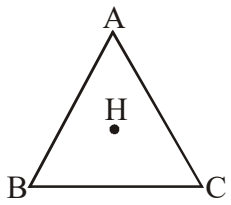
माना एक त्रिभुज की दो भुजाऐं x-अक्ष तथा रेखा $x + y + 1 = 0$ पर स्थित हैं। यदि Δ का लम्बकेन्द्र $(1, 1)$ है, तब त्रिभुज के शीर्षों से गुजरने वाले वृत्त का समीकरण है : **[JEE(Adv.)2021]**

(A) $x^2 + y^2 - 3x + y = 0$ (B) $x^2 + y^2 + x + 3y = 0$

(C) $x^2 + y^2 + 2y - 1 = 0$ (D) $x^2 + y^2 + x + y = 0$

Ans. B

Sol.



equation of AC $x + y + 1 = 0$

equation of BC $y = 0$

orthocenter $H(1, 1)$

equation of AH $x = 1$

equation of BH $x - y = 0$

Get $A = (1, -2)$; $B(0, 0)$ $C(-1, 0)$

Let circum center $P(h, k)$

$PA = PB$ $PB = PC$

$(h - 1)^2 + (k + 2)^2 = h^2 + k^2$ $h^2 + k^2 = (h + 1)^2 + k^2$

$\Rightarrow h = \frac{-1}{2}; \; k = -\frac{3}{2}$

equation of circum circle $x^2 + y^2 + x + 3y = 0$

Question Stem for Question Nos. 15 and 16

Let $M = \{(x, y) \in R \times R : x^2 + y^2 \le r^2\}$, where $r > 0$. Consider the geometric progression $a_n = \frac{1}{2^{n-1}}, n = 1, 2, 3, \ldots$. Let $S_0 = 0$ and, for $n \ge 1$, let S_n denote the sum of the first n terms of this progression. For $n \ge 1$, let C_n denote the circle with center $(S_{n-1}, 0)$ and radius a_n, and D_n denote the circle with center (S_{n-1}, S_{n-1}) and radius a_n.

माना $M = \{(x, y) \in R \times R : x^2 + y^2 \le r^2\}$, जहाँ $r > 0$ है। गुणोत्तर श्रेणी $a_n = \frac{1}{2^{n-1}}, n = 1, 2, 3, \ldots$ विचार कीजिए। माना $S_0 = 0$ तथा $n \ge 1$ के लिए, S_n इस श्रेणी के प्रथम n पदों के योगफल को निरूपित करता है। $n \ge 1$ के लिए माना C_n उस वृत्त को निरूपित करता है जिसका केन्द्र $(S_{n-1}, 0)$ है तथा त्रिज्या a_n है तथा D_n उस वृत्त को निरूपित करता है जिसका केन्द्र (S_{n-1}, S_{n-1}) तथा त्रिज्या a_n है।

15. Consider M with $r = \frac{1025}{513}$. Let k be the number of all those circles C_n that are inside M. Let *l* be the maximum possible number of circles among these k circles such that no two circles intersect. Then :

[JEE(Adv.)2021]

M पर विचार कीजिए जहाँ $r=\frac{1025}{513}$ है। माना k उन सभी वृत्तों की C_n की संख्या है जो M के अंदर स्थित हैं। माना l इन k वृत्तों में से उन वृत्तों की अधिकतम संभावित संख्या है जिसमें कोई दो वृत प्रतिच्छेद नहीं करते हैं। तब : **[JEE(Adv.)2021]**

(A) $k+2l=22$ (B) $2k+l=26$ (C) $2k+3l=34$ (D) $3k+2l=40$

Ans. D

Sol. $a_n=\frac{1}{2^{n-1}}$

$S_n=1+\frac{1}{2}+.....\frac{1}{2^{n-1}}$

$S_n=2-\frac{1}{2^{n-1}}$

hence centre of C_n is $\left(2-\frac{1}{2^{n-2}},0\right)$ and radius of C_n is $\frac{1}{2^{n-1}}$

$r=\frac{1025}{513}<2$

here C_n inside M then

$d<|r_1-r_2|$

$2-\frac{1}{2^{n-2}}<2-\frac{1}{2^{n-1}}$

$2-\frac{1}{2^{n-2}}+\frac{1}{2^{n-1}}<2$

$\Rightarrow K=10$

also we observe that $l=5$ then option (D) is correct.

16. Consider M with $r=\frac{(2^{199}-1)\sqrt{2}}{2^{198}}$. The number of all those circles D_n that are inside M is :

[JEE(Adv.)2021]

M पर विचार कीजिए, जहाँ $r=\frac{(2^{199}-1)\sqrt{2}}{2^{198}}$, तब उन सभी वृत्तों D_n की संख्या कौनसी है जो M के अंदर स्थित है :

[JEE(Adv.)2021]

(A) 198 (B) 199 (C) 200 (D) 201

Ans. B

Sol. Centre of D_n here $S_n = 2 - \frac{1}{2^{n-1}}$

$\left(S_{n-1}, S_{n-1}\right)$ and radius is $\frac{1}{2^{n-1}}$

D_n lies inside M then

$d < |r_1 - r_2|$

$$\sqrt{2}\, S_{n-1} < \frac{\left(2^{199}-1\right)\sqrt{2}}{2^{198}} - \frac{1}{2^{n-1}}$$

$$\sqrt{2}\left(2 - \frac{1}{2^{n-2}}\right) < \frac{\left(2^{199}-1\right)\sqrt{2}}{2^{198}} - \frac{1}{2^{n-1}}$$

$$\frac{\sqrt{2}}{2^{n-2}} > \frac{\sqrt{2}}{2^{198}} + \frac{1}{2^{n-1}}$$

$\Rightarrow$ n = 199.

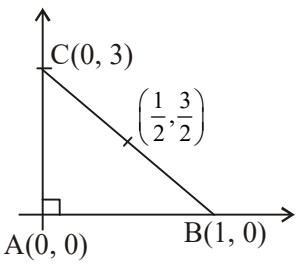
17. Let ABC be the triangle with AB = 1, AC = 3 and $\angle BAC = \frac{\pi}{2}$. If a circle of radius r > 0 touches the sides AB, AC and also touches internally the circumcircle of the triangle ABC, then the value of r is _____.

[JEE(Adv.)2022]

माना कि ABC एक त्रिभुज (triangle) है जिसमें AB = 1, AC = 3 एवं $\angle BAC = \frac{\pi}{2}$ है। यदि त्रिज्या (radius) r > 0 का एक वृत्त (circle) भुजाओं AB, AC को स्पर्श करता है एवं त्रिभुज ABc के परिवृत (circumcircle) को अंदर से स्पर्श (touches internally) करता है, तो r का मान _____ है। **[JEE(Adv.)2022]**

Ans. $4 - \sqrt{10}$

Sol. Let A be the origin B on x-axis, C on y-axis as shown below



$\therefore$ Equation of circumcircle is

$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{3}{2}\right)^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2 = \frac{5}{2} \quad \text{_____(1)}$$

Required circle touches AB and AC, have radius r

$\therefore$ Equation be $(x - r)^2 + (y - r)^2 = r^2$ _____(2)

If circle in equation (2) touches circumcircle internally, we have

$d_{c_1c_2} = |r_1 - r_2|$

$$\Rightarrow \left(\frac{1}{2}-r\right)^2+\left(\frac{3}{2}-r\right)^2=\left(\left|\sqrt{\frac{5}{2}}-r\right|\right)^2$$

$$\Rightarrow \frac{1}{4}+r^2-r+\frac{9}{4}+r^2-3r=\left(\sqrt{\frac{5}{2}}-r\right)^2 \text{ or } \left(r-\sqrt{\frac{5}{2}}\right)^2$$

$$\Rightarrow 2r^2-4r+\frac{5}{2}=\frac{5}{2}+r^2-\sqrt{10}r$$

$$\Rightarrow r=0 \text{ or } 4-\sqrt{10}$$

$$\Rightarrow r=0.837$$

18. Let G be a circle of radius $R>0$. Let $G_1, G_2, \ldots, G_n$ be n circles of equal radius $r>0$. Suppose each of the n circles $G_1, G_2, \ldots G_n$ touches the circle G externally. Also, for $i=1, 2, \ldots, n-1$, the circle G_i touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are TRUE? **[JEE(Adv.)2022]**

(A) If $n=4$, then $(\sqrt{2}-1)r<R$

(B) If $n=5$, then $r<R$

(C) If $n=8$, then $(\sqrt{2}-1)r<R$

(D) If $n=12$, then $\sqrt{2}\ (\sqrt{3}+1)\,r>R$

माना कि त्रिज्या $R>0$ का एक वृत्त G है। माना कि समान त्रिज्या $r>0$ वाले n वृत्त $G_1, G_2, \ldots, G_n$ है। मान लिजिए कि n वृत्तों $G_1, G_2, \ldots G_n$ में से प्रत्येक वृत्त, वृत्त G को बाह्यस्पर्श करता है। इसके अलावा $i=1, 2, \ldots, n-1$ के लिए वृत्त G_i, G_{i+1} को बाह्यस्पर्श करता है, एवं G_n, G_1 को बाह्यस्पर्श करता है। तब निम्न कथनों में से कौनसा सत्य हैं? **[JEE(Adv.)2022]**

(A) यदि $n=4$ है, तब $(\sqrt{2}-1)r<R$ है

(B) यदि $n=5$ है, तब $r<R$ है

(C) यदि $n=8$ है, तब $(\sqrt{2}-1)r<R$ है

(D) यदि $n=12$ है, तब $\sqrt{2}\ (\sqrt{3}+1)\,r>R$ है

Ans. CD

Sol. $2(R+r)\sin\frac{\pi}{n}=2r$

$$\frac{R+r}{r}=\text{cosec}\frac{\pi}{n}$$

(A) $n=4, R+r=\sqrt{2}r$

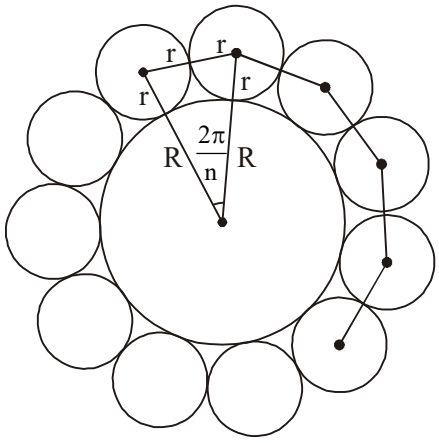
(B) $n=5, \frac{R+r}{r}=\text{cosec}\frac{\pi}{5}<\text{cosec}\frac{\pi}{6}$

$R+r<2r \Rightarrow r>R$

(C) $n=8, \frac{R+r}{r}=\text{cosec}\frac{\pi}{8}>\text{cosec}\frac{\pi}{4}$

$R+r>\sqrt{2}r$

(D) $n=12, \frac{R+r}{r}=\text{cosec}\frac{\pi}{12}=\sqrt{2}(\sqrt{3}+1)$



$$R + r = \sqrt{2}(\sqrt{3}+1)r$$

$$\sqrt{2}(\sqrt{3}+1)r > R$$

19. Let C_1 be the circle of radius 1 with center at the origin. Let C_2 be the circle of radius r with center at the point A = (4, 1), where $1 < r < 3$. Two distinct common tangents PQ and ST of C_1 and C_2 are drawn. The tangent PQ touches C_1 at P and C_2 at Q. The tangent ST touches C_1 at S and C_2 at T. Mid points of the line segments PQ and ST are joined to form a line which meets the x–axis at a point B. If $AB = \sqrt{5}$, then the value of r^2 is _____.

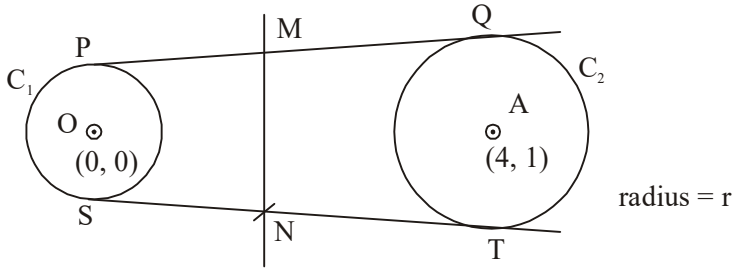
[JEE(Adv.)2023]

माना कि C_1 एक वृत्त है जिसकी त्रिज्या 1 और केन्द्र मूल बिन्दु हैं। माना कि C_2 एक वृत्त है जिसकी त्रिज्या r, जहाँ $1 < r < 3$ है और केन्द्र बिन्दु A = (4,1) है। C_1 एवं C_2 की दो उभयनिष्ठ स्पर्श रेखाएँ (distinct common tangents) PQ एवं ST खींची जाती हैं। स्पर्श रेखा PQ, वृत्त C_1 को P पर और वृत्त C_2 को Q पर स्पर्श करती हैं। स्पर्श रेखा ST, वृत्त C_1 को S पर और वृत्त C_2 को T पर स्पर्श करती है। रेखा खंडों PQ एवं ST के मध्य बिन्दुओं को मिलाकर एक रेखा बनाई जाती है जो x–अक्ष को बिन्दु B पर मिलती है। यदि $AB = \sqrt{5}$, तब r^2 का मान है _____

[JEE(Adv.)2023]

Ans. 2

Sol.



Let the tangents are DCT's

line MN will be radical Axis as PM = PQ and SN = NT

$C_1 : x^2 + y^2 = 1$

$C_2 : (x-4)^2 + (y-1)^2 = r^2$

RA : $S_1 - S_2 = 0$

$\Rightarrow 8x + 2y = 18 - r^2$

Point B is on x-axis

$$\text{put } y = 0 \qquad \Rightarrow x = \frac{18 - r^2}{8}$$

So co-ordinates of $B\left(\frac{18-r^2}{8}, 0\right)$

Given $AB = \sqrt{5}$ $\Rightarrow AB^2 = 5$

$$\Rightarrow \left(\frac{18-r^2}{8}-4\right)^2+1=5$$

$$\Rightarrow \frac{18-r^2-32}{8}=\pm 2$$

$\Rightarrow r^2 + 14 = \pm 16$

$r^2 = -30$ (rejected)

$r^2 = 2$

Ans. 2

20. Let the straight line $y = 2x$ touch a circle with center $(0,\alpha)$, $\alpha > 0$, and radius r at a point A_1. Let B_1 be the point on the circle such that the line segment A_1B_1 is a diameter of the circle. Let $\alpha + r = 5 + \sqrt{5}$.

Match each entry in **List-I** to the correct entry in **List-II**.

List-I	**List-II**
(P) α equals	(1) $(-2,4)$
(Q) r equals	(2) $\sqrt{5}$
(R) A_1 equals	(3) $(-2,6)$
(S) B_1 equals	(4) 5
	(5) $(2,4)$

The correct option is : **[JEE(Adv.)2024]**

माना कि सरल रेखा $y = 2x$, एक वृत्त जिसका केन्द्र $(0,\alpha)$, $\alpha > 0$, है और जिसकी त्रिज्या r है, को एक बिन्दु A_1 पर स्पर्श करती है। माना कि B_1 वृत्त पर वह बिन्दु है कि रेखाखंड A_1B_1 वृत्त का एक व्यास है। माना कि $\alpha + r = 5 + \sqrt{5}$ है।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कीजिए।

सूची-I	**सूची-II**
(P) α बराबर	(1) $(-2,4)$
(Q) r बराबर	(2) $\sqrt{5}$
(R) A_1 बराबर	(3) $(-2,6)$
(S) B_1 बराबर	(4) 5
	(5) $(2,4)$

सही विकल्प है : **[JEE(Adv.)2024]**

(A) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)
(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)
(C) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (3)
(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)

Ans. C

Sol. Circle $(x - 0)^2 + (y - \alpha)^2 = r^2$

tangent $2x - y = 0$

by condition of tangency

$$\left|\frac{0-\alpha}{\sqrt{5}}\right| = r$$

$$|-\alpha| = r\sqrt{5}$$

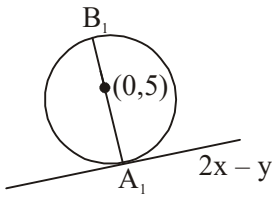
$$\alpha = \pm r\sqrt{5}$$

$\alpha = r\sqrt{5}$ $(\alpha > 0)$ _____(1)

given $\alpha + r = 5 + \sqrt{5}$ _____(2)

by (1) & (2)

$\alpha = 5,\ r = \sqrt{5}$



Circle $x^2 + (y-5)^2 = 5$

tangent $2x - y = 0$

then A_1 (2,4)

Clearly B_1 (–2,6)

Hence, correct option is (C).

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. If $a > 2b > 0$ then the positive value of m for which $y = mx - b\sqrt{1+m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x-a)^2 + y^2 = b^2$ is :

यदि $a > 2b > 0$ तो m का धनात्मक मान जिसके लिये रेखा $y = mx - b\sqrt{1+m^2}$ वृत्तों $x^2 + y^2 = b^2$ व $(x-a)^2 + y^2 = b^2$ की उभयनिष्ठ स्पर्श रेखा है, होगा :

(A) $\frac{2b}{\sqrt{a^2-4b^2}}$ (B) $\frac{\sqrt{a^2-4b^2}}{2b}$ (C) $\frac{2b}{a-2b}$ (D) $\frac{b}{a-2b}$

Ans. A

Sol. $y = mx - b\sqrt{1+m^2}$

$x^2 + y^2 = b^2$

$y = mx - b\sqrt{1+m^2}$ touches $(x-a)^2 + y^2 = b^2$

C.O.T $\Rightarrow P = r$

$$\Rightarrow \left|\frac{ma - b\sqrt{1+m^2}}{\sqrt{1+m^2}}\right| = b$$

$$\Rightarrow \left|ma - b\sqrt{1+m^2}\right| = b\sqrt{1+m^2}$$

$$\Rightarrow m^2a^2 + b^2\left(1+m^2\right) - 2mab\sqrt{1+m^2} = b^2\left(1+m^2\right)$$

$$\Rightarrow ma\left[ma - 2b\sqrt{1+m^2}\right] = 0$$

$$\Rightarrow ma = 2b\sqrt{1+m^2}$$

Given slope $\left(\because ma \neq 0; a \neq 0 \,\&\, m \neq 0\right)$

$$\Rightarrow m^2a^2 = 4b^2 + 4b^2m^2 \Rightarrow m = \frac{2b}{\sqrt{a^2-4b^2}}$$

2. A circle touches a straight line $lx + my + n = 0$ & cuts the circle $x^2 + y^2 = 9$ orthogonally. The locus of centres of such circles is :

(A) $(lx+my+n)^2 = (l^2+m^2)(x^2+y^2-9)$ (B) $(lx+my-n)^2 = (l^2+m^2)(x^2+y^2-9)$

(C) $(lx+my+n)^2 = (l^2+m^2)(x^2+y^2+9)$ (D) none of these

एक वृत्त सरल रेखा $lx + my + n = 0$ को स्पर्श करता है तथा वृत्त $x^2 + y^2 = 9$ को लम्बकोणीय काटता है, तो इसके केन्द्र का बिन्दुपथ हैं :

(A) $(lx+my+n)^2 = (l^2+m^2)(x^2+y^2-9)$ (B) $(lx+my-n)^2 = (l^2+m^2)(x^2+y^2-9)$

(C) $(lx+my+n)^2 = (l^2+m^2)(x^2+y^2+9)$ (D) इनमें से कोई नहीं

Ans. A

Sol. Let required equation of circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

it cuts the circle $x^2 + y^2 - 9 = 0$ orthogonally (By using condition of orthogonality)

$\therefore\ 2g(0) + 2f(0) = c - 9 \Rightarrow c = 9$

It also touches straight line $\ell x + my + n = 0$

$$\therefore\ \left|\frac{\ell(-g)+m(-f)+n}{\sqrt{\ell^2+m^2}}\right| = \sqrt{g^2+f^2-9}$$

(By using conditionof tangency)

Locus of centre

(–g, –f) is $(\ell x + my + n)^2 = (x^2 + y^2 - 9)(\ell^2 + m^2)$

3 The circle $x^2+y^2 = 4$ cuts the circle $x^2+y^2+2x+3y-5 = 0$ in A & B. Then the equation of the circle on AB as a diameter is :

(A) $13(x^2+y^2)-4x-6y-50 = 0$ (B) $9(x^2 + y^2) + 8x - 4y + 25 = 0$

(C) $x^2 + y^2 - 5x + 2y + 72 = 0$ (D) none of these

यदि वृत्त $x^2+y^2 = 4$ एक अन्य वृत्त $x^2+y^2+2x+3y-5 = 0$ को बिन्दु A एवं B पर काटता है, तो AB व्यास वाले वृत्त का समीकरण हैं :

(A) $13(x^2+y^2)-4x-6y-50 = 0$ (B) $9(x^2 + y^2) + 8x - 4y + 25 = 0$

(C) $x^2 + y^2 - 5x + 2y + 72 = 0$ (D) इनमें से कोई नहीं

Ans. A

Sol. Let $S_2 = x^2 + y^2 - 4 = 0$

$S_1 = x^2 + y^2 + 2x + 3y - 5 = 0$

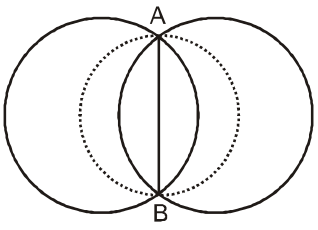
Common chord of given circle

$2x + 3y - 1 = 0 \quad (S_1 - S_2 = 0)$

family of circle passing through point of intersection of given circle

$(x^2 + y^2 + 2x + 3y - 5) + \lambda(x^2 + y^2 - 4) = 0 \Rightarrow (\lambda + 1)x^2 + (\lambda + 1)y^2 + 2x + 3y - (4\lambda + 5) = 0$

$x^2 + y^2 + \dfrac{2x}{\lambda+1} + \dfrac{3}{\lambda+1}y - \dfrac{(4\lambda+5)}{\lambda+1} = 0$



centre $\left(-\dfrac{1}{\lambda+1}, \dfrac{-3}{2(\lambda+1)}\right)$

This centre lies on AB

$2\left(-\dfrac{1}{\lambda+1}\right) + 3\left(\dfrac{-3}{2(\lambda+1)}\right) - 1 = 0 \Rightarrow -4 - 9 - 2\lambda - 2 = 0 \Rightarrow 2\lambda = -15 \Rightarrow \lambda = -15/2$

$\left(-\dfrac{15}{2}+1\right)x^2 + \left(-\dfrac{15}{2}+1\right)y^2 + 2x + 3y - \left(-4\times\dfrac{15}{2}+5\right) = 0 \Rightarrow -\dfrac{13x^2}{2} - \dfrac{13y^2}{2} + 2x + 3y + 25 = 0$

$\Rightarrow 13(x^2 + y^2) - 4x - 6y - 50 = 0.$

4. The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If Q and R have co–ordinates (3, 4) and (–4, 3) respectively, the $\angle QPR$ is equal to :

वृत्त $x^2 + y^2 = 25$ के अन्दर त्रिभुज PQR बनाया जाता है। यदि Q एवं R के निर्देशांक क्रमशः (3,4) एवं (– 4,3) हो, तो $\angle QPR$ बराबर हैं :

(A) $\dfrac{\pi}{2}$ (B) $\dfrac{\pi}{3}$ (C) $\dfrac{\pi}{4}$ (D) $\dfrac{\pi}{6}$

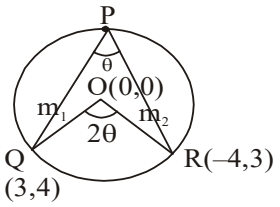
Ans. C

Sol. $x^2 + y^2 = 25$, Q(3,4), R(–4,3)

Angle subtended by an arc at the centre is double the angle subtended by same arc any point on the circle.

$m_1 = \dfrac{4}{3}$

$m_2 = -\dfrac{3}{4}$



$\Rightarrow m_1 m_2 = -1$

$\Rightarrow 2\theta = 90^\circ$

$\theta = 45^\circ$

5. If two distinct chords, drawn from the point (p, q) on the circle $x^2 + y^2 = px + qy$ (where $pq \neq 0$) are bisected by the x–axis, then :

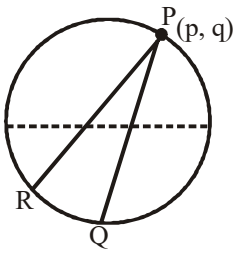
(A) $p^2 = q^2$ (B) $p^2 = 8q^2$ (C) $p^2 < 8q^2$ (D) $p^2 > 8q^2$

किसी बिन्दु (p,q) से वृत्त $x^2 + y^2 = px + qy$, जहाँ $p.q \neq 0$ है, पर दो जीवाएँ खींची जाती है, जो x–अक्ष पर समद्विभाजित होती है, तो :

(A) $p^2 = q^2$ (B) $p^2 = 8q^2$ (C) $p^2 < 8q^2$ (D) $p^2 > 8q^2$

Ans. D

Sol. $x^2 + y^2 - px - qy = 0$



$y = -q$

$\Rightarrow x^2 + q^2 - px + q^2 = 0$

$\Rightarrow x^2 - px + 2q^2 = 0$

this equation must have two distinct roots

PQ and PR are bisected by x-axis

Hence mid point of PQ and PR lies on x-axis.

PQ mid point $\left(\frac{p+x}{2}, \frac{q+y}{2}\right) = (x, 0)$

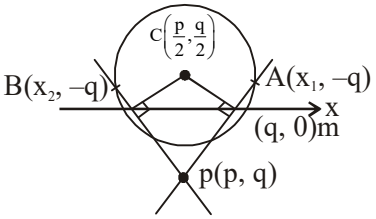
$\Rightarrow q = -y$

Hence, $D > 0 \Rightarrow p^2 - 8q^2 > 0$

$p^2 > 8q^2$

Alternative :

$CM \perp AP$



$$\frac{\frac{q}{2}}{\frac{p}{2} - a} \times \frac{q}{p - a} = -1$$

$\Rightarrow 2a^2 - 3pa + p^2 + q^2 = 0$

Quad. in 'a' which must have two real and distinct solution

$D > 0$

$\Rightarrow p^2 > 8q^2$

6. Let PQ and RS be tangents at the extremities of the diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle then 2r equals :

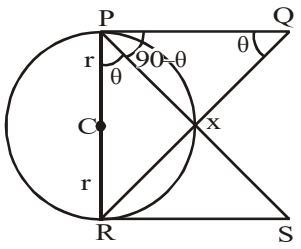
माना कि PQ व RS किसी वृत जिसकी त्रिज्या r है के व्यास PR के सिरों पर स्पर्श रेखायें है। यदि PS व RQ वृत्त की परिधि पर स्थित किसी बिन्दु X पर प्रतिच्छेद करते है तो 2r का मान होगा :

(A) $\sqrt{PQ \cdot RS}$ (B) $\frac{PQ + RS}{2}$ (C) $\frac{2PQ \cdot RS}{PQ + RS}$ (D) $\sqrt{\frac{(PQ)^2 + (RS)^2}{2}}$

Ans. A

Sol. Let $\angle RPS = \theta$

$\angle XPQ = 90 - \theta$



$\therefore$ $\angle PQX = \theta$ ($\because \angle PXQ = 90°$)

$\therefore$ $\Delta PRS \sim \Delta QPR$ (AAA similarity)

$\therefore$ $\frac{PR}{QP} = \frac{RS}{PR} \Rightarrow PR^2 = PQ.RS$

$\Rightarrow$ $PR = \sqrt{PQ.RS}$

7. A circle is given by $x^2 + (y-1)^2 = 1$, another circle C touches it externally and also the x–axis, then the locus of its centre is :

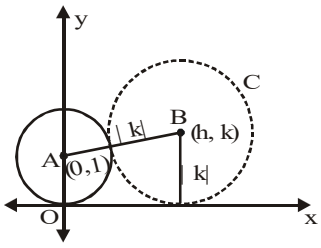
एक वृत्त एक दिये गये वृत्त $x^2 + (y-1)^2 = 1$ को बाह्य स्पर्श करता है तथा x अक्ष को भी स्पर्श करता है, तो उसके केन्द्र का बिन्दुपथ है :

(A) $\{(x, y) : x^2 = 4y\} \cup \{(x, y) : y \le 0\}$ (B) $\{(x, y) : x^2 + (y-1)^2 = 4\} \cup \{x, y) : y \le 0\}$

(C) $\{(x, y) : x^2 = y\} \cup \{(0, y) : y \le 0\}$ (D) $\{(x, y) : x^2 = 4y\} \cup \{(0, y) : y \le 0\}$

Ans. D

Sol. Let the centre of circle C be (h, k). Then as this circle touches axis of x its radius = |k|



Also it touches the given circle $x^2 + (y-1)^2 = 1$, centre (0, 1) radius 1, externally

Therefore

The distance between centres = sum of radii

$\Rightarrow$ $\sqrt{(h-0)^2 + (k-1)^2} = 1 + |k|$

$\Rightarrow$ $h^2 + k^2 - 2k + 1 = (1 + |k|)^2$

$\Rightarrow$ $h^2 + k^2 - 2k + 1 = 1 + 2|k| + k^2$

$\Rightarrow$ $h^2 = 2k + 2|k|$

$\therefore$ Locus of (h, k) is, $x^2 = 2y + 2|y|$

Case - 1 if $y > 0$, it becomes $x^2 = 4y$

and if $y \le 0$, it becomes $x = 0$

$\therefore$ on taking union of both the cases, the required locus is

$\{(x, y) : x^2 = 4y\} \cup \{(0, y) : y \le 0\}$

8. A circle with center (2,2) touches the coordinate axes and a straight line AB where A and B lie on positive direction of coordinate axes such that the circle lies between origin and the line AB. If O be the origin then the locus of circumcenter of ΔOAB will be :

एक वृत जिसका केन्द्र (2,2) है, निर्देशी अक्षों को स्पर्श करता है तथा एक सरल रेखा AB, जहाँ A तथा B निर्देशी अक्षों पर धनात्मक दिशा में स्थित है, इस प्रकार है कि वृत मूल बिन्दु तथा रेखा AB के मध्य स्थित है। यदि O मूल बिन्दु हो, तो ΔOAB के परिकेन्द्र का बिन्दुपथ होगा :

(A) $xy = x + y + \sqrt{x^2 + y^2}$ (B) $xy = x + y - \sqrt{x^2 + y^2}$

(C) $xy + x + y = \sqrt{x^2 + y^2}$ (D) $xy + x + y + \sqrt{x^2 + y^2} = 0$

Ans. A

Sol. $AB :\rightarrow \frac{x}{2h} + \frac{y}{2k} = 1$

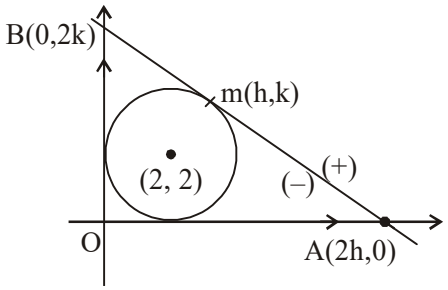
$AB : kx + hy = 2hk$

$p = r = 2$

$$\left|\frac{2k + 2h - 2hk}{\sqrt{h^2 + k^2}}\right| = 2$$

$$hk - h - k = 2\sqrt{h^2 + k^2}$$

$$xy = x + y + 2\sqrt{x^2 + y^2}$$



9. From a point 'P' on the line $2x + y + 4 = 0$; which is nearest to the circle $x^2 + y^2 - 12y + 35 = 0$, tangents are drawn to given circle. The area of quadrilateral PACB (where 'C' is the center of circle and PA & PB are the tangents) is:

(A) 8 (B) $\sqrt{110}$ (C) $\sqrt{19}$ (D) None of these

रेखा $2x + y + 4 = 0$ पर स्थित एक बिन्दु 'P' से वृत $x^2 + y^2 - 12y + 35 = 0$ पर स्पर्श रेखाऐं खींची जाती है, जहाँ बिन्दु P वृत से निकटतम दूरी पर है। चतुर्भुज PACB (जहाँ 'C' वृत का केन्द्र तंथा PA व PB स्पर्श रेखाऐं है) का क्षेत्रफल होगा :

(A) 8 (B) $\sqrt{110}$ (C) $\sqrt{19}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $2x + y + 4 = 0 \rightarrow$ 'P' Circle : $x^2 + y^2 - 12y + 35 = 0$

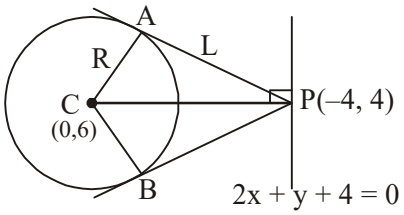
P is foot of perpendicular from centre C(0,6) to the line $2x + y + 4 = 0$

$$\frac{x - 0}{2} = \frac{y - 6}{1} = -\frac{(10)}{5} = -2 \Rightarrow P(-4, 4)$$

$L = \sqrt{S_1} = \sqrt{16+16-48+35} = \sqrt{19}$

$R = \sqrt{0+36-35} = 1$

Area of $PACB = 2 \times \frac{1}{2} \times L \times R = \sqrt{19}$



10. A square OABC is formed by line pairs $xy = 0$ and $xy + 1 = x + y$ where 'O' is the origin. A circle with center C_1 inside the square is drawn to touch the line pair $xy = 0$ and another circle with centre C_2 and radius twice that of C_1, is drawn to touch the circle C_1 and the other line pair. The radius of the circle with centre C_1 is :

रेखा युग्मों $xy = 0$ तथा $xy + 1 = x + y$ से एक वर्ग OABC बनाया जाता है, जहाँ 'O' मूल बिन्दु है। एक वृत जिसका केन्द्र C_1 है, दिये गये वर्ग के अन्दर रेखा युग्म $xy = 0$ को स्पर्श करते हुये बनाया जाता है तथा दूसरा वृत जिसका केन्द्र C_2 तथा त्रिज्या C_1 की दुगुनी है, वृत C_1 तथा दूसरे रेखा युग्म को स्पर्श करते हुये बनाया जाता है। C_1 केन्द्र वाले वृत की त्रिज्या होगी :

(A) $\frac{\sqrt{2}}{\sqrt{3}\left(\sqrt{2}+1\right)}$ (B) $\frac{2\sqrt{2}}{3\left(\sqrt{2}+1\right)}$ (C) $\frac{\sqrt{2}}{3\left(\sqrt{2}+1\right)}$ (D) $\frac{\sqrt{2}+1}{3\sqrt{2}}$

Ans. C

Sol. $xy = 0, xy + 1 = x + y$

$\Rightarrow x - xy + y - 1 = 0$

$\Rightarrow -x(y-1) + (y-1) = 0$

$\Rightarrow (y-1)(1-x) = 0$

$\Rightarrow x = 1, y = 1$

$C_1(r, r)$ & $C_2(1 - 2r, 1 - 2r)$

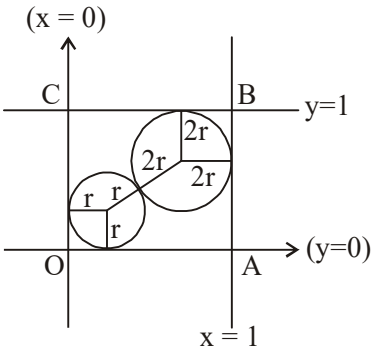
$C_1C_2 = 3r$

$\Rightarrow \left(1-3r^2\right) + (1-3r)^2 = 9r^2$

$\Rightarrow 9r^2 - 12r + 2 = 0$

$\Rightarrow r = \frac{12 \pm \sqrt{144-72}}{18} = \frac{12 \pm \sqrt{72}}{18} = \frac{12 \pm 6\sqrt{2}}{18}$

$\Rightarrow r = \frac{2 \pm \sqrt{2}}{3}$



$r = \frac{2+\sqrt{2}}{3}$ $r = \frac{2-\sqrt{2}}{3}$

$r = \frac{\sqrt{2}\left(\sqrt{2}+1\right)}{3}$ $r = \frac{\sqrt{2}\left(\sqrt{2}-1\right)}{3}$

$r = \frac{\sqrt{2}}{3\left(\sqrt{2}-1\right)}$ $r = \frac{\sqrt{2}}{3\left(\sqrt{2}+1\right)}$

11. A circle of radius unity is centred at origin. Two particles start moving at the same time from the point (1,0) and move around the circle in opposite direction. One of the particle moves counter clockwise with constant speed v and the other moves clockwise with constant speed 3v. After leaving (1,0), the two particles meet first at a point P, and continue until they meet next at point Q. The coordinates of the point Q are :

इकाई त्रिज्या वाला एक वृत मूल बिन्दु पर केन्द्रित है। दो कण एक समय पर बिन्दु (1,0) से चलना प्रारम्भ करते है तथा वृत के संगत विपरित दिशा में चलते है। इनमें से एक कण वामावर्त दिशा में नियत चाल v से गति करता है तथा दूसरा कण दक्षिणावर्त दिशा में नियत चाल 3v से गति करता है। बिन्दु (1,0) से निकलने के बाद दोनों कण पहली बार बिन्दु P पर मिलते है तथा उसके पश्चात उसी क्रम में चलते हुये बिन्दु Q पर मिलते है। बिन्दु Q के निर्देशांक होगें :

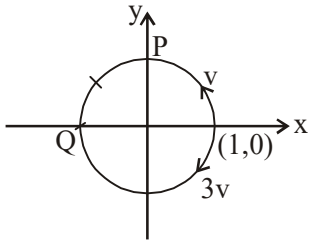
(A) (1,0) (B) (0,1) (C) (0,–1) (D) (–1,0)

Ans. D

Sol. $x^2 + y^2 = 1$

speed of 2nd particle is 3 times so the distance travelled in same time is also 3 times.

Hence, P is (0,1) & Q is (–1,0).



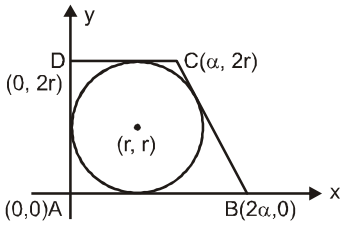
12. Let ABCD be a quadrilateral with area 18, with side AB parallel to the side CD and AB = 2CD. Let AD be perpendicular to AB and CD. If a circle is drawn inside the quadrilateral ABCD touching all the sides, then its radius is :

माना कि चतुर्भुज ABCD का क्षेत्रफल 18 है, AB व CD समान्तर है व AB = 2CD है। माना कि भुजा AD, AB व CD के लम्बवत् है। यदि एक वृत्त चतुर्भुज ABCD के अन्दर इस प्रकार से खींचा जाये कि व चारों भुजाओं को स्पर्श करें तो उसकी त्रिज्या होगी :

(A) 3 (B) 2 (C) 3/2 (D) 1

Ans. B

Sol. $18 = \frac{1}{2}(3\alpha)(2r) \Rightarrow \alpha r = 6$



Line, $y = -\frac{2r}{\alpha}(x - 2\alpha)$ is tangent to circle

$(x - r)^2 + (y - r)^2 = r^2$

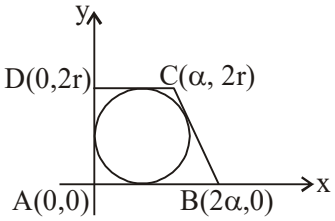
$2\alpha = 3r$ and $\alpha r = 6$

$r = 2$

Alter

$AB = 2CD$

$\text{Equ. of BC} : (y - 0) = \frac{(2r - 0)}{-\alpha}(x - 2\alpha)$



$BC : 2rx + \alpha y - 4\alpha r = 0$

apply C.O.T : $p = r$

$\Rightarrow \frac{|2r^2 + \alpha r - 4\alpha r|}{\sqrt{4r^2 + \alpha^2}} = r$

$\Rightarrow |2r - 3\alpha| = \sqrt{4r^2 + \alpha^2} \qquad \Rightarrow 4r^2 - 12\alpha r + 9\alpha^2 = 4r^2 + \alpha^2$

$\Rightarrow 8\alpha^2 - 12\alpha r = 0 \qquad \Rightarrow \alpha - 3r = 0 \qquad \Rightarrow \alpha = \frac{3}{2}r \qquad$ _____(1)

$\text{Area of ABCD} = \frac{1}{2} \times 3\alpha \times 2r = 18$

$\Rightarrow \alpha r = 6 \qquad \Rightarrow 3r^2 = 12 \qquad \Rightarrow r = 2 \qquad$ _____(2)

13. A line meets the co-ordinate axes in A and B. A circle is circumscribed about the triangle OAB. If d_1 and d_2 are the distances of the tangent to the circle at the origin O from the points A and B respectively, the diameter of the circle is :

एक रेखा निर्देशांक अक्षो को A तथा B पर मिलता है। त्रिभुज OAB का एक परिवृत है। यदि d_1 तथा d_2 बिन्दुओं A तथा B पर क्रमशः मूल बिन्दु पर स्पर्श रेखा से दूरी है तो वृत का व्यास है :

(A) $\frac{2d_1 + d_2}{2}$ (B) $\frac{d_1 + 2d_2}{2}$ (C) $d_1 + d_2$ (D) $\frac{d_1 d_2}{d_1 + d_2}$

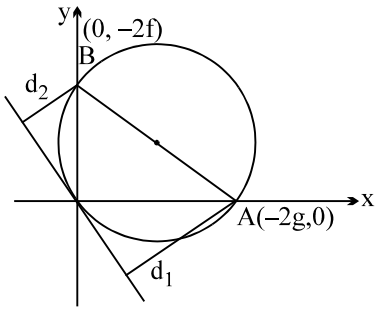
Ans. C

Sol. Let the circle be $x^2 + y^2 + 2gx + 2fy = 0$

Tangent at the origin is $gx + fy = 0$

$$\left|\frac{g(0)+f(-2f)}{\sqrt{g^2+f^2}}\right|$$

$$\left.\begin{aligned} d_1 &= \frac{2g^2}{\sqrt{g^2+f^2}} \\ d_2 &= \frac{2f^2}{\sqrt{g^2+f^2}} \end{aligned}\right\} \Rightarrow d_1 + d_2 = 2\sqrt{g^2+f^2} = \text{diameter}$$



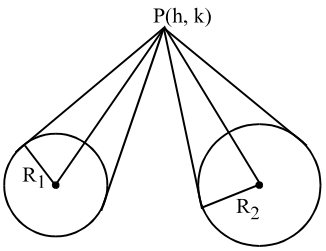
14. The locus of the point from which two given unequal circle subtend equal angles is :

(A) a straight line (B) a circle (C) a parabola (D) none

उस बिन्दु का बिन्दुपथ की दिये गये दो वृत उस पर समान कोण अंतरित करते है :

(A) सरल रेखा (B) वृत्त (C) परवलय (D) कोई नहीं

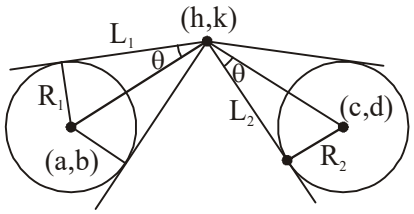
Ans. B



Sol. Use $\frac{R_1}{L_1} = \frac{R_2}{L_2}$

$R_1L_2 = R_2L_1$

Aliter :



$$\frac{R_1}{L_1} = \frac{R_2}{L_2}$$

$$\Rightarrow R_1^2L_2^2 = R_2^2L_1^2$$

$$\Rightarrow R_2^2\left[(h-a)^2+(k-b)^2-R_1^2\right] = R_1^2\left[(n-c)^2+(k-d)^2-R_2^2\right]$$

$$\Rightarrow R_2^2(h-a)^2 + R_2^2(k-b)^2 = R_1^2(h-c)^2 + R_1^2(k-d)^2$$

$\Rightarrow$ Circle

15. Triangle ABC is right angled at A. The circle with centre A and radius AB cuts BC and AC internally at D and E respectively. If BD = 20 and DC = 16 then the length AC equals :

त्रिभुज ABC कोण A पर समकोणीय है। AB त्रिज्या तथा A केन्द्र वाला वृत BC तथा AC को D एवं E पर क्रमशः अन्तः विभाजित करता है। यदि BD = 20 तथा DC = 16 तो AC की लम्बाई है :

(A) $6\sqrt{21}$ (B) $6\sqrt{26}$ (C) 30 (D) 32

Ans. B

Sol. $b^2 + r^2 = (36)^2$ ....(1)

Also $CD \cdot CB = CE \cdot CX$ (using power of the point C)

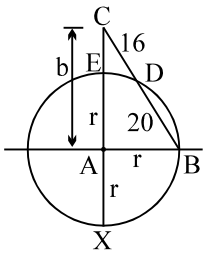
$16 \cdot 36 = (b - r)(b + r)$

$\therefore \quad b^2 - r^2 = 16 \cdot 36$

from (1) and (2)

$2b^2 = 36(36 + 16) = 36 \cdot 52$

$b^2 = 36 \cdot 26 \quad \Rightarrow \quad b = 6\sqrt{26}$ Ans.



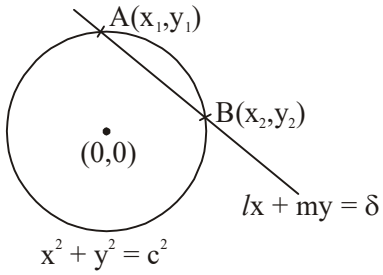
16. If H represent the harmonic mean between the abscissae, and K that between the ordinates of the points, in which a circle $x^2 + y^2 = c^2$ is cut by a chord $lx + my = \delta$, where l and m are the direction cosines of the unit vector in the xy plane, then lH + mK has the value equal to : (Take: $l^2 + m^2 = 1$)

यदि H भुजों का हरात्मक माध्य एवं K उन बिन्दुओं पर कोटि का हरात्मक माध्य है जिन पर वृत $x^2 + y^2 = c^2$ एवं जीवा $lx + my = \delta$ प्रतिच्छेदन करते है, जहाँ l तथा m, x-y तल में स्थित एक इकाई सदिश की दिक् कोज्याऐं है तो lH + mK का मान है : (Take: $l^2 + m^2 = 1$)

(A) $2\delta - \frac{c^2}{\delta}$ (B) $\delta - \frac{c^2}{2\delta}$ (C) $\delta - \frac{2c^2}{\delta}$ (D) $2\delta - \frac{c^2}{2\delta}$

Ans. A

Sol.



$\ell^2 + m^2 = 1$

$x = \frac{\delta - my}{\ell}, \; y = \frac{\delta - \ell x}{m}$

$$\left(\frac{\delta - my}{\ell}\right)^2 + y^2 = c^2 \qquad x^2 + \left(\frac{\delta - \ell x}{m}\right)^2 = c^2$$

$$\Rightarrow m^2y^2 - 2\delta my + \delta^2 + \ell^2y^2 = \ell^2c^2 \qquad \Rightarrow m^2x^2 + \ell^2x^2 - 2\ell\delta x + \delta^2 - c^2m^2 = a$$

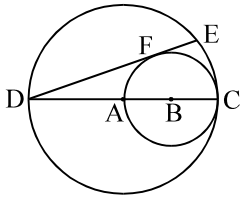
$$\Rightarrow K = \frac{2y, y_2}{y_1 + y_2} = \frac{2\left(\delta^2 - \ell^2c^2\right)}{2\delta m} \qquad \Rightarrow H = \frac{2x_1x_2}{x_1 + x_2} = \frac{2\left(\delta^2 - c^2m^2\right)}{2\ell\delta}$$

$$\ell H + mk = \frac{\delta^2 - c^2m^2}{\delta} + \frac{\delta^2 - \ell^2c^2}{\delta} = \frac{2\delta^2 - c^2\left(\ell^2 + m^2\right)}{\delta}$$

$$\Rightarrow 2\delta - \frac{c^2}{\delta}$$

17. In the diagram, DC is a diameter of the large circle centered at A, and AC is a diameter of the smaller circle centered at B. If DE is tangent to the smaller circle at F and DC = 12 then the length of DE is :

दिखाये गये चित्रनुसार बड़े तथा छोटे वृत का व्यास DC एवं AC है तथा जिनका केन्द्र क्रमशः A एवं B है यदि DE छोटे वृत पर खीची गयी स्पर्श रेखा है तथा DC = 12 तो DE की लम्बाई बताओं :



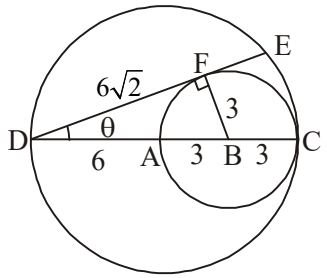
(A) $8\sqrt{2}$ (B) 16 (C) $9\sqrt{2}$ (D) $10\sqrt{2}$

Ans. A

Sol. $DF = \sqrt{9^2 - 3^2} = \sqrt{72} = 6\sqrt{2}$

$$\cos\theta = \frac{9}{6\sqrt{2}} = \frac{12}{DE}$$

$$\Rightarrow DE = 8\sqrt{2}$$



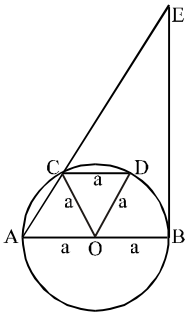
18. AB is a diameter of a circle. CD is a chord parallel to AB and 2 CD = AB. The tangent at B meets the line AC produced at E then AE is equal to :

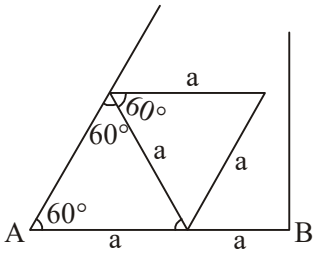
वृत की व्यास AB तथा इसके समान्तर जीवा CD हो यदि 2CD = AB तथा B पर खीची गयी स्पर्श रेखा यदि AC को बढ़ाने पर E पर मिलती है तो AE की लम्बाई होगी :

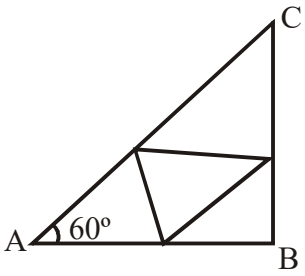
(A) AB (B) $\sqrt{2}$ AB (C) $2\sqrt{2}$ AB (D) 2 AB

Ans. D

Sol.







$AB = AE \cos 60°$

$AB = \frac{AE}{2} \Rightarrow AE = 2AB$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

19. If $a = \max\{(x+2)^2 + (y-3)^2\}$ and $b = \min\{(x+2)^2 + (y-3)^2\}$ where x,y satisfying $x^2 + y^2 + 8x - 10y - 40 = 0$, then :

यदि $a = \max\{(x+2)^2 + (y-3)^2\}$ तथा $b = \min\{(x+2)^2 + (y-3)^2\}$ जहाँ x,y, $x^2 + y^2 + 8x - 10y - 40 = 0$ को संतुष्ट करते है, तो :

(A) a + b = 18 (B) a + b = 178 (C) a – b = $4\sqrt{2}$ (D) a – b = $72\sqrt{2}$

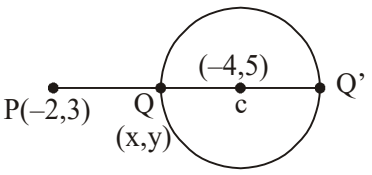
Ans. BD

Sol. C(–4,5)

$r = \sqrt{16 + 25 + 40} = \sqrt{81}$

r = 9

$a = PQ^2\big|_{max}$ & $b = PQ^2\big|_{max}$



$a = (cp + r^2)$ $b = (cp - r)^2$

$CP = \sqrt{4+4} = 2\sqrt{2}$

(B) $a + b = 2[CP^2 + r^2] = 2[8 + 81] = 2 \times 89 = 178$

(D) $a - b = 2 \times 2CP \times r = 4 \times 2\sqrt{2} \times 9 = 72\sqrt{2}$

20. A square is inscribed in the circle $x^2 + y^2 - 2x + 4y - 93 = 0$ with the sides parallel to the co–ordinate axes. The co–ordinate of the vertices are :

वृत $x^2 + y^2 - 2x + 4y - 93 = 0$ के अन्दर निर्देशी अक्षों के समान्तर भुजाओं वाला एक वृत बनाया जाता है। इसके शीर्षो के निर्देशांक होगें :

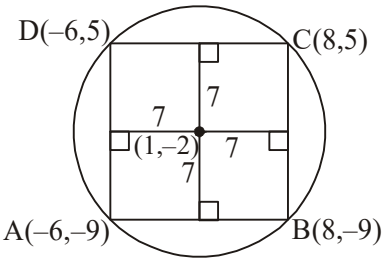
(A) (8, 5) (B) (8, 9) (C) (–6, 5) (D) (–6, –9)

Ans. ACD

Sol. $x^2 + y^2 - 2x + 4y - 93 = 0 \Rightarrow (1, -2),\ r = \sqrt{1+4+93} \Rightarrow r = 7\sqrt{2}$

$BD = 14\sqrt{2}$

$AB = 14$



21. The centre(s) of the circle(s) passing through the points (0, 0), (1, 0) and touching the circle $x^2 + y^2 = 9$ is/are :

(0, 0) (1, 0) से गुजरने वाली तथा वृत $x^2 + y^2 = 9$ को स्पर्श करने वाले वृत का केन्द्र होगा :

(A) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, \frac{3}{2}\right)$ (C) $\left(\frac{1}{2}, 2^{1/2}\right)$ (D) $\left(\frac{1}{2}, -2^{1/2}\right)$

Ans. CD

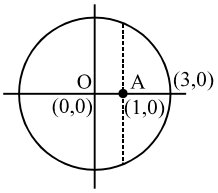
Sol. consider family of circles through (0, 0) and (1, 0)

$x(x - 1) + y^2 + \lambda y = 0$

touches $x^2 + y^2 = 9$

$\therefore$ common chord $= -x + hy + 9 = 0$ ....(1)

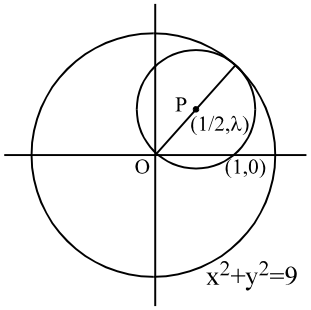
$\therefore$ perpendicular from (0, 0) on (1) is equal to 3.



$$\left|\frac{9}{\sqrt{1+\lambda^2}}\right| = 3 \quad \Rightarrow \quad \lambda^2 = 8 \quad \Rightarrow \quad \lambda = \pm 2\sqrt{2}$$

circle $x(x - 1) + y^2 + 2\sqrt{2}\, y = 0$

$\therefore$ centre $\left(\frac{1}{2}, -2^{-1/2}\right)$



Alternatively : OP = 3/2

$$\sqrt{\frac{1}{4}+\lambda^2} = \frac{3}{2} \; ; \; \frac{1}{4}+\lambda^2 = \frac{9}{4} \Rightarrow \lambda = \sqrt{2} \text{ or } -\sqrt{2}$$

22. Locus of the intersection of the two straight lines passing through (1, 0) and (–1, 0) respectively and including an angle of 45° can be a circle with :

(A) centre (1, 0) and radius $\sqrt{2}$ (B) centre (1, 0) and radius 2

(C) centre (0, 1) and radius $\sqrt{2}$ (D) centre (0, – 1) and radius $\sqrt{2}$

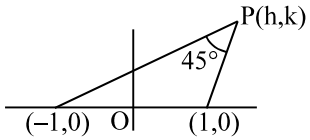
(1, 0) तथा (–1, 0) से जाने वाली दो रेखाओं के बीच यदि 45° का कोण बनता है उनका प्रतिच्छेद बिन्दु एक वृत है जिसका केन्द्र तथा त्रिज्या :

(A) (1, 0), $\sqrt{2}$ (B) (1, 0), 2 (C) (0, 1), $\sqrt{2}$ (D) (0, – 1), $\sqrt{2}$

Ans. CD

Sol. $m_1 = \frac{k}{h-1}$; $m_2 = \frac{k}{h+1}$; $\therefore \tan 45° = \left|\frac{\frac{k}{h-1} - \frac{k}{h+1}}{1+\frac{k^2}{h^2-1}}\right| = \left|\frac{k(h+1-h+1)}{h^2-1+k^2}\right|$

$h^2 + k^2 - 1 = \pm 2k]$



locus is $x^2 + y^2 \pm 2y - 1 = 0$

$x^2 + (y \pm 1)^2 = \left(\sqrt{2}\right)^2 \Rightarrow$ (0, 1) or (0, – 1) and radius = $\sqrt{2} \Rightarrow$ (C) and (D) Ans.

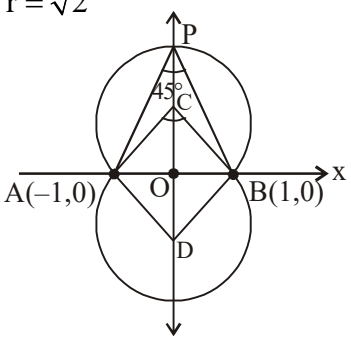
Alter :

O is circumcentre of ΔAOB

OA = OB = OC = 1

C(0,1) or C(0,–1)

$r = \sqrt{2}$



23. The circles $x^2 + y^2 + 2x + 4y - 20 = 0$ and $x^2 + y^2 + 6x - 8y + 10 = 0$:

(A) are such that the number of common tangents on them is 2

(B) are not orthogonal

(C) are such that the length of their common tangent is $5(12/5)^{1/4}$

(D) are such that the length of their common chord is $5\sqrt{\frac{3}{2}}$

दिया है $x^2 + y^2 + 2x + 4y - 20 = 0$ तथा $x^2 + y^2 + 6x - 8y + 10 = 0$ तो सही विकल्प चुनों।

(A) उभयनिष्ठ स्पर्श रेखा की संख्या 2

(B) लाम्बिक नही होगा

(C) उभयनिष्ठ स्पर्श रेखा की लम्बाई $5\,(12/5)^{1/4}$

(D) उभयनिष्ठ जीवा की लम्बाई $5\sqrt{\frac{3}{2}}7$

Ans. ACD

Sol. $r_1 = 5;\ \ r_2 = \sqrt{15};\ \ d = \sqrt{40} \ \Rightarrow$ ABCD

Alter :

$x^2 + y^2 + 2x + 4y - 20 = 0 \qquad C,(-1,-2), r_1 = \sqrt{25} = 5$

$x^2 + y^2 + 6x - 8y + 10 = 0 \qquad \Rightarrow C_2(-3,4),\ r_2 = \sqrt{15}$

(A) $C_1C_2 = \sqrt{4+36} = 2\sqrt{10}$

$r_1 + r_2 = 5 + \sqrt{15} \qquad r_1 - r_2 = 5 - \sqrt{15}$

$r_1 - r_2 < C_1C_2 < r_1 + r_2$

(B) $2g_1g_2 + 2f_1f_2 = c_1 + c_2$

LHS = 6 – 16 = –10

RHS = –20 + 10 = –10

m–2

$r_1^2 + r_2^2 = d^2$

$25 + 15 = 40$

(C) $L_T = \sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{40 - \left(25 + 15 - 10\sqrt{15}\right)}$

$$= \sqrt{10\sqrt{15}} = (1500)^{\frac{1}{4}} = 25\left(\frac{12}{5}\right)^{\frac{1}{4}}$$

24. Consider the circle $x^2 + y^2 - 8x - 18y + 93 = 0$ with centre 'C' and point P(2, 5) outside it. From the point P, a pair of tangents PQ and PR are drawn to the circle with S as the midpoint of QR. The line joining P to C intersects the given circle at A and B. Which of the following hold(s) good?

(A) CP is the arithmetic mean of AP and BP (B) PR is the geometric mean of PS and PC

(C) PS is the harmonic mean of PA and PB (D) The angle between the two tangents from P is $\tan^{-1}\left(\frac{3}{4}\right)$

वृत $x^2 + y^2 - 8x - 18y + 93 = 0$ का केन्द्र 'C' तथा बिन्दु P(2, 5) वृत के बाहर है PQ तथा PR वृत की स्पर्श रेखाऐं है जहाँ QR का मध्य बिन्दु S है P को C से मिलाने वाली रेखा वृत को A एवं B पर मिलती है। तो सही विकल्प चुनों :

(A) AP एवं BP का समान्तर माध्य CP होगा (B) PS एवं PC का गुणोत्तर माध्य PR होगा

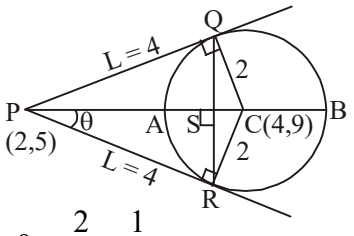
(C) PA एवं PC का हरात्मक माध्य PS होगा (D) P से खीचे गये दोनों स्पर्श रेखाओं के बीच का कोण $\tan^{-1}\left(\frac{3}{4}\right)$ होगा

Ans. ABC

Sol. Radius = $\sqrt{16+81-93} = 2$

$CP = \sqrt{20}$; $AP = \sqrt{20} - 2$; $BP = \sqrt{20} + 2$

$\Rightarrow$ $CP = \frac{AP + BP}{2}$ $\Rightarrow$ (A) is correct.



Now, let $L = PR = \sqrt{(PC)^2 - r^2} = \sqrt{20-4} = 4 = PQ$; $\tan\theta = \frac{2}{4} = \frac{1}{2}$

Also $\cos\theta = \frac{PS}{PR}$ $\Rightarrow$ $PS = PR\cos\theta = 4 \cdot \left(\frac{2}{\sqrt{5}}\right) = \frac{8}{\sqrt{5}}$

Harmonic Mean between PA and PB = $\frac{2\left(\sqrt{20}-2\right)\left(\sqrt{20}+2\right)}{2\sqrt{20}} = \frac{16}{2\sqrt{5}} = \frac{8}{\sqrt{5}} = PS \Rightarrow$ (C) is correct

Hence $(PS)(PC) = \left(\frac{8}{\sqrt{5}}\right)\left(\sqrt{20}\right) = 16 = (PR)^2$

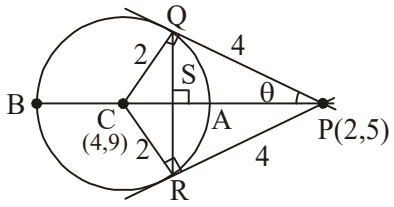
$\Rightarrow$ PR is the Geometric Mean of PS and PC $\Rightarrow$ (B) is correct

Now angle between the two tangents is 2θ, then (As $\tan\theta = \frac{1}{2}$)

$$2\tan^{-1}\left(\frac{1}{2}\right) = \tan^{-1}\left(\frac{2\cdot\frac{1}{2}}{1-\frac{1}{4}}\right) = \tan^{-1}\left(\frac{4}{3}\right) \quad \Rightarrow \quad \text{(D) is incorrect}$$

Alter :

$x^2 + y^2 - 8x - 18y + 93 = 0 \quad \Rightarrow C(4,9), \quad r = \sqrt{16+81-93} = 2$



(A) $\left.\begin{matrix} AP = d - r \\ CP = d \\ BP = d + r \end{matrix}\right\} \Rightarrow$ A.P.

(B) $PQ = PR = \sqrt{S,} = \sqrt{4+25-16-90+93} = 4$

$PC = \sqrt{4+16} = 2\sqrt{5}$

$\Delta PSQ \sim \Delta PQC$

$\frac{PS}{PQ} = \frac{PQ}{PL} \quad \Rightarrow PS = \frac{4}{2\sqrt{5}} x^4 = \frac{8}{\sqrt{5}}$

$GM(PS, PC) = \sqrt{\frac{8}{\sqrt{5}} \times 2\sqrt{5}} = 4 = PR$

PR is GM of PS and PC

(C) $PA = CP - r = 2\sqrt{5} - 2$

$PB = CP + r = 2\sqrt{5} + 2$

$HM(PA \text{ and } PB) = \frac{2(16)}{4\sqrt{5}} = \frac{8}{\sqrt{5}} = PS$

PS is HM of PA and PB

(D) $\tan\theta = \frac{1}{2}, \ \tan 2\theta = \frac{2 \times \frac{1}{2}}{1 - \frac{1}{4}} = \frac{4}{3}$

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. C_1 & C_2 are two concentric circles, the radius of C_2 being twice that of C_1. From a point P on C_2, tangents PA & PB are drawn to C_1. Prove that the centroid of the triangle PAB lies on C_1 :

दो सहकेन्द्रीय वृत्त C_1 एवं C_2 है। वृत्त C_2 की त्रिज्या वृत्त C_1 की त्रिज्या की दो गुनी है। C_2 के बिन्दु P से वृत्त C_1 पर दो स्पर्श रेखाएँ PA एवं PB खींची जाती है। सिद्ध कीजिए कि त्रिभुज PAB का केन्द्रक वृत्त C_1 पर स्थित हैं :

Ans.

Sol. ΔPAB

$$\sin \angle APM = \frac{1}{2} \Rightarrow \theta = 30^o$$

$$2\theta = 60^o$$

ΔAPB is equilateral Δ

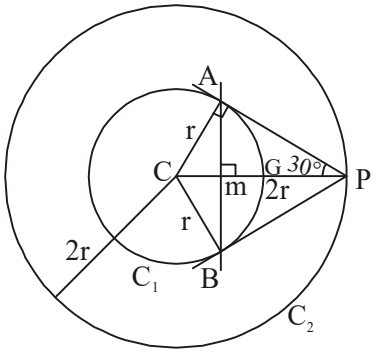
$$AP = 2r\cos 30^o = \sqrt{3}r$$

$$PM = \sqrt{3}r\cos 30^o = \frac{3r}{2}$$

$$PG = CP - CG = r$$

So, PM is median of ΔAPB and G divides median in ratio 2 :1,

Hence, G lies on Circle C_1.



2. Two vertices of an equilateral triangle are (– 1, 0) and (1, 0) and its third vertex lies above the x–axis. The equation of its circumcirlce is _____.

समबाहु त्रिभुज के दो शीर्ष (– 1, 0) एवं (1, 0) है तथा इसका तृतीय शीर्ष x–अक्ष के ऊपर है। इस त्रिभुज के परिवृत्त की समीकरण _____ हैं।

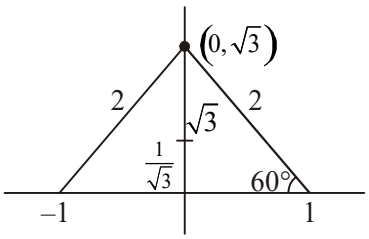
Ans. $x^2 + y^2 - \frac{2y}{\sqrt{3}} - 1 = 0$

Sol. $G \equiv 0 = \left(0, \frac{1}{\sqrt{3}}\right)$

$$r = \sqrt{1 + \frac{1}{3}} = \frac{2}{\sqrt{3}}$$

Circle $\Rightarrow x^2 + \left(y - \frac{1}{\sqrt{3}}\right)^2 = \frac{4}{3}$

$\Rightarrow x^2 + y^2 - \frac{2y}{\sqrt{3}} + \frac{1}{3} - \frac{4}{3} = 0$



3. Let $S : x^2 + y^2 - 8x - 6y + 24 = 0$ be a circle and O is the origin. Let OAB is the line intersecting the circle at A and B. On the chord AB a point P is taken. The locus of the point P in each of the following cases.

(i) OP is the arithmetic mean of OA and OB

(ii) OP is the geometric mean of OA and OB

(iii) OP is the harmonic mean between OA and OB

माना $S : x^2 + y^2 - 8x - 6y + 24 = 0$ एक वृत है तथा O मूल बिन्दु है। माना OAB एक रेखा है जो वृत को A तथा B पर प्रतिच्छेद करती है। जीवा AB पर एक बिन्दु P लिया गया है। P का बिन्दु पथ ज्ञात कीजिये।

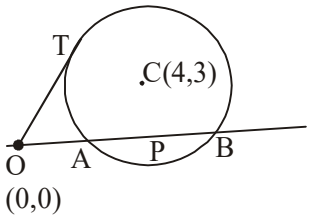
(i) यदि OA तथा OB का समानान्तर माध्य OP है।

(ii) यदि OA तथा OB का गुणोत्तर माध्य OP है।

(iii) यदि OA तथा OB का हरात्मक माध्य OP है।

Ans. (i) $x^2 + y^2 - 4x - 3y = 0$ (ii) $x^2 + y^2 = 24$ (iii) $4x + 3y = 24$

Sol. $S : x^2 + y^2 - 8x - 6y + 24 = 0$



(a) $OP = \frac{OA + OB}{2}$

P is mid point of AB

$CP \perp OP$

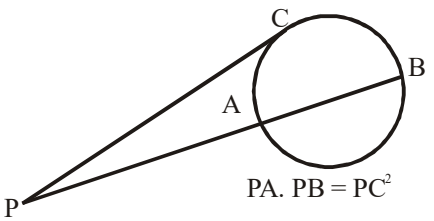
$(x - 4)x - (y - 3)y = 0$

$x^2 + y^2 - 4x - 3y = 0$

(b) $OP = \sqrt{OA \times OB}$

$= \sqrt{OT^2} = |OT| = \sqrt{S_1} = \sqrt{24}$

$x^2 + y^2 = 24$

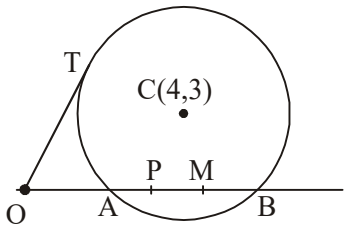


(c) $OP = \dfrac{2OA \cdot OB}{OA + OB}$

$\Rightarrow OA + OB = 2OM$

$OP = \dfrac{OA \cdot OB}{OM}$

$OP \cdot OM = OA \times OB$



$\therefore$ A and M are harmonic conjugates of P & B.

$\dfrac{AM}{PM} = \dfrac{AB}{MB} \Rightarrow \dfrac{AM}{PM} = 2$

$\therefore$ P is mid point of A and M.

$\therefore$ Locus O & P $x^2 + y^2 - 8x - 6y + 24 - (x^2 + y^2 - 4x - 3y) = 0$

4. A circle $x^2 + y^2 + 4x - 2\sqrt{2}\ y + c = 0$ is the director circle of circle S_1 and S_1 is the director circle of circle S_2 and so on. If the sum of radii of all these circles is 2, then the value of c is equal to $\sqrt{n}$ where $n \in N$. Find the value of n.

एक वृत $x^2 + y^2 + 4x - 2\sqrt{2}\ y + c = 0$ अन्य वृत S_1 का नियामक वृत है तथा S_1 अन्य वृत S_2 का नियामक वृत है तथा इस प्रकार श्रेणी आगे बढ़ रही है। यदि इन सभी वृतो की त्रिज्याओं का योग 2 है, तो c का मान $\sqrt{n}$ है जहाँ $n \in N$ है। n का मान ज्ञात कीजिए।

Ans. 32

Sol. $x^2 + y^2 + 4x - 2\sqrt{2}y + c = 0$ $\qquad C \equiv \left(-2, \sqrt{2}\right)$

$r = \sqrt{4 + 2 - c} = \sqrt{6 - c}$

$r + \dfrac{r}{\sqrt{2}} + \dfrac{r}{2} + ...\infty = \dfrac{r}{1 - \dfrac{1}{\sqrt{2}}} = 2$

$r = \sqrt{2}\left(\sqrt{2} - 1\right) = 2 - \sqrt{2}$

$\sqrt{6 - c} = 2 - \sqrt{2}$

$6 - c = 4 + 2 - 4\sqrt{2}$

$c = 4\sqrt{2} = \sqrt{32}$

$n = 32$

5. A rhombus ABCD has sides of length 10. A circle with centre 'A' passes through C (the opposite vertex) likewise, a circle with centre B passes through D. If the two circles are tangent to each other, find the area of the rhombus.

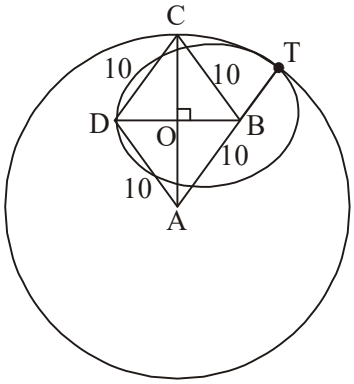
एक समचतुर्भुज ABCD की भुजाओं की लम्बाई 10 है। एक वृत जिसका केन्द्र 'A' है, C (विपरीत शीर्ष) से गुजरता है इसी प्रकार एक वृत जिसका केन्द्र B है, D से गुजरता है यदि दोनों वृत्त एक–दूसरे को स्पर्श करते है तो समचतुर्भुज का क्षेत्रफल ज्ञात कीजिए।

Ans.

Sol. Let AC = R = Radius of larger circle

BD = r = Radius of smaller circle

A, B, T are collinear. So,



$AB = AT - BT$

$10 = R - r$ ...(1)

AC, BD, perpendicular bisect each other.

Hence in $\Delta AOB \Rightarrow OA^2 + OB^2 = AB^2$

$$\Rightarrow \left(\frac{R}{2}\right)^2 + \left(\frac{r}{2}\right)^2 = 10^2$$

$\Rightarrow R^2 + r^2 = 400$ ...(2)

From equation $(1)^2 \rightarrow R^2 + r^2 - 2Rr = 100$

$Rr = 150$ ...(3)

Area of rhombus ABCD = $4 \times \Delta AOB = 4 \times \frac{1}{2} \times \frac{R}{2} \times \frac{r}{2}$

$$= \frac{Rr}{2} = \frac{150}{2} = 75$$

6. An isosceles right angled triangle whose sides are 1, 1, $\sqrt{2}$ lies entirely in the first quadrant with the ends of the hypotenuse on the coordinate axes. If it slides prove that the locus of its centroid is $(3x - y)^2 + (x - 3y)^2 = \frac{32}{9}$.

एक समद्विबाहु समकोणीय त्रिभुज जिसकी भुजायें 1, 1, $\sqrt{2}$ है इस तरह से प्रथम चतुर्थांश में स्थित है कि कर्ण के अंतिम बिन्दु निर्देशांक अक्षों पर स्थित है। यदि कर्ण फिसलता है तो सिद्ध करों कि केन्द्रक का बिन्दुपथ $(3x - y)^2 + (x - 3y)^2 = \frac{32}{9}$ होगा।

Ans. 75 sq. unit

Sol. $3h = \sqrt{2}\cos\theta + \cos(45^\circ - \theta)$

$$3h = \sqrt{2}\cos\theta + \frac{1}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta$$

$$3h = \frac{3}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta$$

$$3\sqrt{2}h = 3\cos\theta + \sin\theta \quad ...(1)$$

$$3k = \sqrt{2}\sin\theta + \sqrt{2}\sin\theta + \sin(45^\circ - \theta)$$

$$3k = 2\sqrt{2}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta - \frac{1}{\sqrt{2}}\sin\theta$$

$$3k = \frac{3}{\sqrt{2}}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta$$

$$3\sqrt{2}k = 3\sin\theta + \cos\theta \quad ...(2)$$

$$3\sqrt{2}h = 9\sqrt{2}k - 9\sin\theta + \sin\theta$$

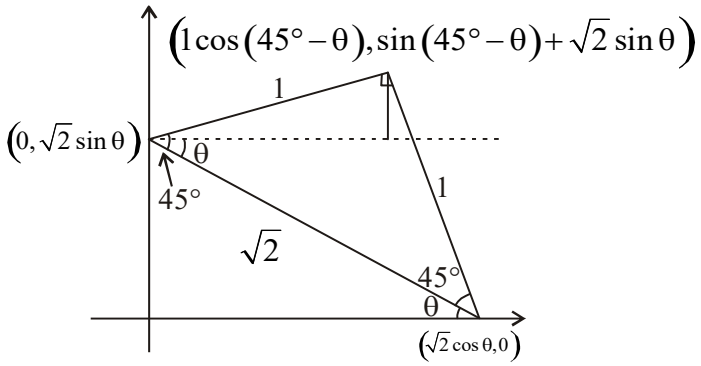
$$8\sin\theta = 9\sqrt{2}k - 3\sqrt{2}h \quad ...(3)$$

$$3\sqrt{2}k = 9\sqrt{2}h - 9\cos\theta + \cos\theta$$

$$8\cos\theta = 9\sqrt{2}h - 3\sqrt{2}k \quad ...(4)$$

$$64 = 18(3k - h)^2 + 18(3h - k)^2$$

$$\Rightarrow (3x - y)^2 + (x - 3y)^2 = \frac{32}{9}$$



7. The radical axis of the circles $x^2 + y^2 + 2gx + 2fy + c = 0$ and $2x^2 + 2y^2 + 3x + 8y + 2c = 0$ touches the circle $x^2 + y^2 + 2x - 2y + 1 = 0$. Show that either $g = 3/4$ or $f = 2$.

वृतों $x^2 + y^2 + 2gx + 2fy + c = 0$ तथा $2x^2 + 2y^2 + 3x + 8y + 2c = 0$ का मूलाक्ष अक्ष वृत्त $x^2 + y^2 + 2x - 2y + 1 = 0$ को स्पर्श करता है। प्रदर्शित करो कि या तो $g = 3/4$ या $f = 2$ होगा।

Ans.

Sol. $S_1 = x^2 + y^2 + 2gx + 2fy + c = 0$

$$S_2 = x^2 + y^2 + \frac{3}{2}x + 4y + c = 0$$

$$S_1 - S_2 = 0 \Rightarrow \left(2g - \frac{3}{2}\right)x + (2f - 4)y = 0$$

C(–1,1), $\qquad r = \sqrt{1+1-1} = 1$

$P = r$

$$\Rightarrow \frac{\left|-2g + \frac{3}{2} + 2f - 4\right|}{\sqrt{\left(2g - \frac{3}{2}\right)^2 + (2f-4)^2}} = 1$$

$$\Rightarrow \left(2g - \frac{3}{2}\right)^2 + (2f-4)^2 + 2\left(-2g + \frac{3}{2}\right)(2f-4) = \left(2g - \frac{3}{2}\right)^2 + (2f-4)^2$$

$$g = \frac{3}{4} \text{ or } f = 2$$

8. Consider a family of circles passing through two fixed points A (3, 7) & B(6, 5). The the chords in which the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ cuts the members of the family are concurrent at a point. Find the coordinates of this point.

एक वृतों का समूह बिन्दुओं A (3, 7) तथा B(6, 5) से गुजरता है तो वे जीवायें जिनमें वृत $x^2 + y^2 - 4x - 6y - 3=0$ वृतों के समूह के सदस्यों को प्रतिच्छेद करता है, एक बिन्दु पर संगामी है। उस बिन्दु के निर्देशांक ज्ञात कीजिए।

Ans. $\left(2, \frac{23}{3}\right)$

Sol. A (3,7) & B (6,5)

$S_D \equiv (x - 3)(x - 6) + (y - 7)(y - 5) = 0$

$x^2 + y^2 - 9x - 12y + 18 + 35 = 0$

$x^2 + y^2 - 9x - 12y + 53 = 0$

$$L = (y-5) = -\frac{2}{3}(x-6)$$

$\Rightarrow 3y - 15 = -2x + 12$

$2x + 3y = 27$

$S_1 \equiv x^2 + y^2 + (2\lambda - 9)x + (3\lambda - 12)y + 53 - 27\lambda = 0$

$S_2 \equiv x^2 + y^2 - 4x - 6y - 3 = 0$

$S_1 - S_2 = 0 \Rightarrow (2\lambda - 5)x + (3\lambda - 6)y + 56 - 27y = 0$

$\lambda(2x + 3y - 27) - (5x + 6y - 56) = 0$

$L_1 : 2x + 3y - 27 = 0 \Rightarrow 4x + 6y - 54 = 0$

$L_2 : 5x + 6y - 56 = 0 \Rightarrow 5x + 6y - 56 = 0$

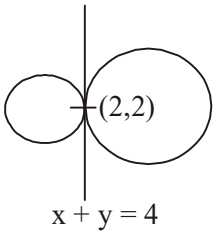
$\left(2, \frac{23}{3}\right)$

9. Find the equation of circle passing through (1, 1) belonging to the system of co–axal circles that are tangent at (2, 2) to the locus of the point of intersection of mutually perpendicular tangent to the circle $x^2 + y^2 = 4$.

बिन्दु (1, 1) से गुजरने वाले उस वृत का समीकरण ज्ञात करें जो समाक्षी निकाय से सम्बन्धित है जो वृत $x^2 + y^2 = 4$ की परस्पर लम्बवत् स्पर्श रेखाओं के प्रतिच्छेद बिन्दुओं के बिन्दुपथ को (2, 2) पर स्पर्श करता है।

Ans. $x^2 + y^2 - 3x - 3y + 4 = 0$

Sol. D.C. : $x^2 + y^2 = 8$ tangent at (2,2)



$2x + 2y = 8 \Rightarrow L : x + y = 4$

$S_P = (x - 2)^2 + (y - 2)^2 = 0$

$S : (x - 2)^2 + (y - 2)^2 + \lambda x + \lambda y - 4\lambda = 0 \rightarrow (1,1)$

$\Rightarrow 1 + 1 + \lambda + \lambda - 4\lambda = 0 \Rightarrow \lambda = 1$

$S \equiv x^2 + y^2 - 3x - 3y + 4 = 0$

10. The circle $C : x^2 + y^2 + kx + (1 + k)y - (k + 1) = 0$ passes through two fixed points for every real number k. Find
(i) the coordinates of these two points. (ii) the minimum value of the radius of a circle C.

वृत्त $C : x^2 + y^2 + kx + (1 + k)y - (k + 1) = 0$, k के सभी वास्तविक मानों के लिये दो अचर बिन्दुओं से गुजरता है। ज्ञात करें।
(i) उन दो बिन्दुओं के निर्देशांक (ii) वृत C की त्रिज्या का न्यूनतम मान

Ans. (1, 0) & (1/2,1/2); $r = \frac{1}{2\sqrt{2}}$

Sol. $C : x^2 + y^2 + kx + (1 + k)y - (k + 1) = 0$

$C : (x^2 + y^2 + y - 1) + k(x + y - 1) = 0$

$x = 1 - y$

$(1-y)^2 + y^2 + y - 1 = 0 \Rightarrow 2y^2 - y = 0$

$\Rightarrow y = 0, y = \frac{1}{2}$

$x = 1, x = \frac{1}{2}$

(i) $A(1, 0),\ B\left(\frac{1}{2}, \frac{1}{2}\right)$

(ii) $2r = \sqrt{\frac{1}{4} + \frac{1}{4}} \Rightarrow r = \frac{1}{2\sqrt{2}}$

11. Show that the locus of the centres of a circle which cuts two given circles orthogonally is a straight line & hence deduce the locus of the centers of the circles which cut the circles $x^2 + y^2 + 4x - 6y + 9=0$ and $x^2 + y^2 - 5x + 4y + 2 = 0$ orthogonally. Interpret the locus.

प्रदर्शित करो कि दो दिये गये वृतों को लम्बकोणीय प्रतिच्छेद करने वाले वृतों के केन्द्र का बिन्दुपथ एक सरलरेखा होता है। फलस्वरूप वृतों $x^2 + y^2 + 4x - 6y + 9=0$ तथा $x^2 + y^2 - 5x + 4y + 2 = 0$ को लम्बकोणीय प्रतिच्छेद करने वाले वृतों के केन्द्र का बिन्दुपथ ज्ञात कीजिए। बिन्दुपथ की विवेचना कीजिए।

Ans. $9x - 10y + 7 = 0$; radical axis

Sol.

$S_1 = x^2 + y^2 + 4x - 6y + 9 = 0$	$x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$
$S_2 : x^2 + y^2 - 5x + 4y + 2 = 0$	$x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$
$S = x^2 + y^2 + 2gx + 2fy + c = 0$	$x^2 + y^2 + 2gx + 2fy + c = 0$
$S \& S_1 \Rightarrow 4g - 6f = c + 9$	$2gg_1 + 2ff_1 = c_1 + c$
$S \& S_2 \Rightarrow -5g + 4f = c + 2$	$2gg_2 + 2ff_2 = c_2 + c$
$9g - 10f = 7$	$2g(g_1 - g_2) + 2f(f_1 - f_2) = c - c_2$
$(-g_1 - f) = (h, k)$	$2x(g_1 - g_2) + 2y(f_1 - f_2) + (c_1 - c_2) = 0$
$-9x + 10y = 7$	$S_1 - S_2 = 0 \rightarrow$ radical axis.
$9x - 10y + 7 = 0$	

12. Find the equation of a circle which touches the lines $7x^2 - 18xy + 7y^2 = 0$ and the circle $x^2 + y^2 - 8x - 8y = 0$ and is contained in the given circle.

रेखाओं $7x^2 - 18xy + 7y^2 = 0$ तथा वृत्त $x^2 + y^2 - 8x - 8y = 0$ को स्पर्श करने वाले वृत का समीकरण ज्ञात करें। यदि यह वृत दिये

गये वृत के अन्दर स्थित है।

Ans. $x^2 + y^2 - 12x - 12y + 64 = 0$

Sol. $7x^2 - 18xy + 7y^2 = 0$

Coefficient of x^2 = coeff. of y^2

Complimentary angles with x–axis.

$\rightarrow$ angle bisectors are $y = x$ or $y = -x$ center lies on angle bisector.

$x^2 + y^2 - 8x - 8y = 0$ (4,4), $r = 4\sqrt{2}$

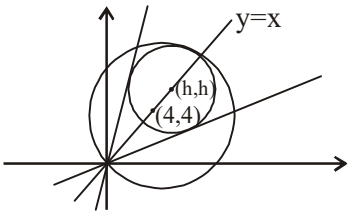
Circle lies in Ist quadrant.

So the circle contained in it must lie in Ist quadrant.

So the centre must lie on $y = x$ so center is (h,h).

Circle : $(x - h)^2 + (y - h)^2 = R^2$

$$y = mx \Rightarrow \text{C.O.T.} \Rightarrow P = R \Rightarrow \left|\frac{mh - h}{\sqrt{1+m^2}}\right| = R$$



$$\Rightarrow h^2 + m^2h^2 - 2mh^2 = R^2 + R^2m^2$$

$$\Rightarrow \left(R^2 - h^2\right)m^2 + 2h^2m + \left(R^2 - h^2\right) = 0$$

$$\Rightarrow m^2 + \left(\frac{2h^2}{R^2 - h^2}\right)m + 1 = 0$$

On comparing with lines we have $\frac{2h^2}{R^2 - h^2} = \frac{-18}{7}$

$$m^2 - \frac{18}{7}m + 1 = 0$$

$$\frac{2h^2}{R^2 - h^2} = -\frac{18}{7} \Rightarrow 14h^2 = -18R^2 + 18h^2$$

$$\Rightarrow 4h^2 = 18R^2 \Rightarrow h^2 = \frac{9R^2}{2} \Rightarrow R^2 = \frac{2h^2}{9}$$

Circle $(x - h)^2 + (y - h)^2 = R^2$ touches $x^2 + y^2 - 8x - 8y = 0$ internally.

(4,4), $r = 4\sqrt{2}$

$d = r_1 - r_2 \Rightarrow \sqrt{(h-4)^2 + (h-4)^2} = 4\sqrt{2} - R$

$2(h-4)^2 \Rightarrow 2\left(\frac{3R}{\sqrt{2}} - 4\right)^2 = \left(4\sqrt{2} - R\right)^2$

$\Rightarrow \left(3R - 4\sqrt{2}\right)^2 - \left(R - 4\sqrt{2}\right)^2 = 0$

$\Rightarrow \left(4R - 8\sqrt{2}\right)^2 - (2R) = 0$

$R = 2\sqrt{2}$, $h = 6$

Circle = $(x-6)^2 + (y-6)^2 = 8$

$x^2 + y^2 - 12x - 12y + 64 = 0$

13. Let $2x^2 + y^2 - 3xy = 0$ be the equation of a pair of tangents drawn from the origin 'O' to a circle of radius 3 with centre in the first quadrant. If A is one of the points of contact, find the length of OA.

माना कि मूल बिन्दु Q से एक वृत जिसकी त्रिज्या 3 व केन्द्र प्रथम पाद में स्थित है, पर खीची गई स्पर्श रेखा युग्म का समीकरण $2x^2 + y^2 - 3xy = 0$ है। यदि A एक स्पर्श बिन्दु है तो OA की लम्बाई ज्ञात कीजिए।

Ans. $OA = 3(3 + \sqrt{10})$

Sol. $2x^2 + y^2 - 3xy = 0$

$y^2 - 2xy - xy + 2x^2 = 0$

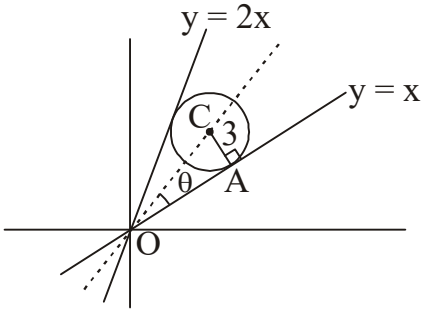
$(y - x)(y - 2x) = 0$

Circle : r = 3, center in I[st] quadrant

$\tan\theta = \frac{CA}{OA} = \frac{3}{OA}$

$OA = \frac{CA}{\tan\theta} = \frac{3}{\tan\theta}$

$\tan 2\theta = \frac{2-1}{1+2} = \frac{1}{3}$



$\frac{2\tan\theta}{1-\tan^2\theta} = \frac{1}{3}$ $\left(\because \tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|\right)$, $m_1 =$, $m_2 = 2$

$\Rightarrow 6\tan\theta = 1 - \tan^2\theta \Rightarrow \tan^2\theta + 6\tan\theta - 1 = 0$

$$\tan\theta = \frac{-6 \pm \sqrt{36+4}}{2} = \frac{\sqrt{40}-6}{2} = \sqrt{10}-3$$

$$OA = \frac{3}{\sqrt{10}-3} = \frac{3\left(\sqrt{10}+3\right)}{10-9} = 3\left(\sqrt{10}+3\right)$$

14 Obtain the equations of the straight lines passing through the point A(2, 0) & making 45° angle with the tangent at A to the circle $(x+2)^2 + (y-3)^2 = 25$. Find the equations of the circles each of radius 3 whose centres are on these straight lines at a distance of $5\sqrt{2}$ from A.

बिन्दु A(2, 0) से गुजरने वाली तथा वृत्त $(x + 2)^2 + (y - 3)^2 = 25$ की A पर स्पर्श रेखा के साथ 45° का कोण बनाने वाली सरल रेखाओं का समीकरण ज्ञात करें। उन वृत्तों का समीकरण ज्ञात करें जिनमें से प्रत्येक की त्रिज्या 3 है तथा केन्द्र उन सरल रेखाओं पर A से $5\sqrt{2}$ दूरी पर स्थित है।

Ans. $x - 7y = 2$, $7x + y = 14$; $(x-1)^2 + (y-7)^2 = 3^2$;
$(x-3)^2 + (y+7)^2 = 3^2$; $(x-9)^2 + (y-1)^2 = 3^2$;
$(x+5)^2 + (y+1)^2 = 3^2$

Sol. A (2,0), $\theta = 45^\circ$

Circle : $x^2 + y^2 + 4x - 6y - 12 = 0$ (–2,3), 5 = r

Tangent at A(2,0), T = 0

$2x + 2(x + 2) - 3(y + 0) - 12 = 0$

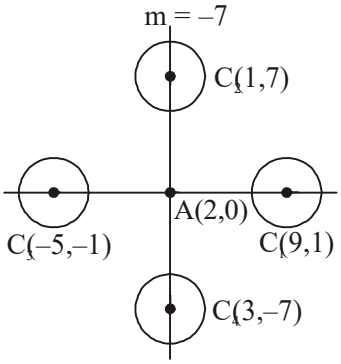
$4x - 3y - 8 = 0$ .....(1) $\Rightarrow$ Slope $= \frac{4}{3}$

Let m be the slope of the line making angle of 45° with the line .....(1)

$$\tan 45^\circ = \left|\frac{m - \frac{4}{3}}{1 + m\frac{4}{3}}\right| = \left|\frac{3m-4}{3+4m}\right| \Rightarrow 1 = \left|\frac{3m-4}{3+4m}\right| \Rightarrow 3 + 4m = \pm(3m-4) \quad m = 7, \frac{1}{7}$$

$(y - 0) = -7(x - 2)$ $\qquad$ $(y-0) = \frac{1}{7}(x-2)$

$7x + y = 14$ $\qquad$ $x - 7y - 2 = 0$



$m = \frac{1}{7}, \sin\theta = \frac{1}{5\sqrt{2}}, \cos\theta = \frac{7}{5\sqrt{2}}$

$\sin\theta = \frac{7}{5\sqrt{2}}$

$\cos\theta = -\frac{1}{5\sqrt{2}}$

$C_1 : (x-9)^2 + (y-1)^2 = 9$

$C_2 : (x-1)^2 + (y-7)^2 = 9$

$C_3 : (x+5)^2 + (y+1)^2 = 9$

$C_4 : (x-3)^2 + (y+7)^2 = 9$

15 A circle C is tangent to the x and y axis in the first quadrant at the points P and Q respectively. BC and AD are parallel tangents to the circle with slope – 1. If the points A and B are on the y-axis while C and D are on the x-axis and the area of the figure ABCD is $900\sqrt{2}$ sq. units then find the radius of the circle.

एक वृत्त x अक्ष व y अक्ष को प्रथम चतुर्थांश में क्रमशः P व Q पर स्पर्श करता है। BC व AD वृत्त की दो समान्तर स्पर्श रेखायें है। जिनकी प्रवणता – 1 है। यदि A व B y-अक्ष पर है, C व D x अक्ष पर है तथा क्षेत्र ABCD का क्षेत्रफल $900\sqrt{2}$ वर्ग ईकाई है तो वृत्त की त्रिज्या ज्ञात कीजिए।

Ans. r = 15

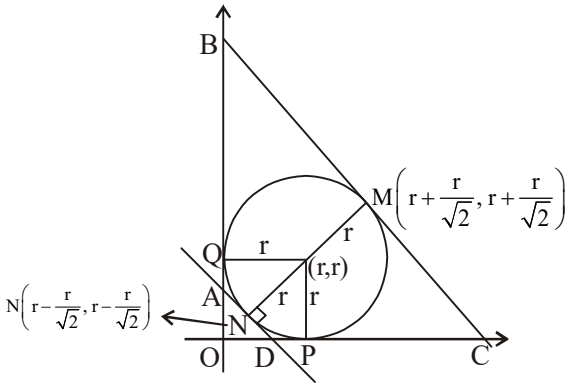
Sol. $AD :\rightarrow x + y = 2r - \sqrt{2}r \qquad \Rightarrow AD = \left(2r - \sqrt{2}r\right)\sqrt{2}$

$BC :\rightarrow x + y = 2r + \sqrt{2}r \qquad \Rightarrow BC = \left(2r + \sqrt{2}r\right)\sqrt{2}$

$\frac{1}{2}(AD + BC) \times 2r = 900\sqrt{2}$

$\Rightarrow (4r)\sqrt{2}r = 900\sqrt{2}$

$$\Rightarrow r^2 = \frac{900}{4} \Rightarrow r = \frac{30}{2} = 15$$



16 Circles C_1 and C_2 are externally tangent and they are both internally tangent to the circle C_3. The radii of C_1 and C_2 are 4 and 10, respectively and the centres of the three circles are collinear. A chord of C_3 is also a common internal tangent of C_1 and C_2. Given that the length of the chord is $\frac{m\sqrt{n}}{p}$ where *m*, *n* and *p* are positive integers, m and p are relatively prime and n is not divisible by the square of any prime, find the value of (m + n + p).

वृत्त C_1 व C_2 बाह्य स्पर्श करते है तथा दोनों C_3 को आन्तरिक स्पर्श करते है। C_1 व C_2 की त्रिज्यायें क्रमशः 4 व 10 है तथा तीनो वृत्तो के केन्द्र संरेखीय है। C_3 की एक जीवा C_1 व C_2 की उभयनिष्ठ आन्तरिक स्पर्श रेखा है। यदि जीवा की लम्बाई $\frac{m\sqrt{n}}{p}$ है जहाँ m, n व p धनात्मक पूर्णांक है, m व p सापेक्ष अभाज्य संख्यायें है व n किसी भी अभाज्य संख्या के वर्ग से भाज्य नही है तो (m + n + p) ज्ञात कीजिए।

Ans. 19

Sol. Radius of bigger circle is 10 + 4 = 14

Length of chord = DE = 2 × MD

ΔCMD is right angle Δ

CM = AM – AC

CM = 10 – 4 = 6

$CM^2 + MD^2 = CD^2$

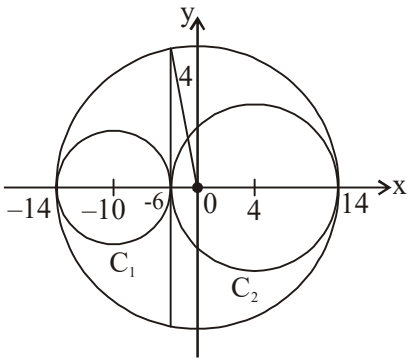
$6^2 + MD^2 = 14^2$

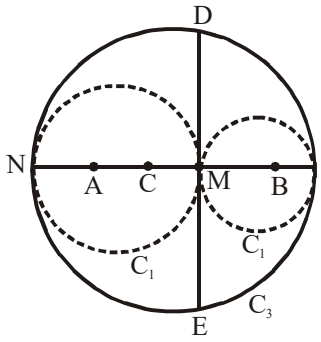
$MD^2 = 16 \Rightarrow MD = 4\sqrt{10}$

$$ED = 8\sqrt{10} = \frac{m\sqrt{n}}{P}$$

$n = 8, n = 10, p = 1$

$\Rightarrow m + n + p = 19$





17 Let $S_1 = 0$ and $S_2 = 0$ be two circles intersecting at P (6, 4) and both are tangent to x-axis and line $y = mx$ (where $m > 0$). If product of radii of the circles $S_1 = 0$ and $S_2 = 0$ is $\frac{52}{3}$, then find the value of m.

यदि $S_1 = 0$ व $S_2 = 0$ बिन्दु P (6, 4) पर प्रतिच्छेद करने वाले वृत है तथा दोनों x अक्ष व रेखा $y = mx$ (जहाँ $m > 0$) को स्पर्श करते है। यदि दोनों वृतो की त्रिज्याओं का गुणनफल $\frac{52}{3}$ है तो m का मान ज्ञात कीजिए।

Ans. $\sqrt{3}$

Sol. $S_1 = 0$ & $S_2 = 0$ P(6,4)

$$r_1 r_2 = \frac{52}{3}$$

$$(x-h)^2 + (y-r)^2 = r^2$$

$x^2 - 2hx + h^2 + y^2 - 2ry + r^2 = r^2$ $\tan\theta = \frac{r}{h} \Rightarrow h = r\cot\theta$

(6,4)

$\Rightarrow 36 - 12h + h^2 + 16 - 8r = 0$

$\Rightarrow \cot^2\theta r^2 - 8r - 12\cot\theta r + 52 = 0$

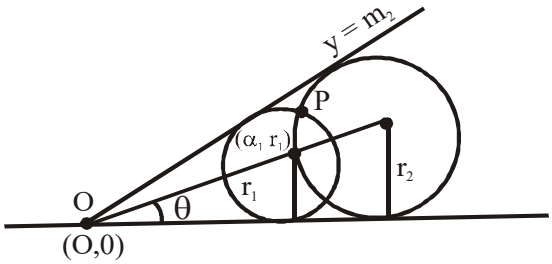
$$r_1 r_2 = \frac{52}{\cot^2\theta} = \frac{52}{3}$$

$\tan^2\theta = \frac{1}{3} \Rightarrow \tan\theta = \pm\frac{1}{\sqrt{3}}$

$\theta = 30^o$

$2\theta = 60^o$

$m = \tan 2\theta = \sqrt{3}$



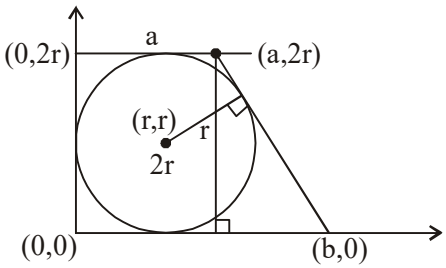
18 Given that a right angled trapezium has an inscribed circle. Prove that the length of the right angled leg is the Harmonic mean of the lengths of bases.

एक समकोणीय समलम्ब चतुर्भुज के अन्दर एक अन्तः वृत खींचा गया है। सिद्ध करें कि समकोणीय भुजा की लम्बाई आधार की भुजाओं की लम्बाईयो के हरात्मक माध्य के बराबर होगी।

Ans.

Sol. $y = \left(\frac{2r}{a-b}\right)(x-b)$

$2rx - (a-b)y - 2rb = 0$



C.O.T. $P = r \Rightarrow \left|\frac{2r^2 - (a-b)r - 2rb}{\sqrt{4r^2 + (a-b)^2}}\right| = r = 1$

$\Rightarrow (2r - a - b)^2 = 4r^2 + (a-b)^2$

$\Rightarrow 4r^2 + a^2 + b^2 - 4ar - 4br + 2ab = 4r^2 + a^2 + b^2 - 2ab$

$4ab = 4r(a+b)$

$2r = \frac{2ab}{a+b}$

19. Tangents are drawn from any point on the circle $x^2 + y^2 = R^2$ to the circle $x^2 + y^2 = r^2$. Show that if the line joining the points of intersection of these tangents with the first circle also touches the second, then $R = 2r$.

वृत $x^2 + y^2 = R^2$ की किसी बिन्दु से वृत $x^2 + y^2 = r^2$ पर स्पर्श रेखायें बनायी जाती है। सिद्ध कीजिये कि प्रथम वृत के साथ इन स्पर्श रेखाओं के प्रतिच्छेदन बिन्दुओं को मिलाने वाली रेखा यदि द्वितीय वृत को स्पर्शकरती है तो $R = 2r$ है।

Ans.

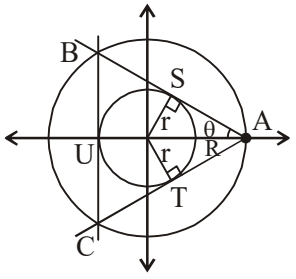
Sol. $x^2 + y^2 = R^2$

$x^2 + y^2 = r^2$

$AS = AT - \sqrt{R^2 - r^2}$

Similarly, $BS = BU = CU = CT = \sqrt{R^2 - r^2}$

Hence, ΔABC is equilateral $\Delta^r \Rightarrow R = 2r$



20. Prove that the length of the common chord of the two circles $x^2 + y^2 = a^2$ and $(x - c)^2 + y^2 = b^2$ is $\frac{1}{c}\sqrt{\{(a+b+c)(a-b+c)(a+b-c)(-a+b+c)\}}$.

सिद्ध कीजिये कि वृतों $x^2 + y^2 = a^2$ तथा $(x - c)^2 + y^2 = b^2$ कि उभयनिष्ठ जीवा की लम्बाई $\frac{1}{c}\sqrt{\{(a+b+c)(a-b+c)(a+b-c)(-a+b+c)\}}$ है।

Ans.

Sol. $S_1 : x^2 + y^2 = a^2$

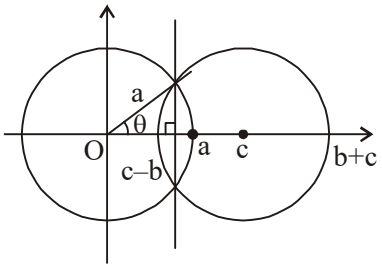
$S_2 : (x - c)^2 + y^2 = b^2$

$S_1 - S_2 = 0$

$2cx - c^2 = a^2 - b^2$

$$x = \frac{a^2 + c^2 - b^2}{2c}$$

$$L = 2\sqrt{a^2 - \left(\frac{a^2 + c^2 - b^2}{2c}\right)^2}$$



$$= \frac{1}{c}\sqrt{4a^2c^2 - \left(a^2 + c^2 - b^2\right)}$$

$$= \frac{1}{c}\sqrt{\left(a^2 + c^2 + 2ac - b^2\right)\left(b^2 - \left(a^2 + c^2 - 2bc\right)\right)}$$

$$L = \frac{1}{c}\sqrt{(a + c - b)(a + c + b)(b - a + c)(b + a - c)}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 21 to 23

Let A, B, C be three sets of real numbers (x, y) defined as

A : $\{(x, y): y \geq 1\}$

B : $\{(x, y): x^2 + y^2 - 4x - 2y - 4 = 0\}$

C : $\{(x, y): x + y = \sqrt{2}\}$

A, B एवं C वास्तविक संख्याओं (x, y) का उपसमुच्चय इस प्रकार से है कि

A : $\{(x, y) : y \geq 1\}$

B : $\{(x, y): x^2 + y^2 - 4x - 2y - 4 = 0\}$

C : $\{(x, y): x + y = \sqrt{2}\}$

21. Number of elements in the $A \cap B \cap C$ is :

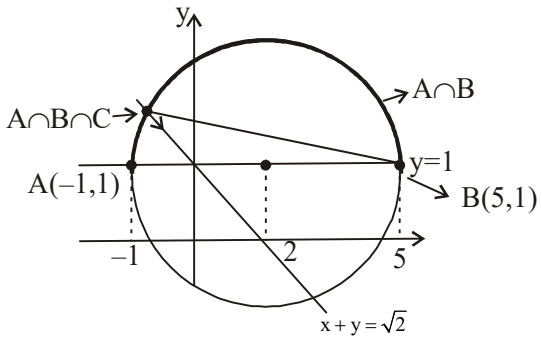
(A) 0 (B) 1 (C) 2 (D) infinite

$A \cap B \cap C$ में कुल अवयवों की संख्या :

(A) 0 (B) 1 (C) 2 (D) अनन्त

Ans. B

Sol. refer figure



(B) Only one point

22. $(x + 1)^2 + (y - 1)^2 + (x - 5)^2 + (y - 1)^2$ has the value equal to :

$(x + 1)^2 + (y - 1)^2 + (x - 5)^2 + (y - 1)^2$ का मान होगा :

(A) 16 (B) 25 (C) 36 (D) 49

Ans. C

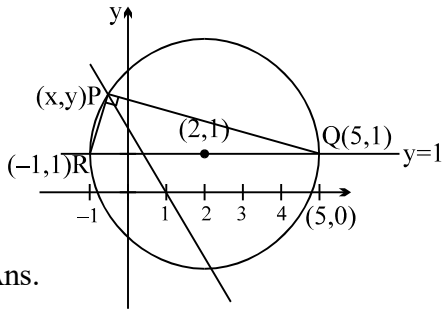
Sol. when $y = 1$

$x^2 - 4x - 5 = 0$

$(x - 5)(x + 1) = 0$

$x = -1$ or $x = 5$

$(x + 1)^2 + (y - 1)^2 + (x - 5)^2 + (y - 1)^2 = (QR)^2 = 36$ Ans.



23. If the locus of the point of intersection of the pair of perpendicular tangents to the circle B is the curve S then the area enclosed between B and S is :

यदि ऐसे बिन्दु का बिन्दुपथ जहाँ से परस्पर लम्बवत् स्पर्श रेखाऐं वृत B पर खीची जाये तो यदि वह आरेख S हो तो B एवं S के बीच परिबद्ध क्षेत्रफल होगा :

(A) 6π (B) 8π (C) 9π (D) 18π

Ans. C

Sol. equation of director circle is

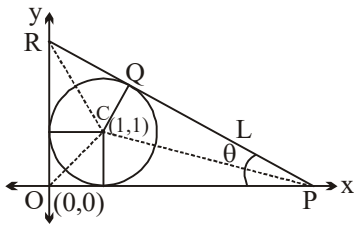
$$(x-2)^2 + (y-1)^2 = \left(3\sqrt{2}\right)^2 = 18$$

Area $= \pi\left[r_1^2 - r_2^2\right] = \pi[18-9] = 9\pi$ Ans.

Paragraph for Question Nos. 24 to 26

In the diagram as shown, a circle is drawn with centre C(1, 1) and radius 1 and a line L. The line L is tangential to the circle at Q. Further L meet the y-axis at R and the x-axis at P in such a way that the angle OPQ equals θ where $0 < \theta < \frac{\pi}{2}$.

चित्र में दर्शाये अनुसार वृत का केन्द्र C(1, 1) है तथा त्रिज्या 1 है एवं रेखा L वृत को Q पर स्पर्शी है। L y-अक्ष को R पर तथा x-अक्ष को P पर इस प्रकार काटती है कि कोण OPQ θ है जहाँ $0 < \theta < \frac{\pi}{2}$.



24. The coordinates of Q are :

Q के निर्देशांक है :

(A) $(1+\cos\theta, 1+\sin\theta)$ (B) $(\sin\theta, \cos\theta)$

(C) $(1+\sin\theta, \cos\theta)$ (D) $(1+\sin\theta, 1+\cos\theta)$

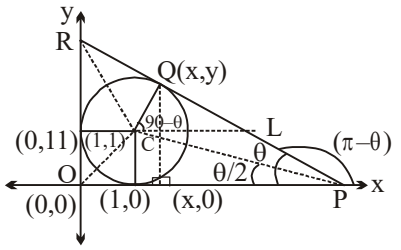
Ans. D

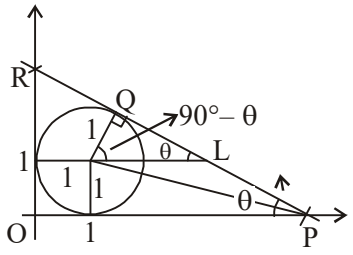
Sol. For the circle with C, using parametric

$x - 1 = \sin\theta$

$y - 1 = \cos\theta$

$\Rightarrow \left.\begin{matrix} x = 1+\sin\theta \\ y = 1+\cos\theta \end{matrix}\right\} \Rightarrow$ $(1 + \sin\theta, 1 + \cos\theta)$ Ans.





25. Equation of the line PR is :

PR का समीकरण है :

(A) $x\cos\theta + y\sin\theta = \sin\theta + \cos\theta + 1$ (B) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta - 1$

(C) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta + 1$ (D) $x\tan\theta + y = 1 + \cot\left(\frac{\theta}{2}\right)$

Ans. C

Sol. $m_{CQ} = \frac{y-1}{x-1} = \cot\theta$

$m_{PR} = -\tan\theta$

equation of PR is

$$y - (1 + \cos\theta) = -\tan\theta\left(x - (1+\sin\theta)\right)$$

$$y + x\tan\theta = (1 + \cos\theta) + \tan\theta\,(1 + \sin\theta) = \frac{\cos\theta(1+\cos\theta) + \sin\theta(1+\sin\theta)}{\cos\theta}$$

$$y + x\tan\theta = \frac{1+\cos\theta+\sin\theta}{\cos\theta}$$

$x\sin\theta + y\cos\theta = (\cos\theta + \sin\theta + 1)$ Ans.

26. If the area bounded by the circle, the x-axis and PQ is $A(\theta)$, then $A\left(\frac{\pi}{4}\right)$ equals :

यदि x-अक्ष तथा वृत तथा PQ द्वारा परिबद्ध क्षेत्रफल $A(\theta)$ है, तो $A\left(\frac{\pi}{4}\right)$ का मान है :

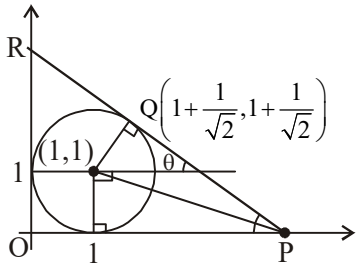
(A) $\sqrt{2}+1-\frac{3\pi}{8}$ (B) $\sqrt{2}-1+\frac{3\pi}{8}$ (C) $\sqrt{2}+1+\frac{\pi}{8}$ (D) $\sqrt{2}-1+\frac{\pi}{8}$

Ans. A

Sol. $A(\theta) = \text{Area of quadrilateral} - \text{area of sector}$

$$= \cot\frac{\theta}{2} - \frac{\pi-\theta}{2}$$

if $\theta = \frac{\pi}{4}$, $\quad A\left(\frac{\pi}{4}\right) = \cot\frac{\pi}{8} - \frac{3\pi}{8} = \sqrt{2}+1-\frac{3\pi}{8}$ Ans.



MATCH THE COLUMN TYPE QUESTION(S)

27. Consider two circles C_1 of radius 'a' and C_2 of radius 'b' ($b > a$) both lying in the first quadrant and touching the coordinate axes. In each of the conditions listed in column–I, the ratio of $\frac{b}{a}$ is given in column-II :

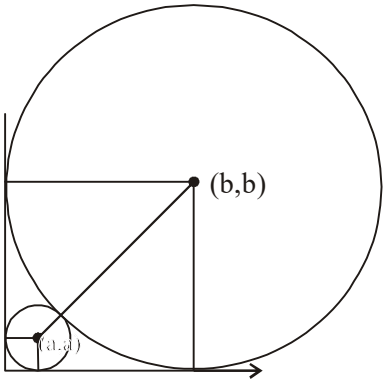
Column–I	**Column–II**
(A) C_1 and C_2 touch each other	(P) $2+\sqrt{2}$
(B) C_1 and C_2 are orthogonal	(Q) 3
(C) C_1 and C_2 intersect so that the common chord is longest	(R) $2+\sqrt{3}$
(D) C_2 passes through the centre of C_1	(S) $3+2\sqrt{2}$
	(T) $3-2\sqrt{2}$

दो वृत्त C_1 जिसकी त्रिज्या 'a' व C_2 जिसकी त्रिज्या 'b' ($b > a$) दोनों प्रथम पाद में निर्देशांक अक्षों को स्पर्श करते हुये स्थित है। स्तम्भ–I में कुछ परिस्थितियों दी गई है तथा स्तम्भ–II में $\frac{b}{a}$ का अनुपात दिया गया है :

Column–I	**Column–II**
(A) C_1 व C_2 परस्पर स्पर्श करते है	(P) $2+\sqrt{2}$
(B) C_1 व C_2 परस्पर लम्बकोणीय है	(Q) 3
(C) C_1 व C_2 इस प्रकार से प्रतिच्छेद करते है कि उभयनिष्ठ जीवा की लम्बाई अधिकतम है	(R) $2+\sqrt{3}$
(D) C_2, C_1के केन्द्र से गुजरता है	(S) $3+2\sqrt{2}$
	(T) $3-2\sqrt{2}$

Ans. (A) → S; (B) → R; (C) → Q; (D) → P

Sol. (A) $C_1 \to r_1 = a$, $C_2 \to r_2 = b$



$$\sqrt{(b-a)^2+(b-a)^2} = a+b$$

$2(a^2 + b^2 - 2ab) = a^2 + b^2 + 2ab$

$a^2 + b^2 - 6ab = 0$

$$K^2 - 6KA = 0 \Rightarrow K = \frac{6+\sqrt{32}}{2}$$

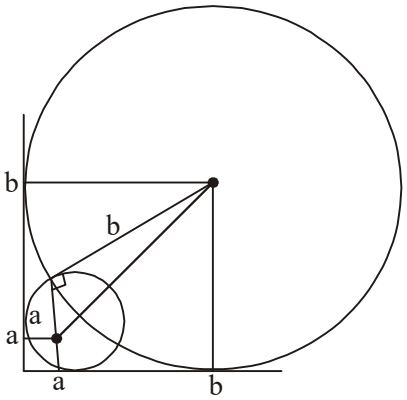
$$K = 3+2\sqrt{2} > 1$$

(B) $a^2 + b^2 = (a-b)^2 + (a-b)^2$

$a^2 + b^2 = 2a^2 + 2b^2 - 4ab$

$a^2 + b^2 - 4ab = 0$

$K^2 - 4K + 1 = 0$



$$K = \frac{4+\sqrt{12}}{2} = 2+\sqrt{3}$$

(C) Longest common chord = 2a

C_2 bisect circumference of C_1.

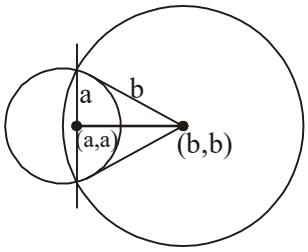
$b^2 = a^2 + (b-a)^2 + (b-a)^2$

$\Rightarrow b^2 = a^2 + 2a^2 + 2b^2 - 4ab$

$\Rightarrow 3a^2 + b^2 - 4ab = 0$

$\Rightarrow K^2 - 4K + 3 = 0$

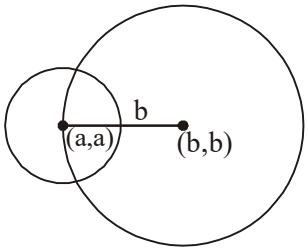
$\Rightarrow K=3,\ K \neq 1$ $\qquad b > a$



(D) $(b-a)^2 + (b-a)^2 = b^2$

$\Rightarrow b^2 + 2a^2 - 4ab = 0$

$\Rightarrow K^2 - 4K + 2 = 0$



$$K = \frac{4 \pm \sqrt{16-8}}{2} = 2+\sqrt{2}$$